Supplementary Material

Solvent Optimization for Caffeine and Tannic Acid Extraction from Guarana Seed: Insights from Molecular Dynamics Simulations

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Table S1 Number of atoms in different solvents with initial system details for system I (1:1)

Solvent	Extraction System using Ethanol	Extraction System using Ethyl Acetate	Extraction System using DES
Initial Box dimension(nm^3)	21 x 21 x 21	32.99 x 32.99 x 32.99	20 x 20 x 20
Final Box Dimension(nm^3)	6.76 x 6.76 x 6.76	6.84 x 6.84 x 6.84	6.72 x 6.72 x 6.72
Total number of Solvent Molecules	1667	1071	944
Total number of Caffeine Molecules	30	30	30
Total number of Catechin Molecules	4	4	4

Total number of Epicatechin Molecules	5	5	5
Total number of Theobromine Molecules	5	5	5
Total number of Theophylline Molecules	3	3	3
Total number of Tanic Acid Molecules	75	75	75
Total number of Procyanidins Molecules	10	10	10

Table S2 Number of atoms in different solvents with initial system details for system II (1:9)

Solvent	Extraction System using Ethanol	Extraction System using Ethyl Acetate	Extraction System using DES
Initial Box dimension(nm^3)	21 x 21 x 21	20 x 20 x 20	20 x 20 x 20
Final Box Dimension(nm^3)	6.86 x 6.86 x 6.86	6.91 x 6.91 x 6.91	6.61 x 6.61 x 6.61
Total number of Solvent Molecules	3000	1929	1700
Total number of Caffeine Molecules	6	6	6
Total number of Catechin Molecules	1	1	1
Total number of Epicatechin Molecules	1	1	1
Total number of Theobromine Molecules	1	1	1
Total number of Theophylline Molecules	1	1	1
Total number of Tanic Acid Molecules	15	15	15
Total number of Procyanidins Molecules	2	2	2

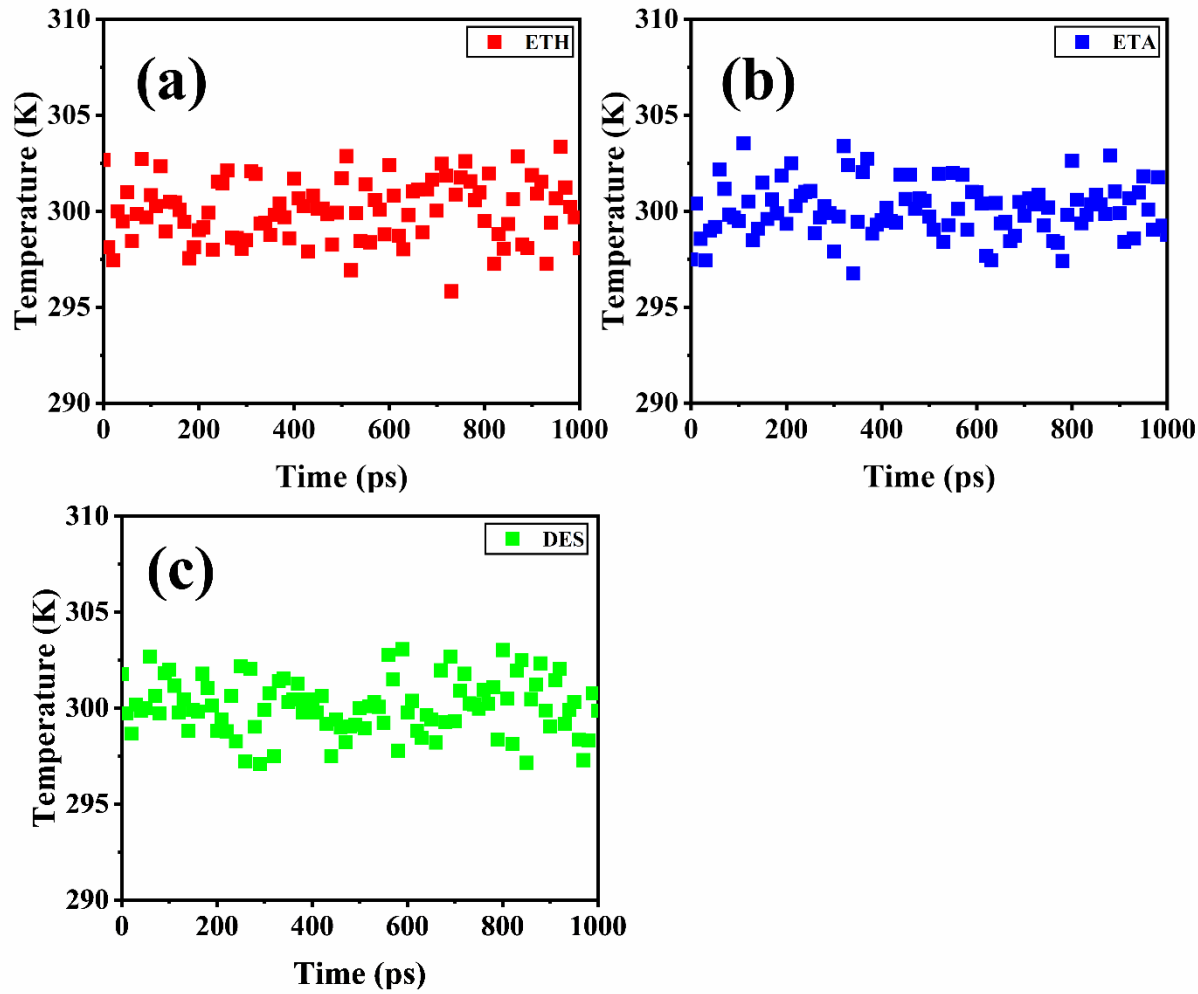


Figure S1. The temperature profile during NVT equilibration for 1:1 molar ratio (a) Ethanol based system (b) Ethyl acetate based system (c) DES based system

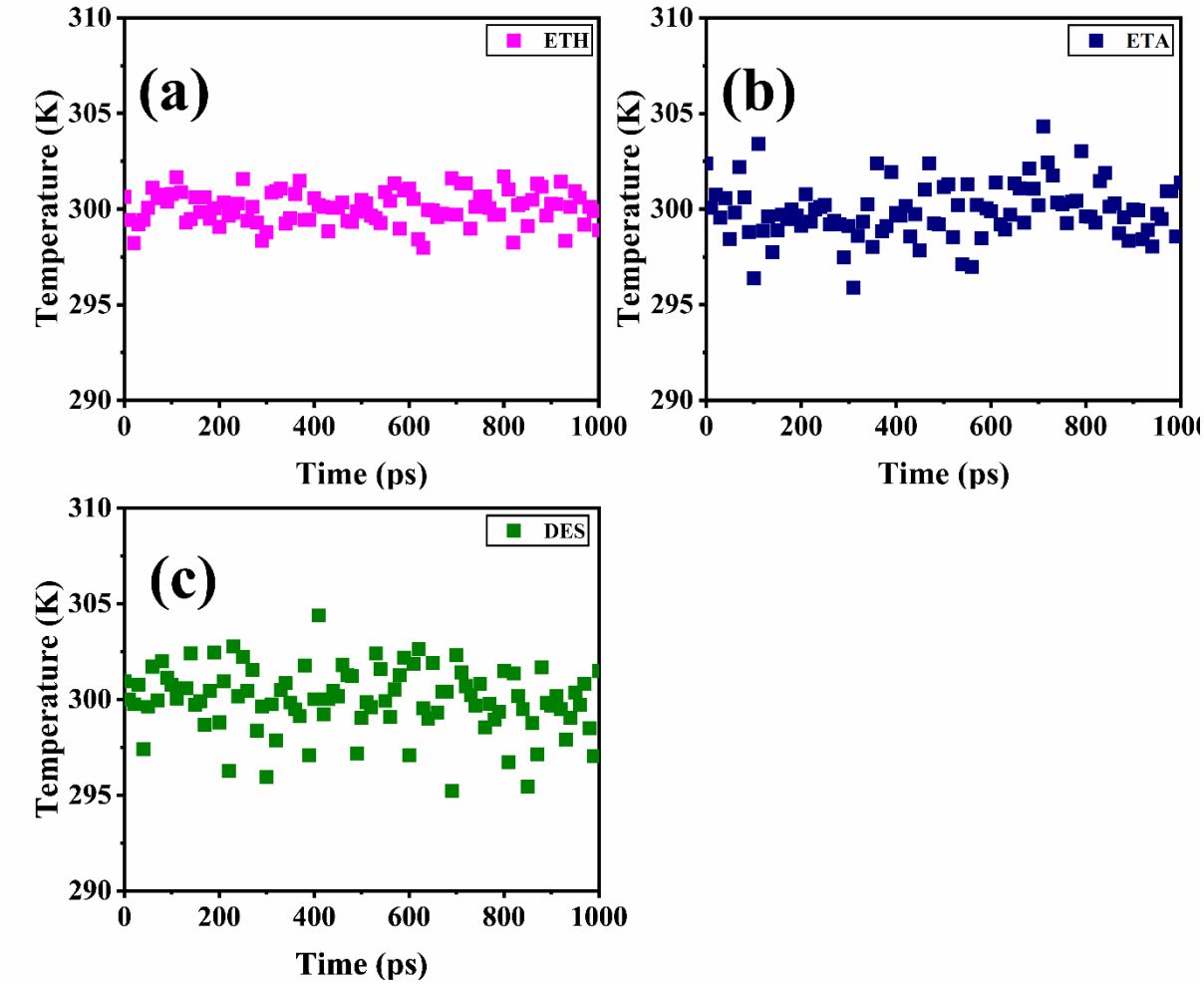
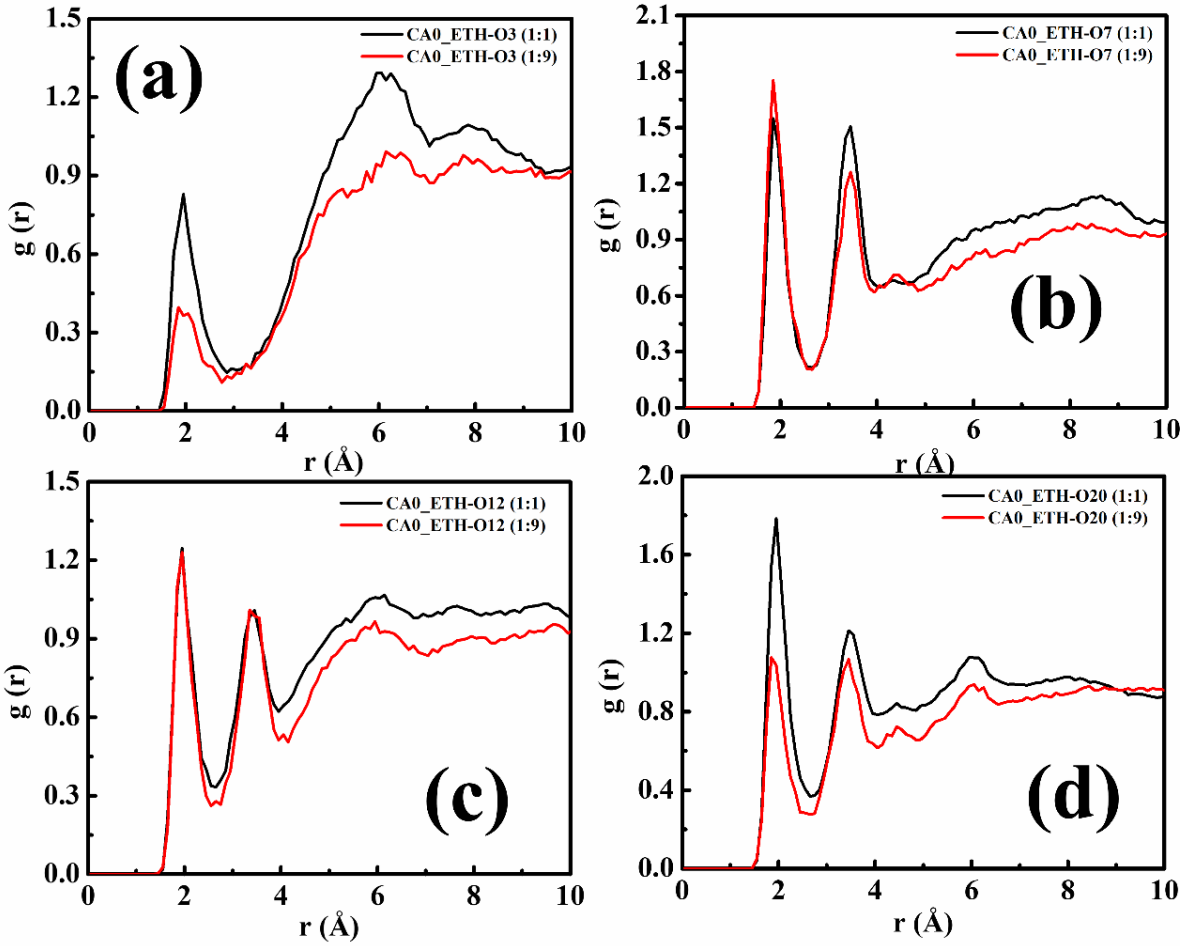


Figure S2. The temperature profile during NVT equilibration for 1:9 molar ratio (a) Ethanol based system (b) Ethyl acetate based system (c) DES based system



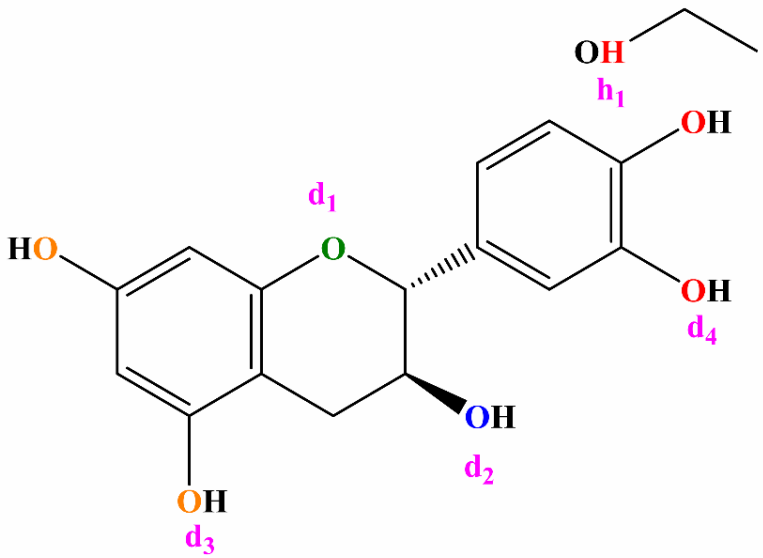


Figure. S3. RDF of catechin in ethanol-based system (a) O3-H (denoted by $d_1 \cdots h_1$) (b) O7-H (denoted by $d_2 \cdots h_1$) (c) O12-H (denoted by $d_3 \cdots h_1$) and (d) O20-H. (denoted by $d_4 \cdots h_1$).

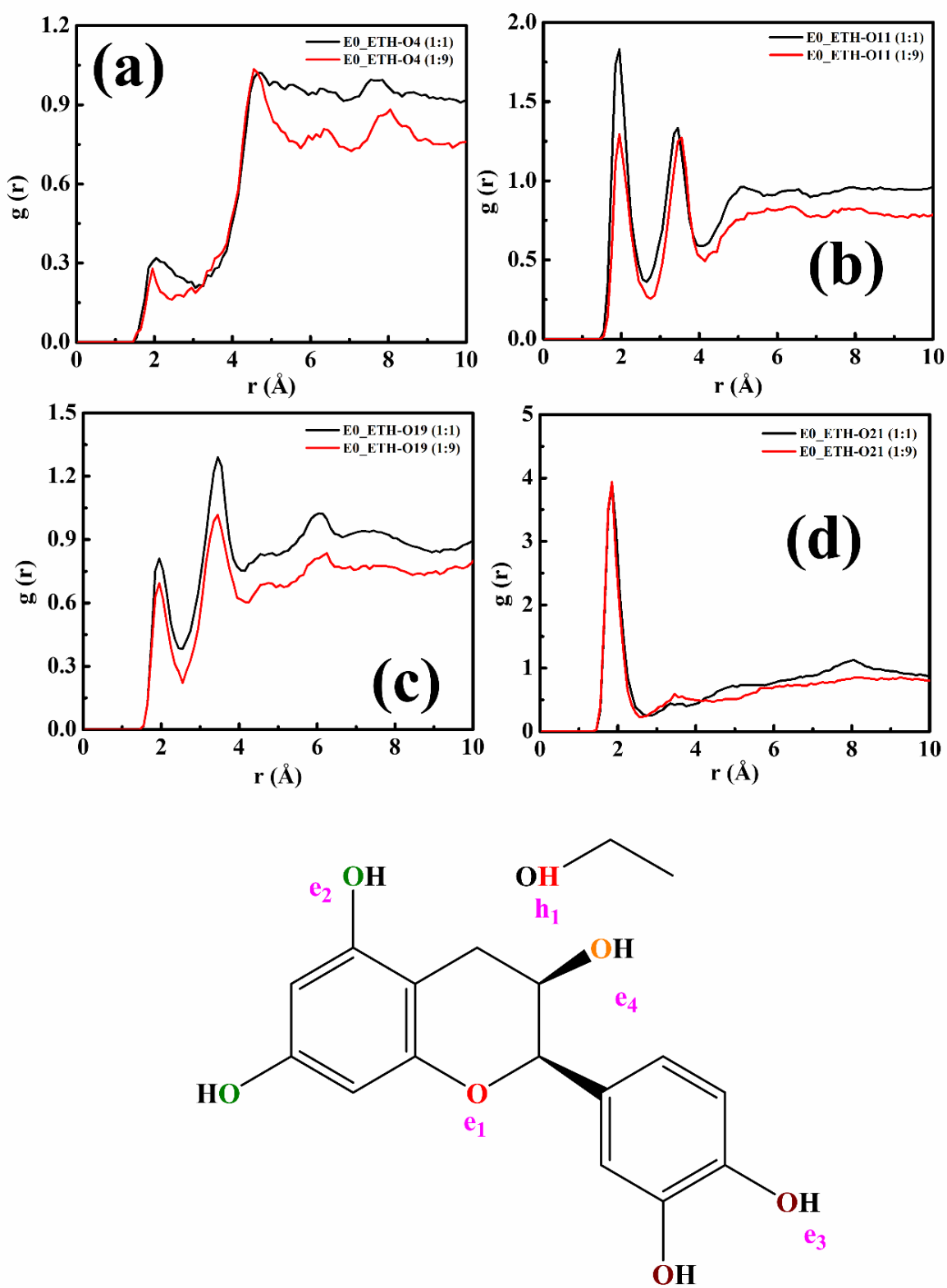


Figure. S4. RDF of epicatechin in ethanol-based system (a) O4-H (denoted by $e_1 \cdots h_1$) (b) O11-H (denoted by $e_2 \cdots h_1$) (c) O19-H(denoted by $e_3 \cdots h_1$) and (d) O21-H (denoted by $e_4 \cdots h_1$).

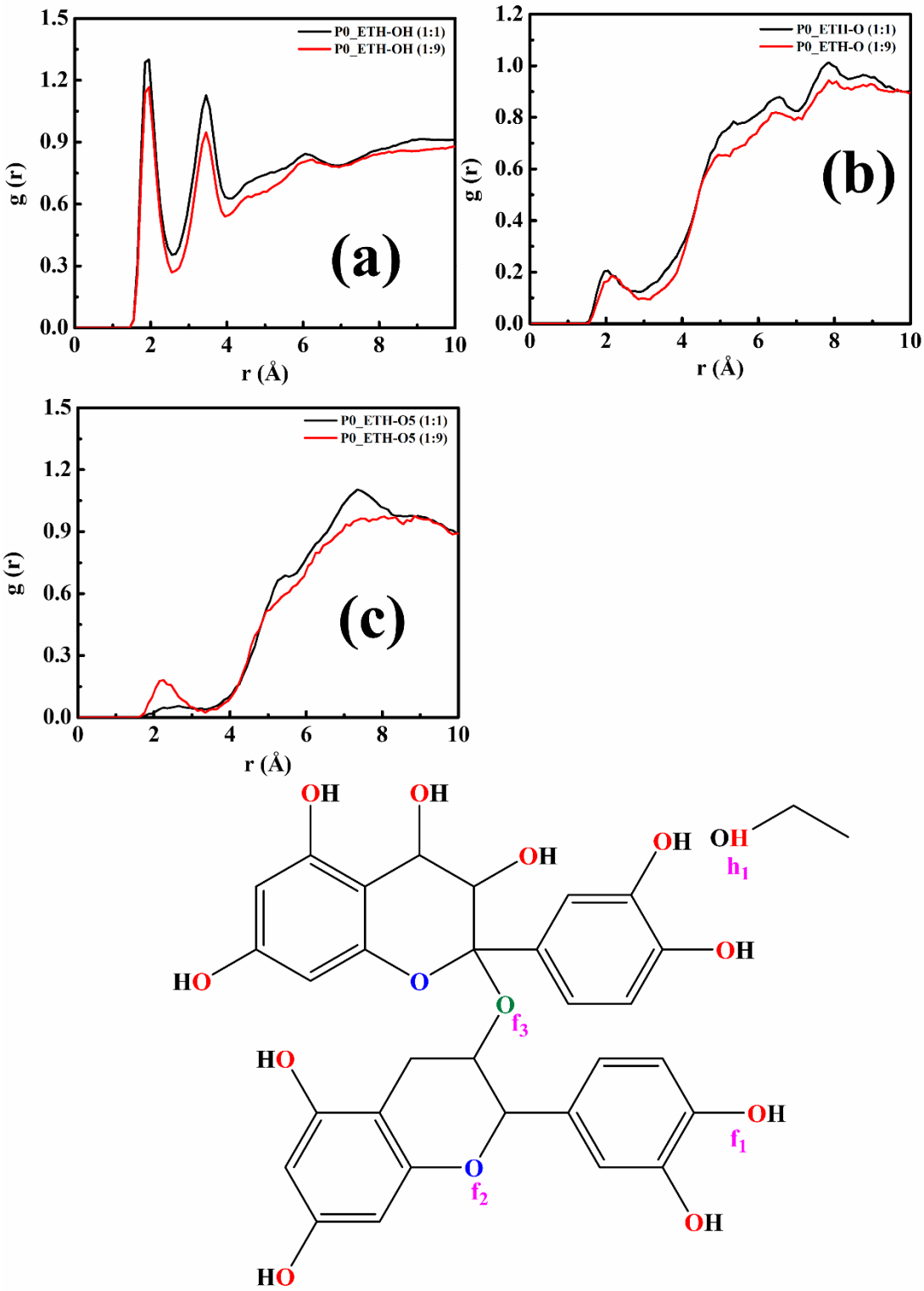


Figure. S5. RDF of Procyanidins in ethanol-based system (a) OH-H (denoted by $f_1 \cdots h_1$) (b) O-H (denoted by $f_2 \cdots h_1$) and (c) O5-H (denoted by $f_3 \cdots h_1$).

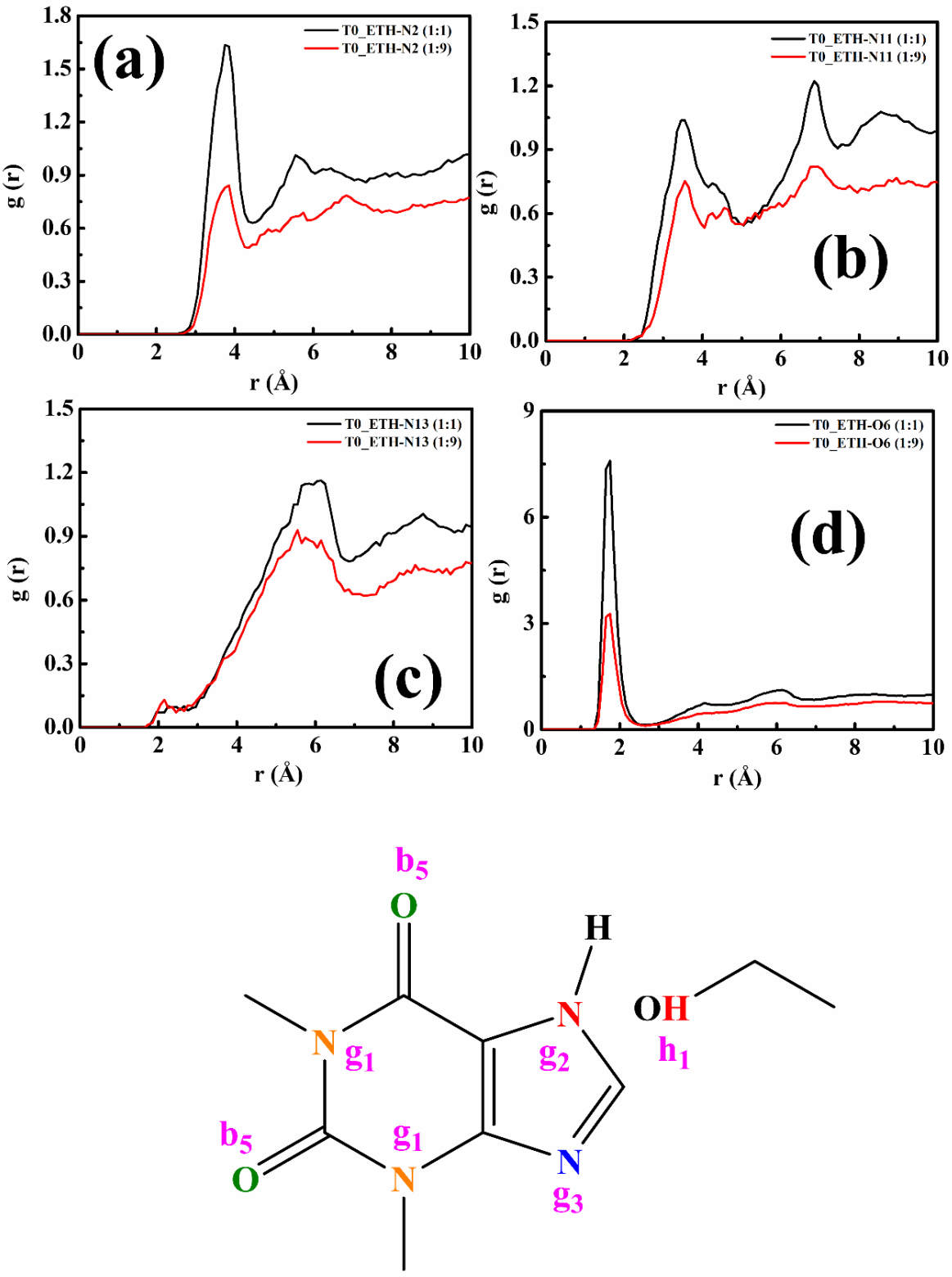


Figure S6. RDF of Theophylline in ethanol-based system (a) N2-H (denoted by $g_1 \cdots h_1$) (b) N11-H (denoted by $g_2 \cdots h_1$) (c) N13-H (denoted by $g_3 \cdots h_1$) and (d) O6-H. (denoted by $g_4 \cdots h_1$).

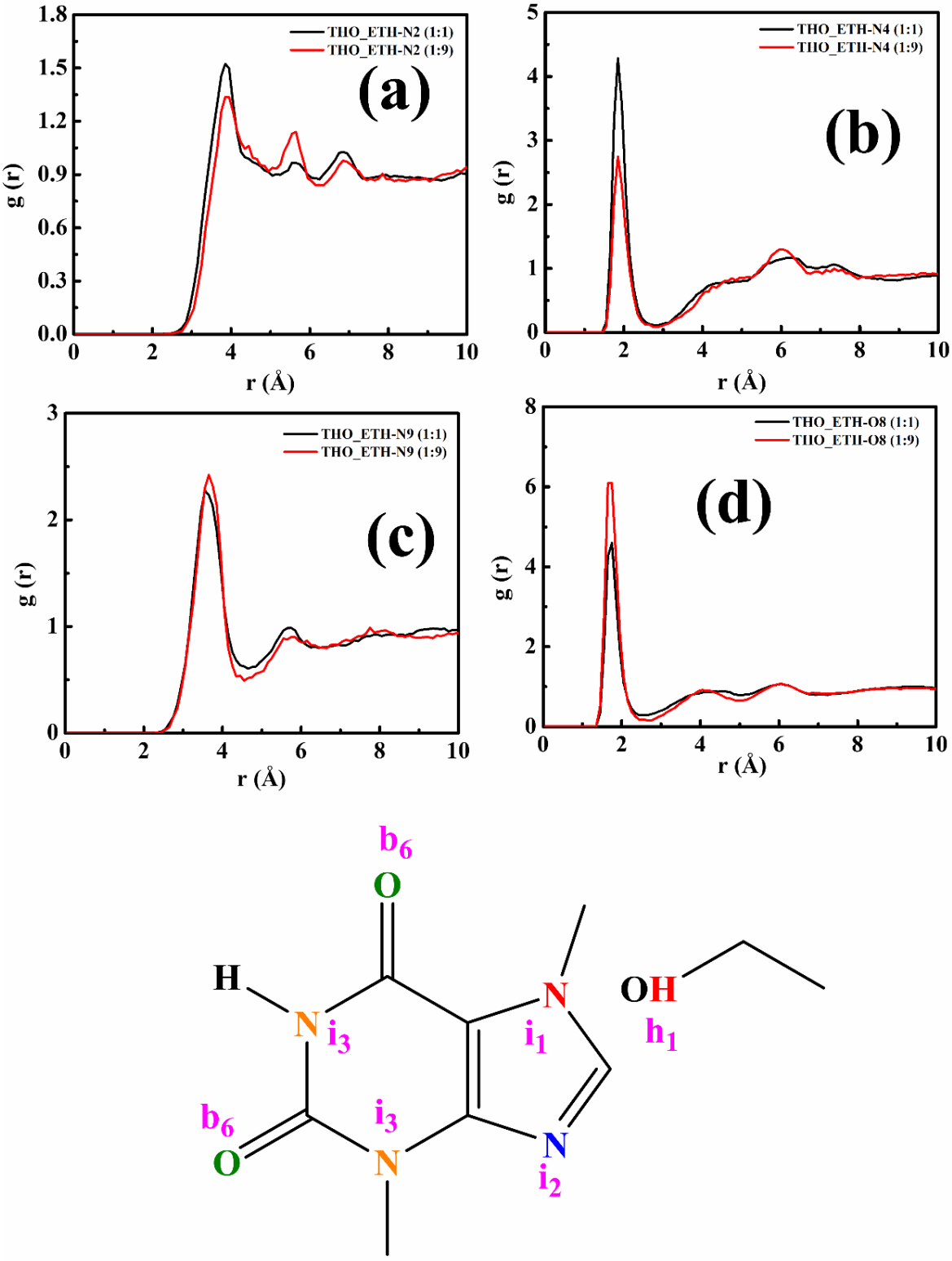


Figure. S7. RDF of theobromine in ethanol-based system (a) N2-H (denoted by $i_1 \cdots h_1$) (b) N4-H(denoted by $i_2 \cdots h_1$) (c) N9-H(denoted by $i_3 \cdots h_1$) and (d) O8-H (denoted by $b_6 \cdots h_1$).

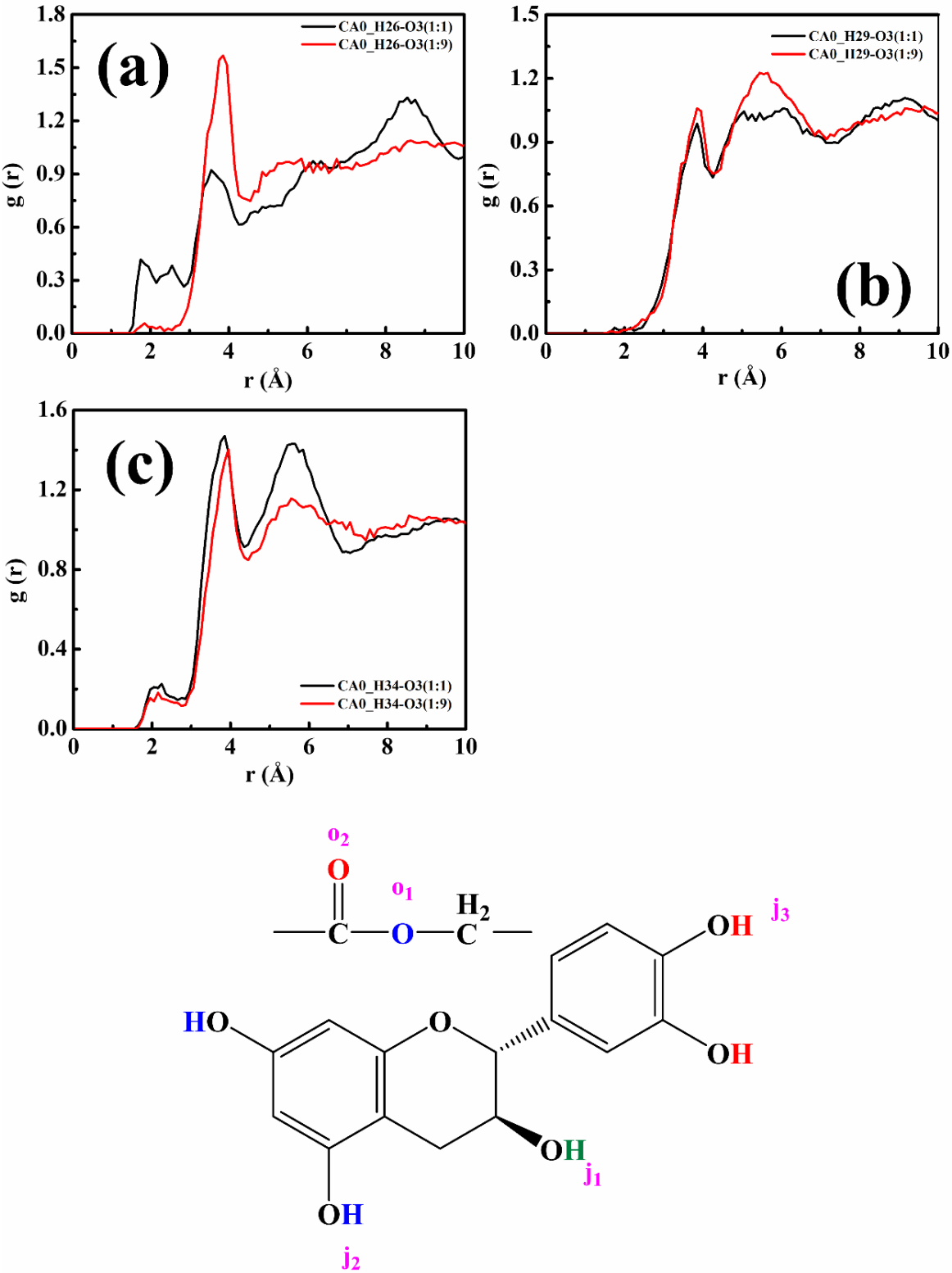


Figure. S8. RDF of Catechin of ethyl-acetate based system (a) H26-O3 (denoted by $j_1 \cdots o_1$) (b) H29-O3(denoted by $j_2 \cdots o_1$) and (c) H34-O3(denoted by $j_3 \cdots o_1$).

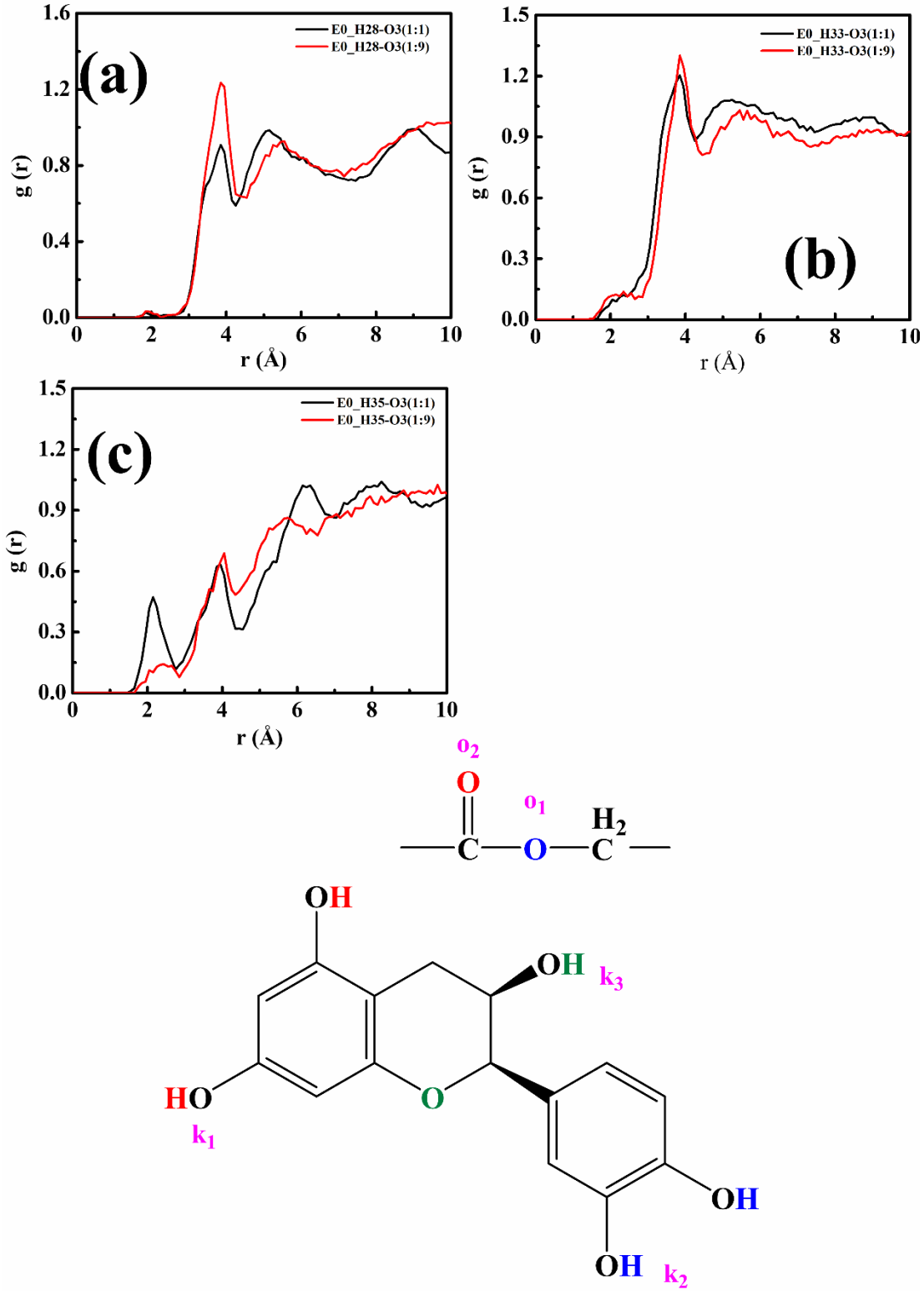
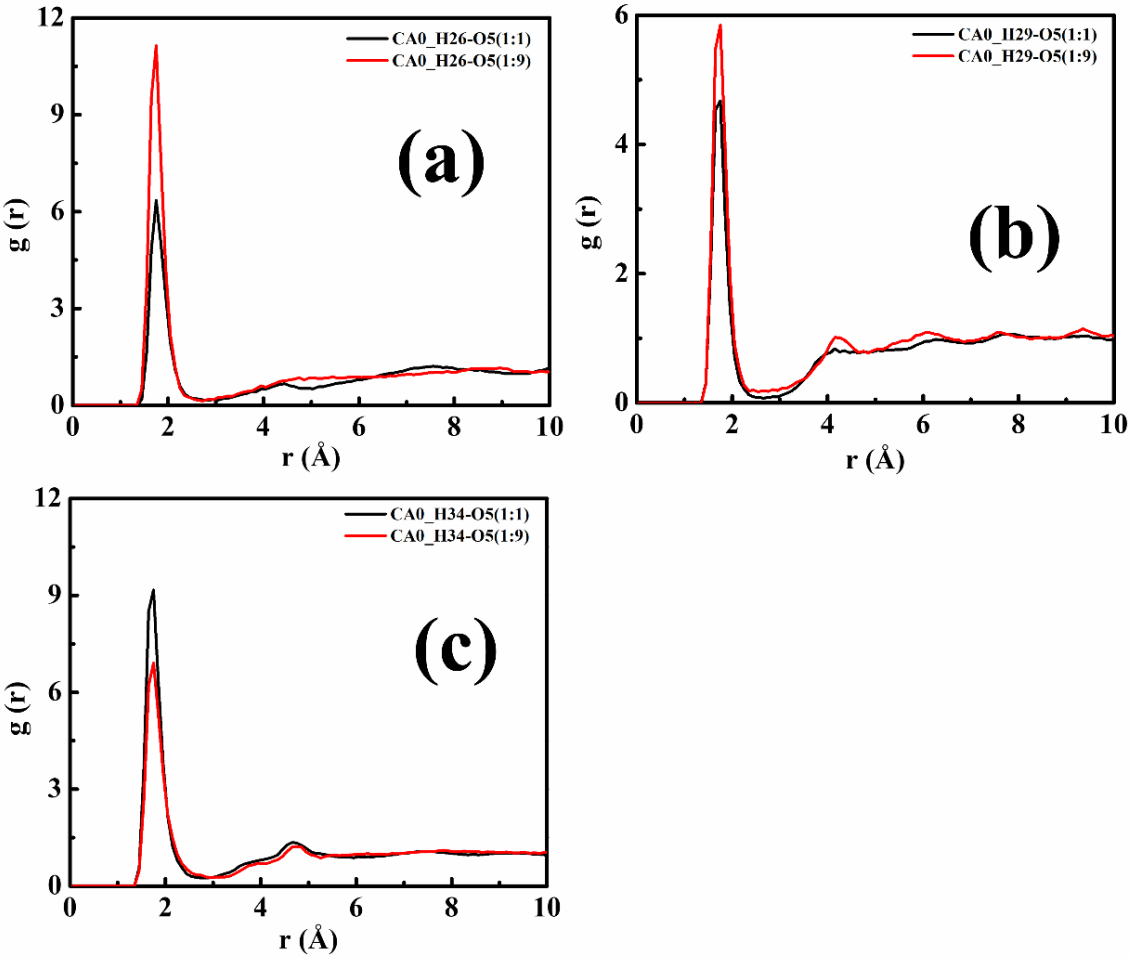


Figure. S9. RDF of Epicatechin in ethyl-acetate based system (a) H28-O3 (denoted by $k_1 \cdots o_1$) (b) H33-O3 (denoted by $k_2 \cdots o_1$) and (c) H35-O3. (denoted by $k_3 \cdots o_1$).



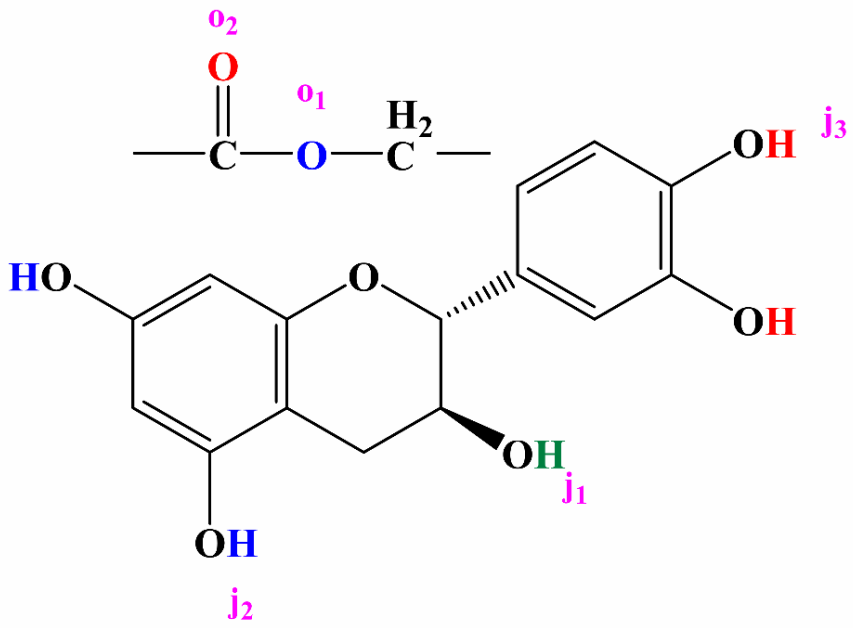


Figure. S10. RDF of Catechin of ethyl-acetate based system (a) H26-O5 (denoted by $j_1 \cdots o_2$) (b) H29-O5(denoted by $j_2 \cdots o_2$) and (c) H34-O5(denoted by $j_3 \cdots o_2$)

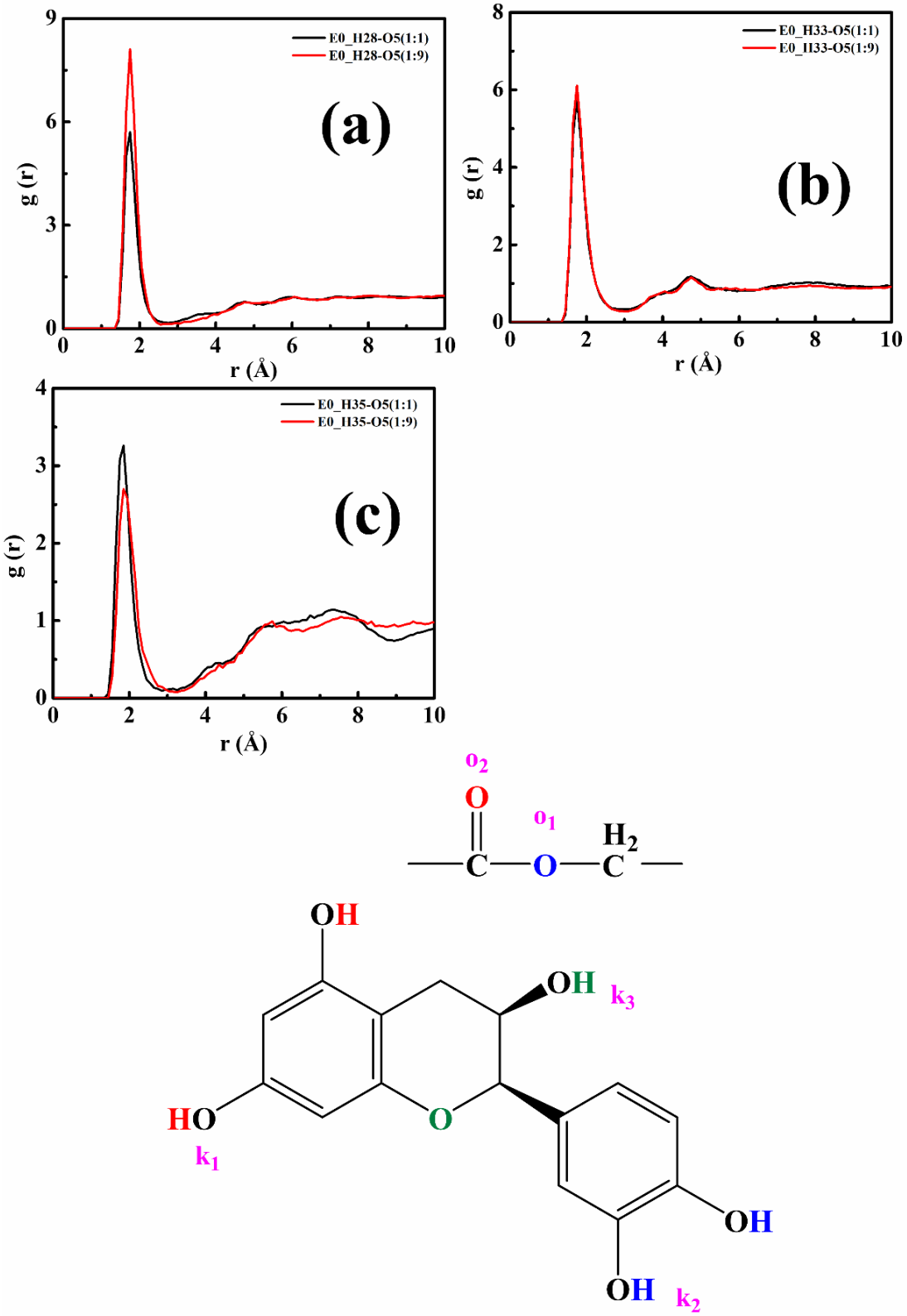


Figure. S11. RDF of Epicatechin of ethyl-acetate based system (a) H28-O5(denoted by $k_1 \cdots o_2$) (b) H33-O5 (denoted by $k_2 \cdots o_2$)and (c) H35-O5 (denoted by $k_3 \cdots o_2$)

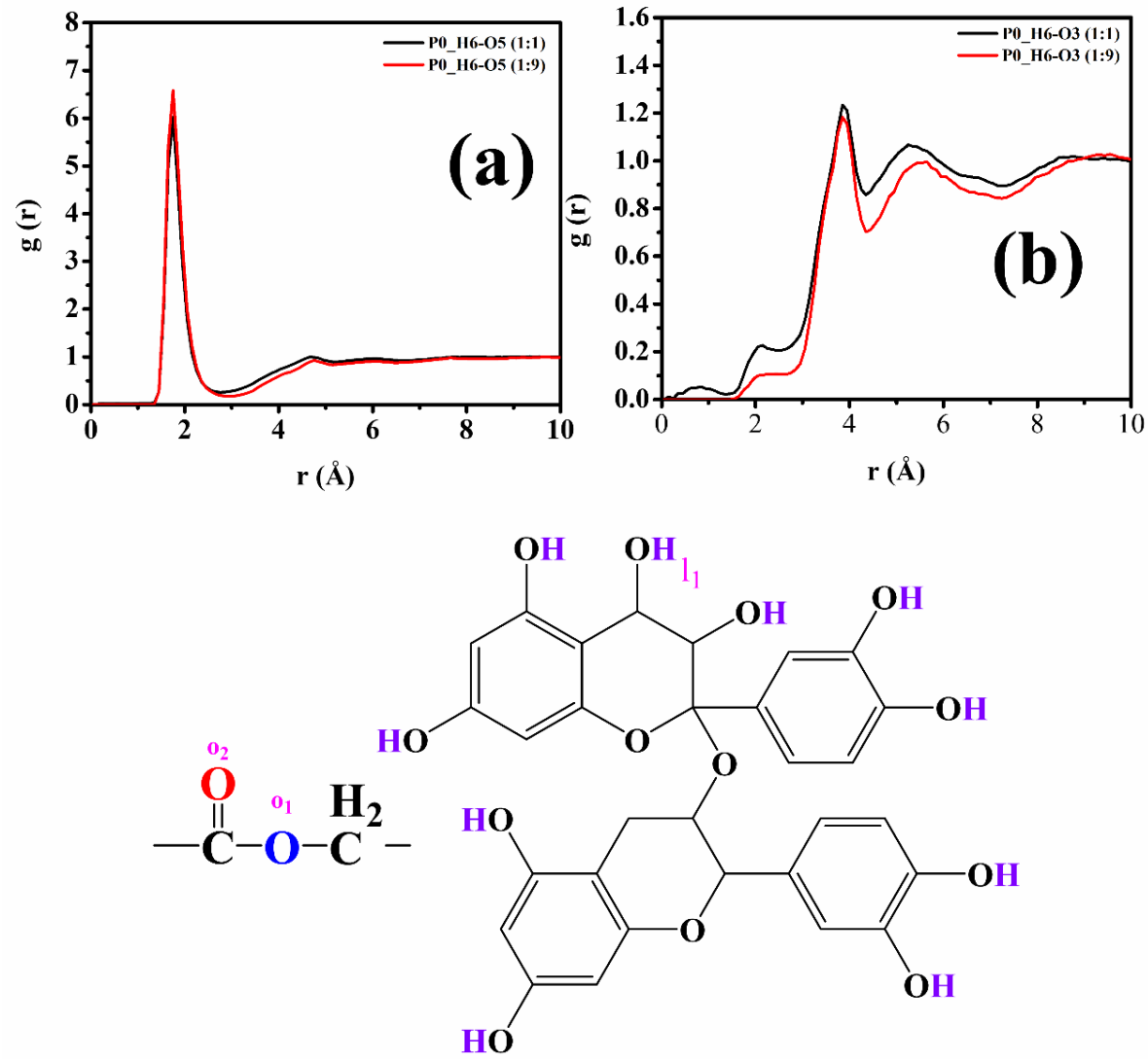


Figure. S12. RDF of Procyanidins of ethyl-acetate based system (a) H6-O5 (denoted by $l_1 \cdots o_2$) and (b) H6-O3 (denoted by $l_1 \cdots o_1$).

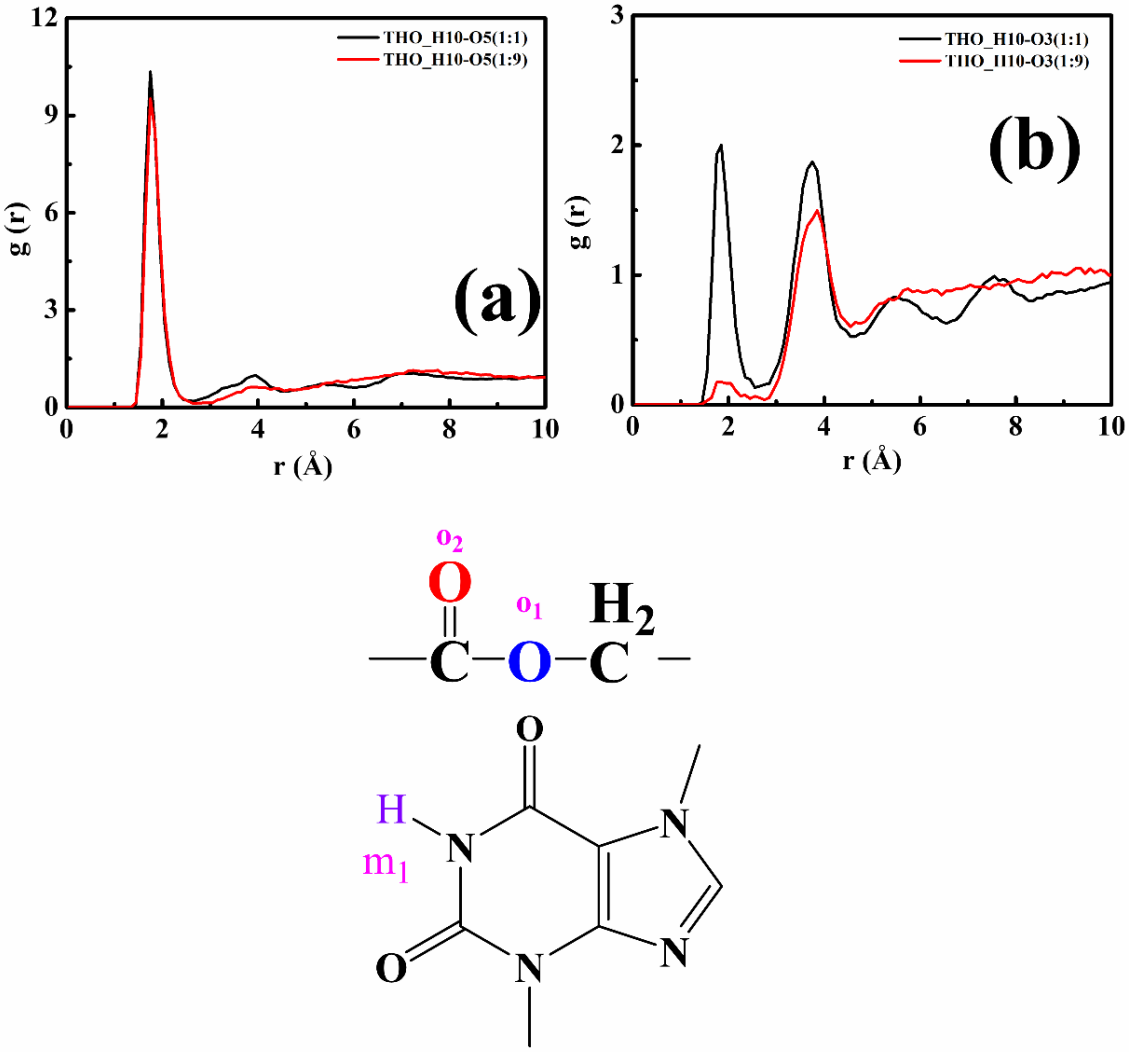


Figure. S13. RDF of Theobromine of ethyl-acetate based system (a) H10-O5 (denoted by $m_1 \cdots o_2$) and (b) H10-O3 (denoted by $m_2 \cdots o_1$).

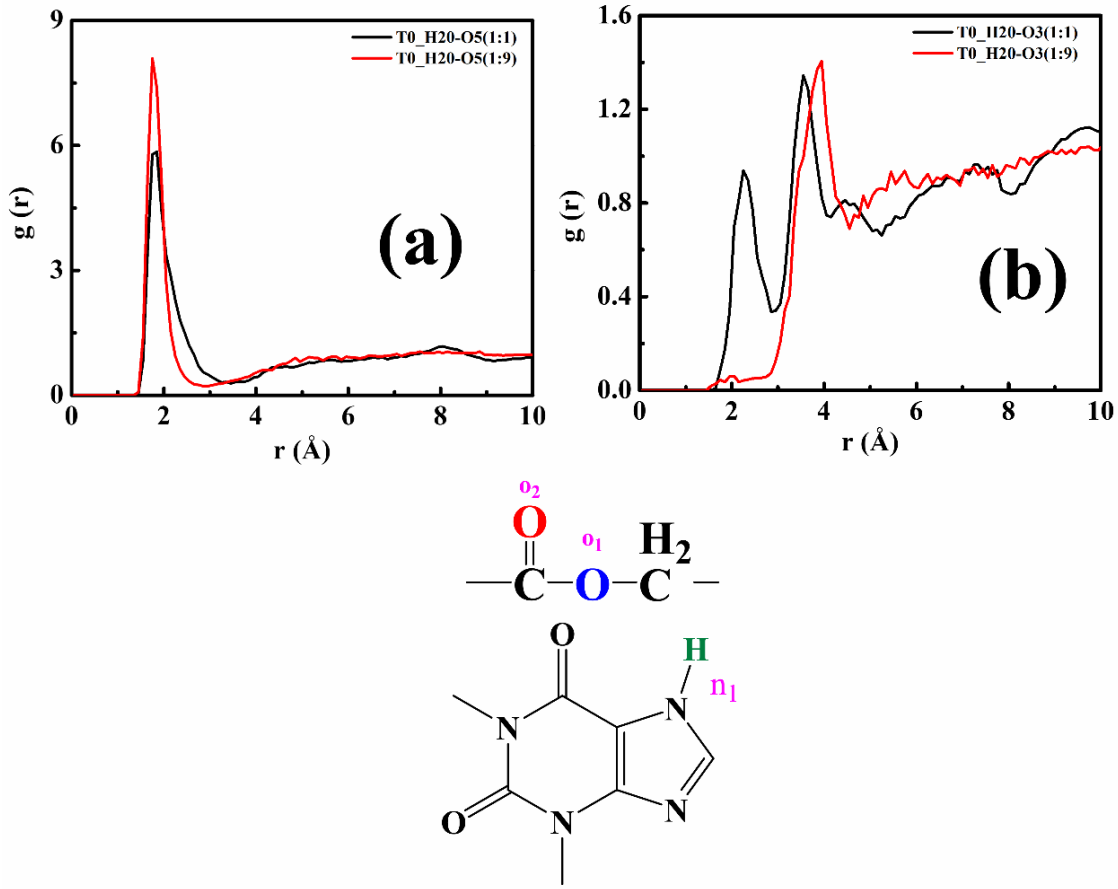
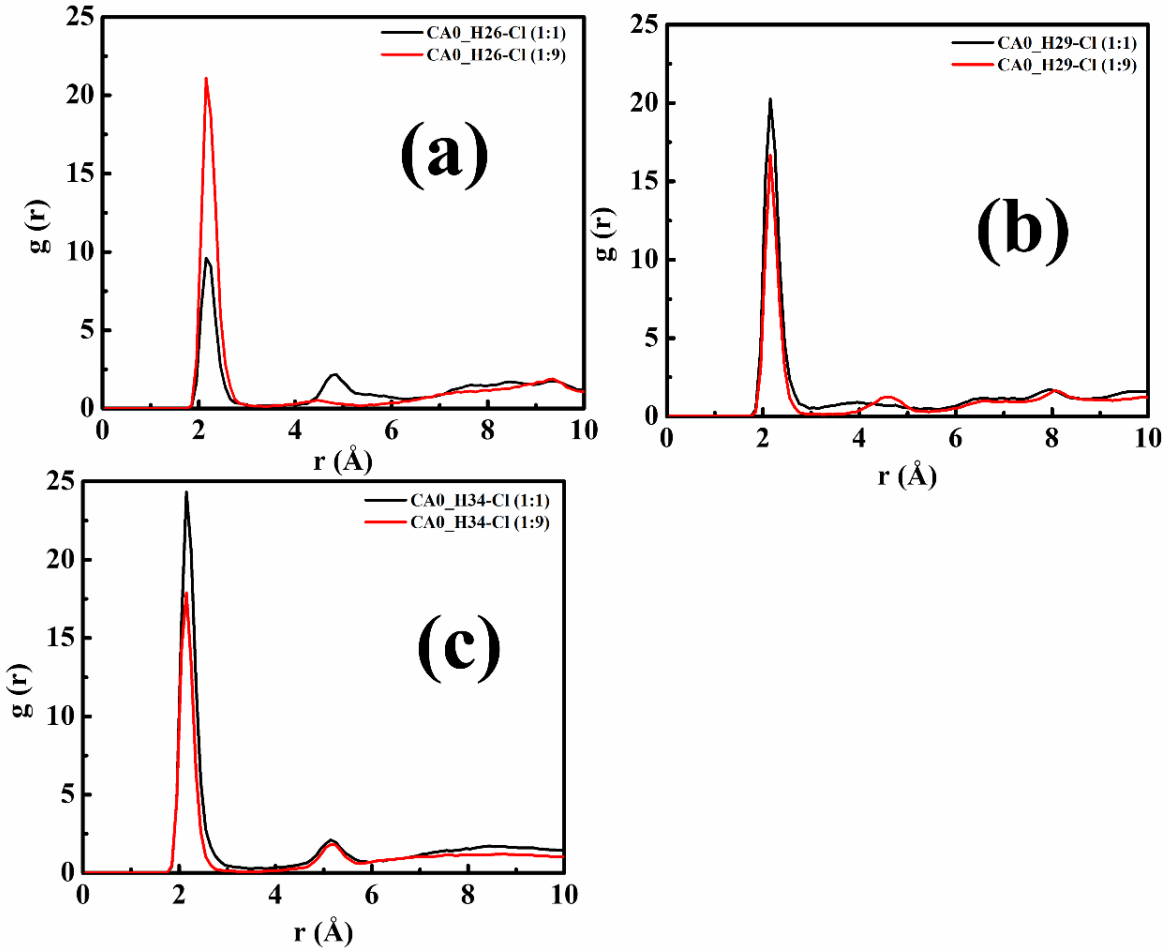


Figure. S14. RDF of Theophylline of ethyl-acetate based system (a) H20-O5 (denoted by $n_1 \cdots o_2$)and (b) H20-O3 (denoted by $n_1 \cdots o_1$).



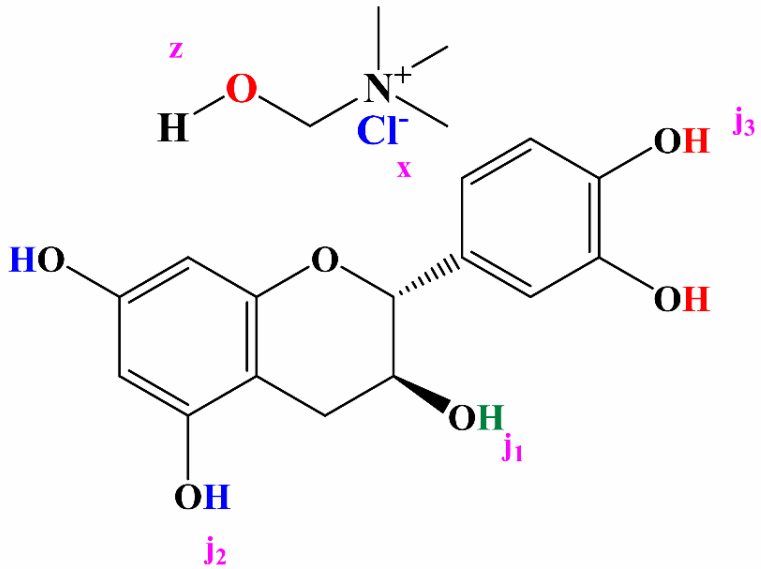


Figure. S15. RDF of Catechin in DES based system (a) H26-Cl (denoted by $j_1 \cdots x$) (b) H29-Cl (denoted by $j_2 \cdots x$) and (c) H34-Cl (denoted by $j_3 \cdots x$).

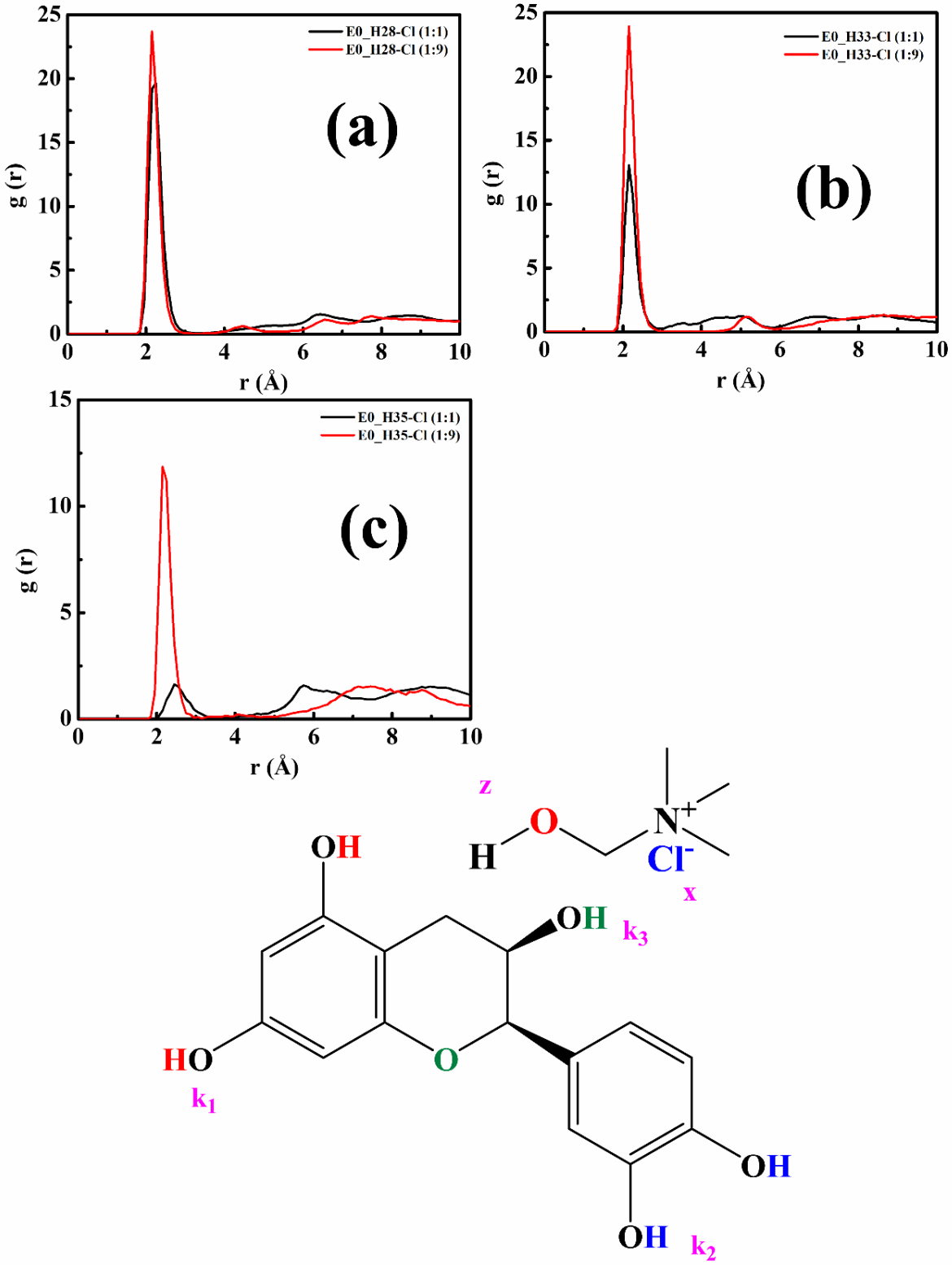
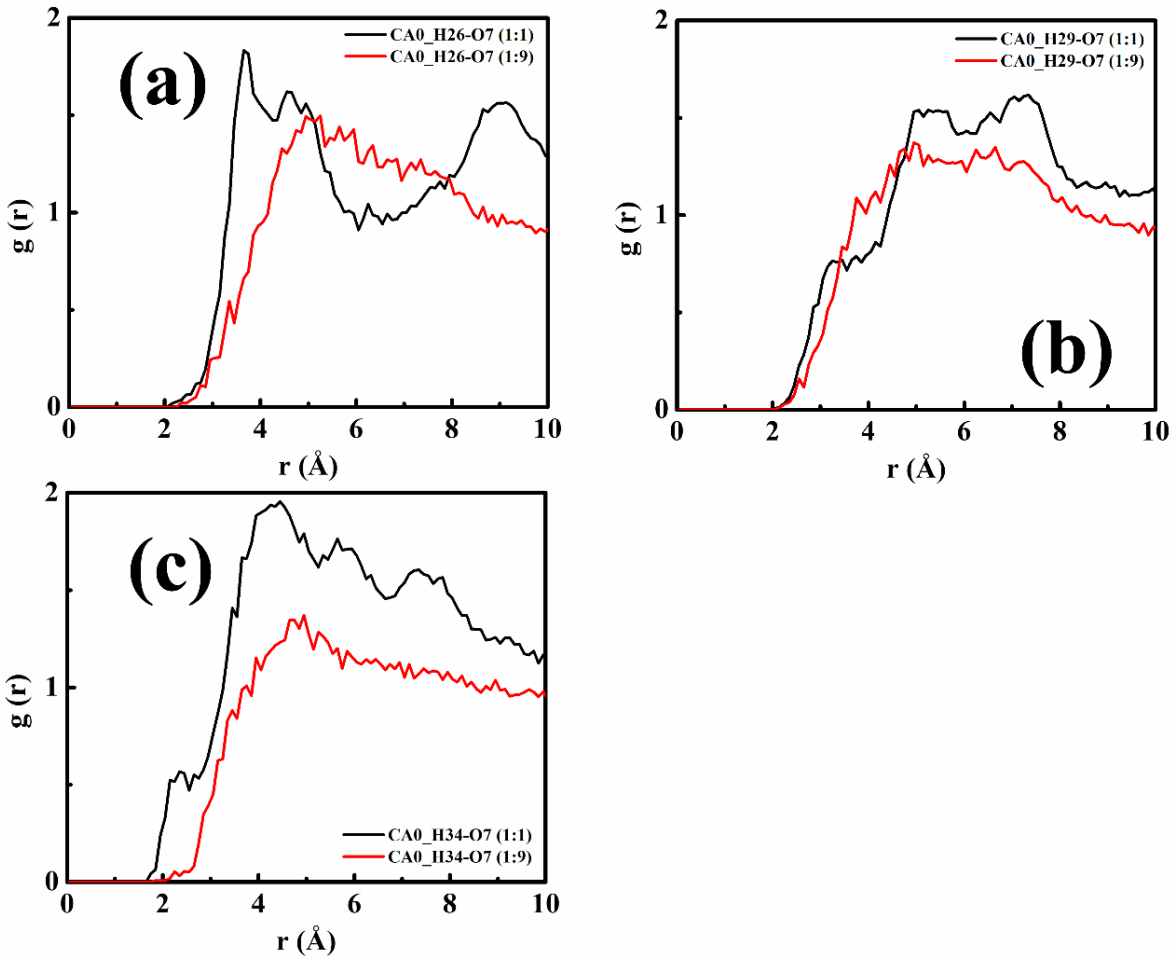


Figure. S16. RDF of epicatechin in DES based system (a) H28-Cl(denoted by $k_1 \cdots x$) (b) H33- Cl (denoted by $k_2 \cdots x$) and (c) H35- Cl (denoted by $k_3 \cdots x$).



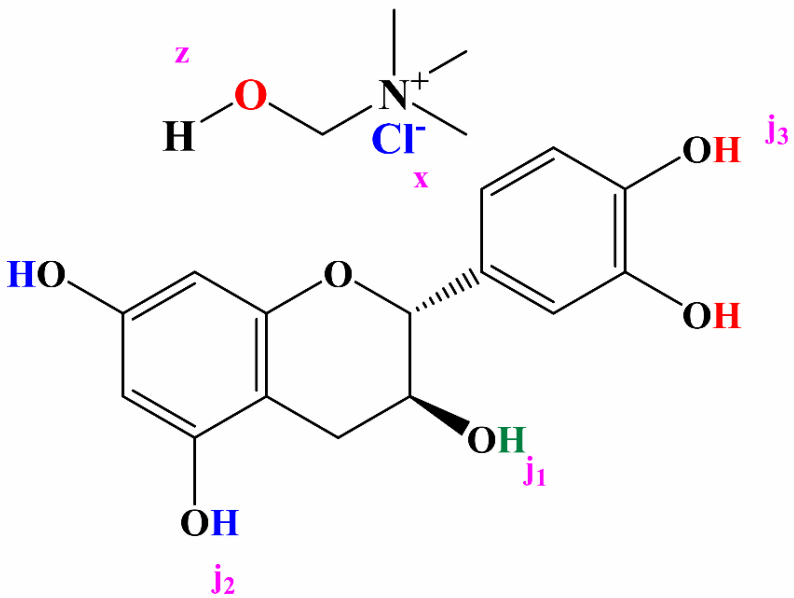


Figure. S17. RDF of Catechin in DES based system (a) H26-O7 (denoted by $j_1 \cdots z$) (b) H29-O7(denoted by $j_2 \cdots z$) and (c) H34-O7 (denoted by $j_3 \cdots z$).

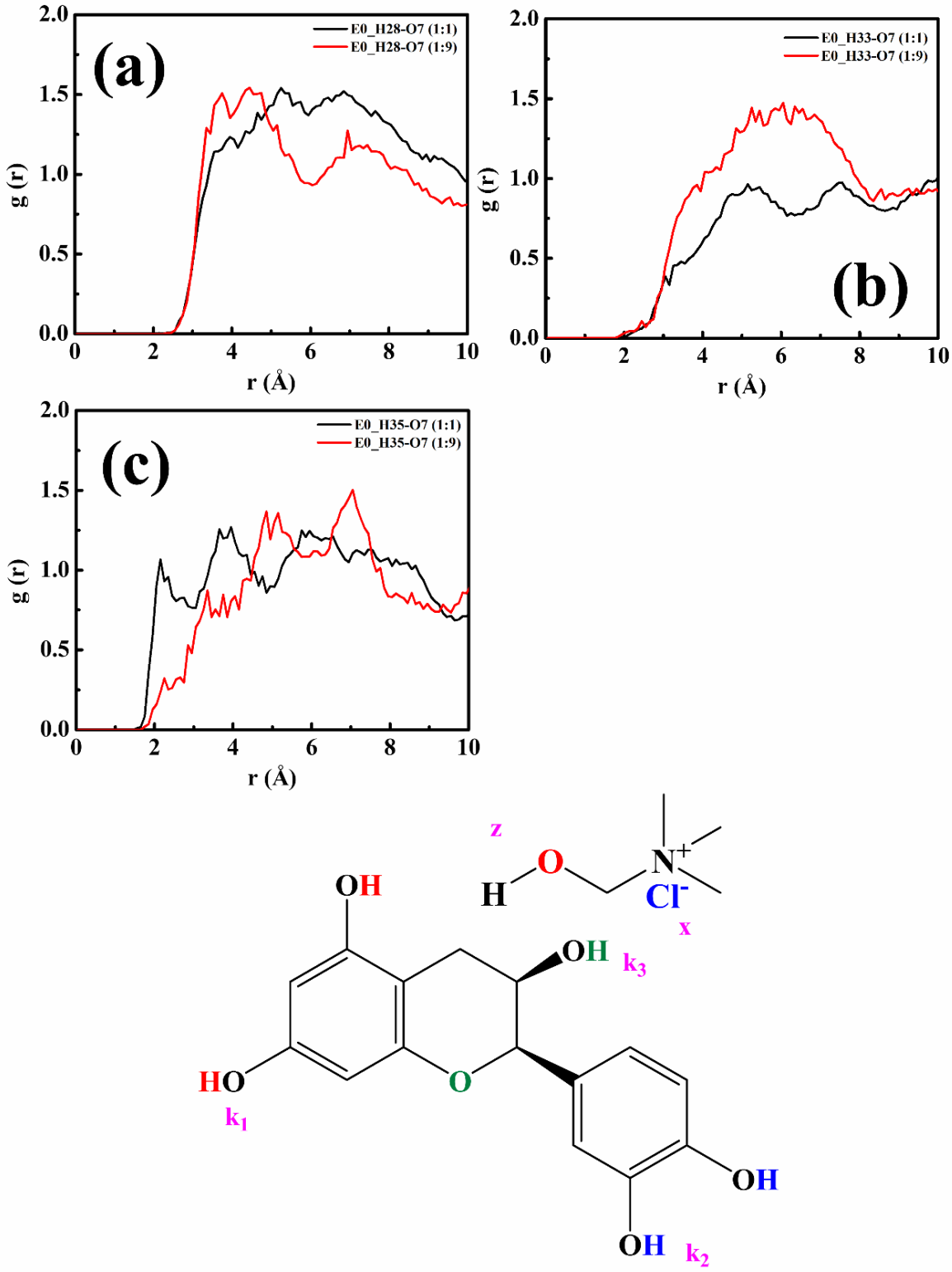


Figure. S18. RDF of epicatechin in DES based system (a) H28- O7 (denoted by $k_1 \cdots z$). (b) H33- O7 (denoted by $k_2 \cdots z$) and (c) H35- O7(denoted by $k_3 \cdots z$).

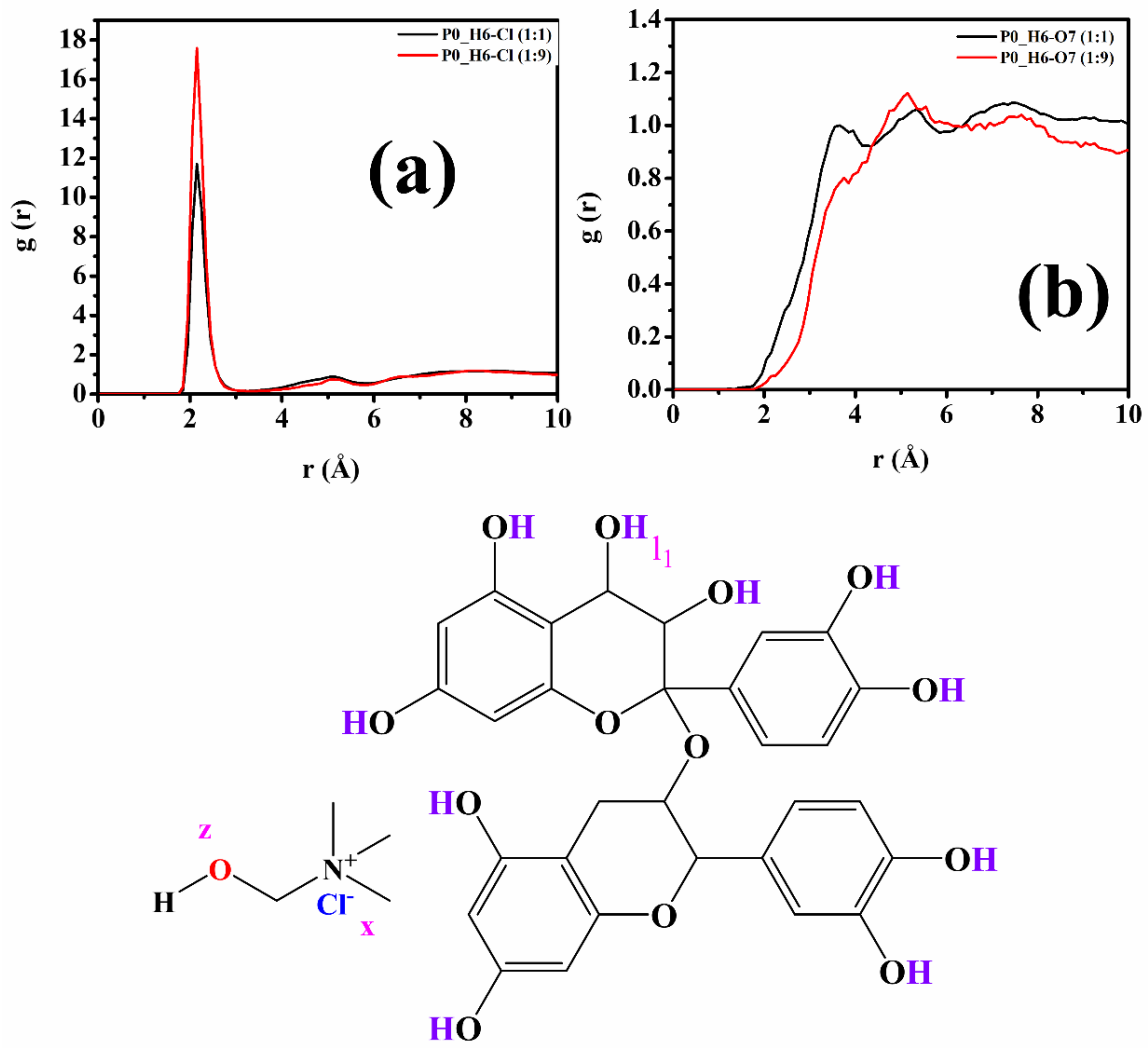


Figure. S19. RDF of Procyanidins in DES based system (a) H6-Cl (denoted by $l_1 \cdots x$) and (b) H6-O7 (denoted by $l_1 \cdots z$).

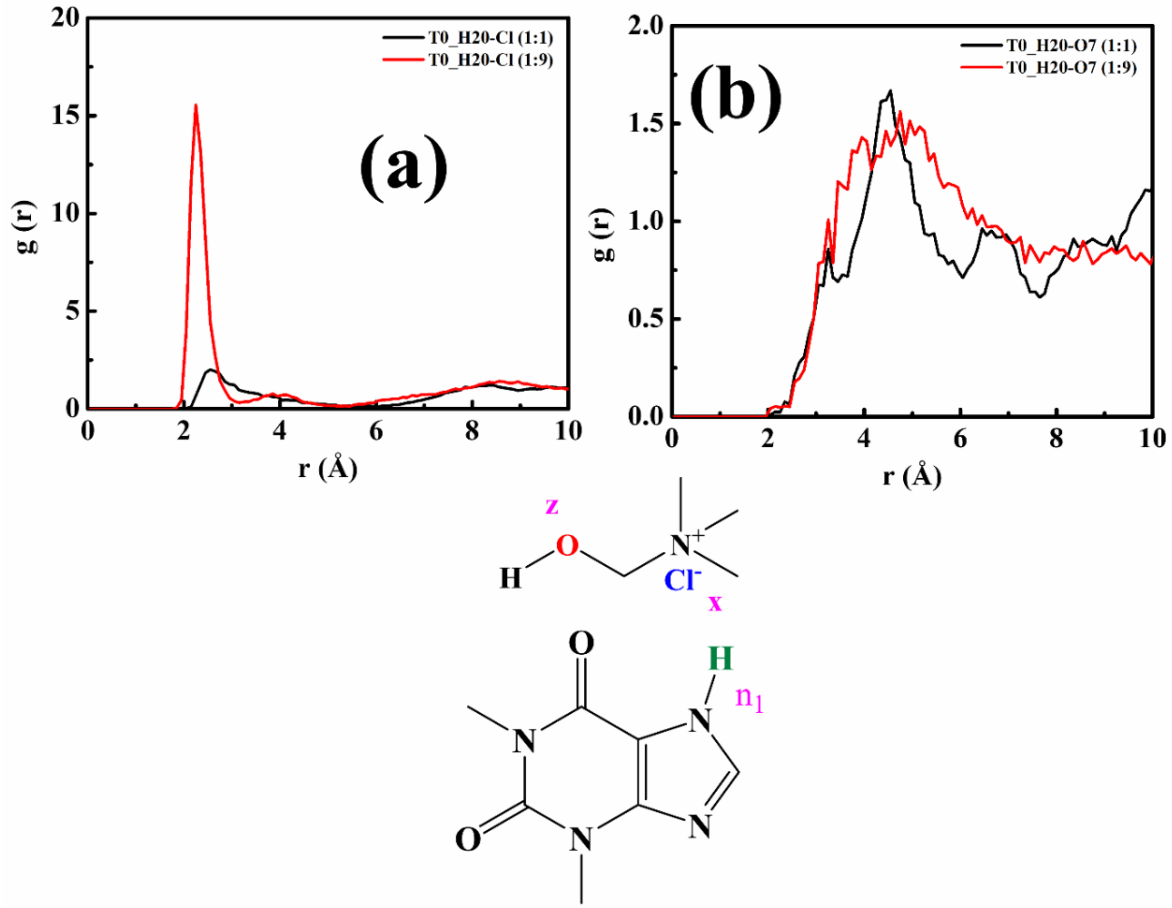


Figure. S20. RDF of Theophylline in DES based system (a) H20-Cl (denoted by $n_1 \cdots x$)and (b) H20-O7 (denoted by $n_1 \cdots z$).

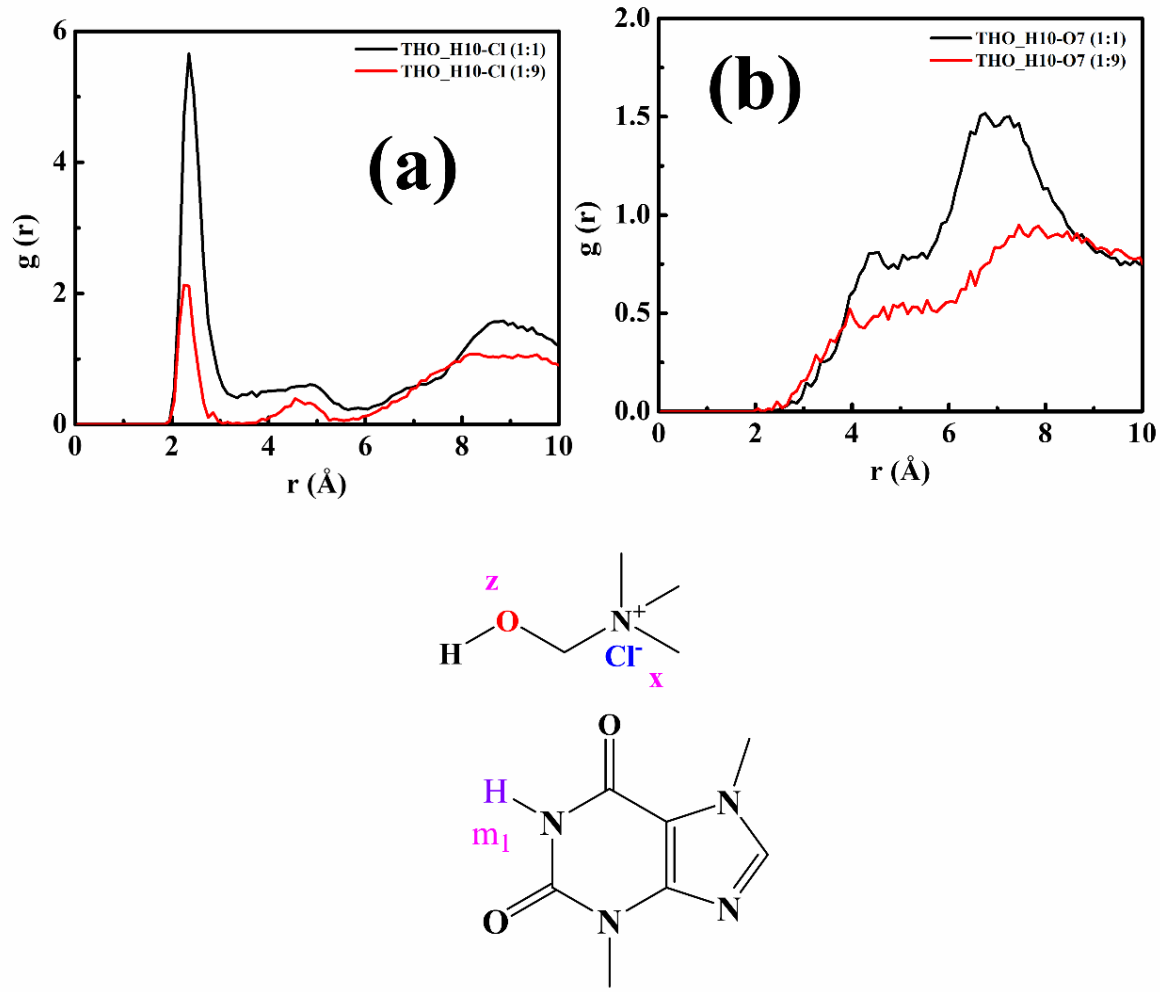
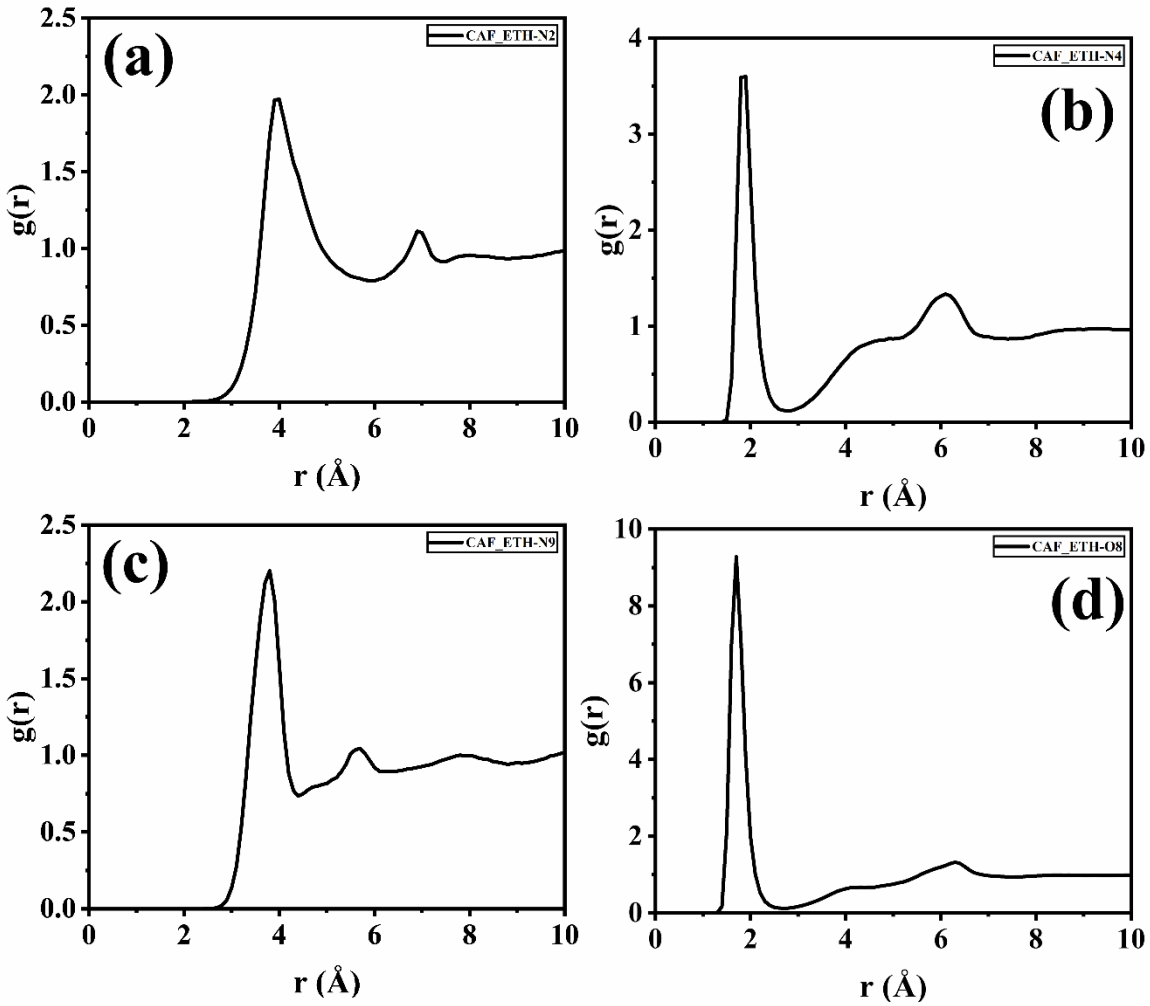


Figure. S21. RDF of Theobromine in DES based system (a) H10-Cl (denoted by $m_1 \cdots x$) and (b) H10-O7 (denoted by $m_2 \cdots z$).



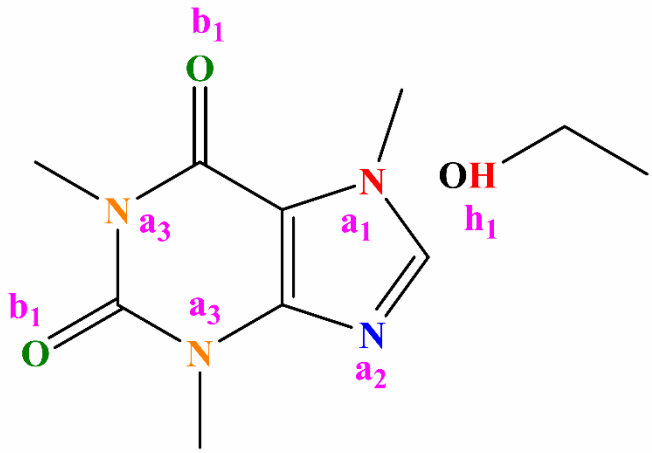


Figure S22. RDF of caffeine in ethanol-based system (a) N2-H (denoted by $a_1 \cdots h_1$) (b) N4-H (denoted by $a_2 \cdots h_1$) (c) N9-H (denoted by $a_3 \cdots h_1$) and (d) O8-H (denoted by $b_1 \cdots h_1$).

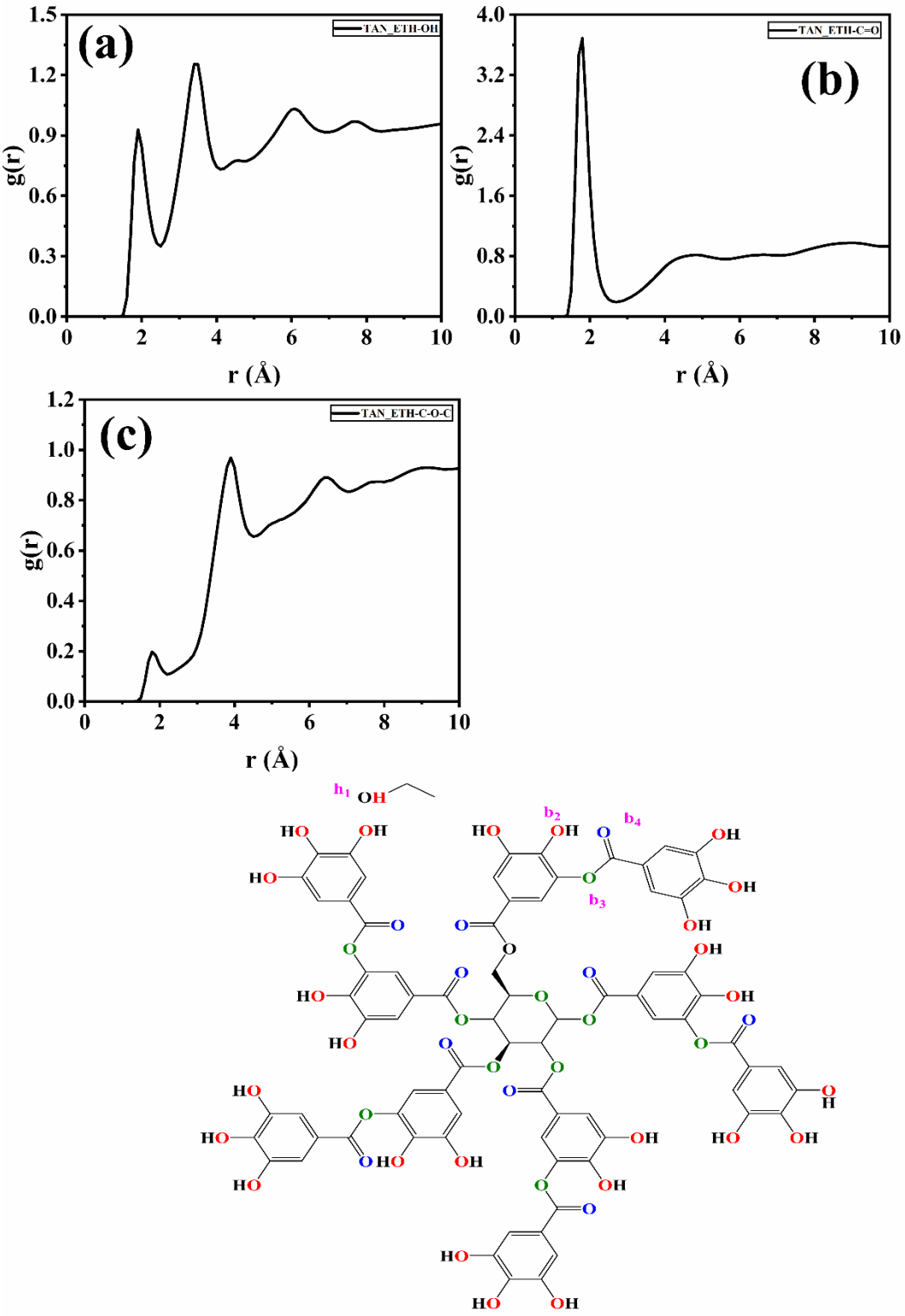


Figure S23. RDF Tannic acid of ethanol-based system (a)O-H (denoted by $b_2 \cdots h_1$) (b) C-O-C-H (denoted by $b_3 \cdots h_1$) and (c) C=O-H (denoted by $b_4 \cdots h_1$)

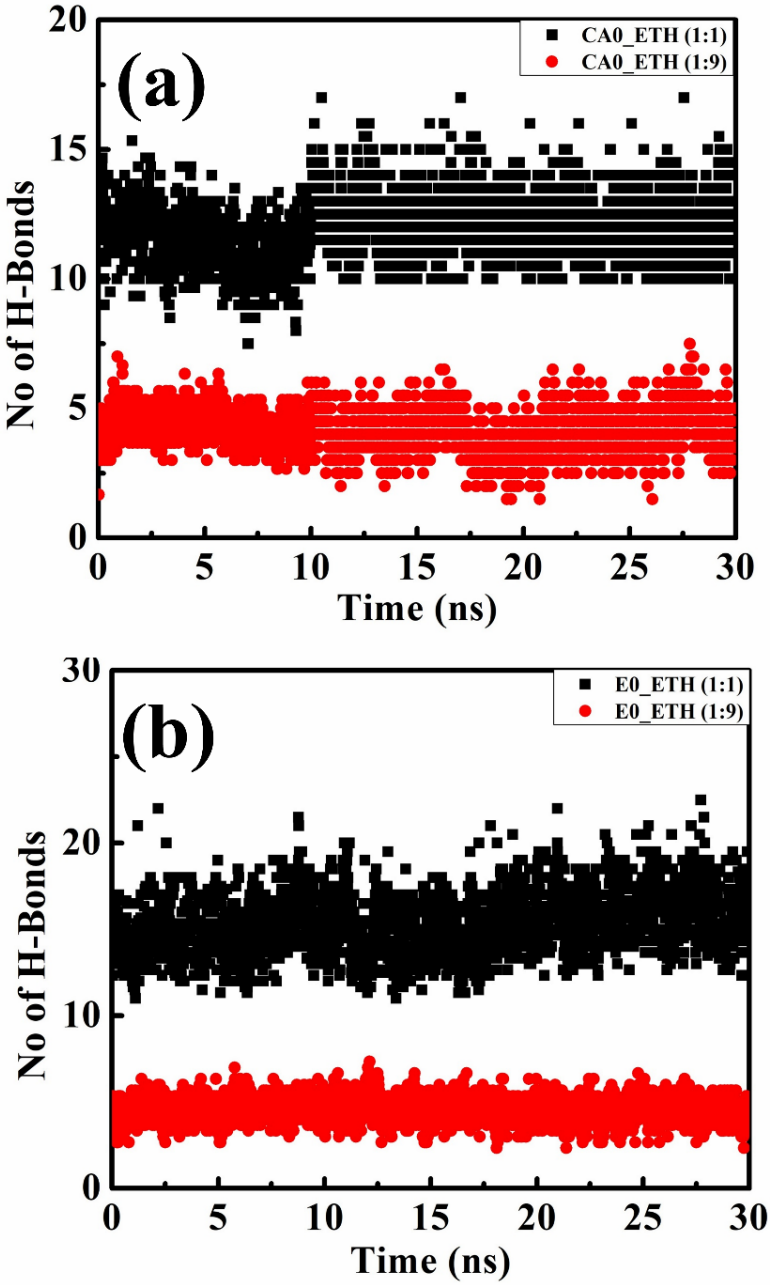


Figure. S24 (a). Number of Hydrogen bonds in catechin in ethanol-based system and (b) Number of Hydrogen bonds in epicatechin in ethanol-based system

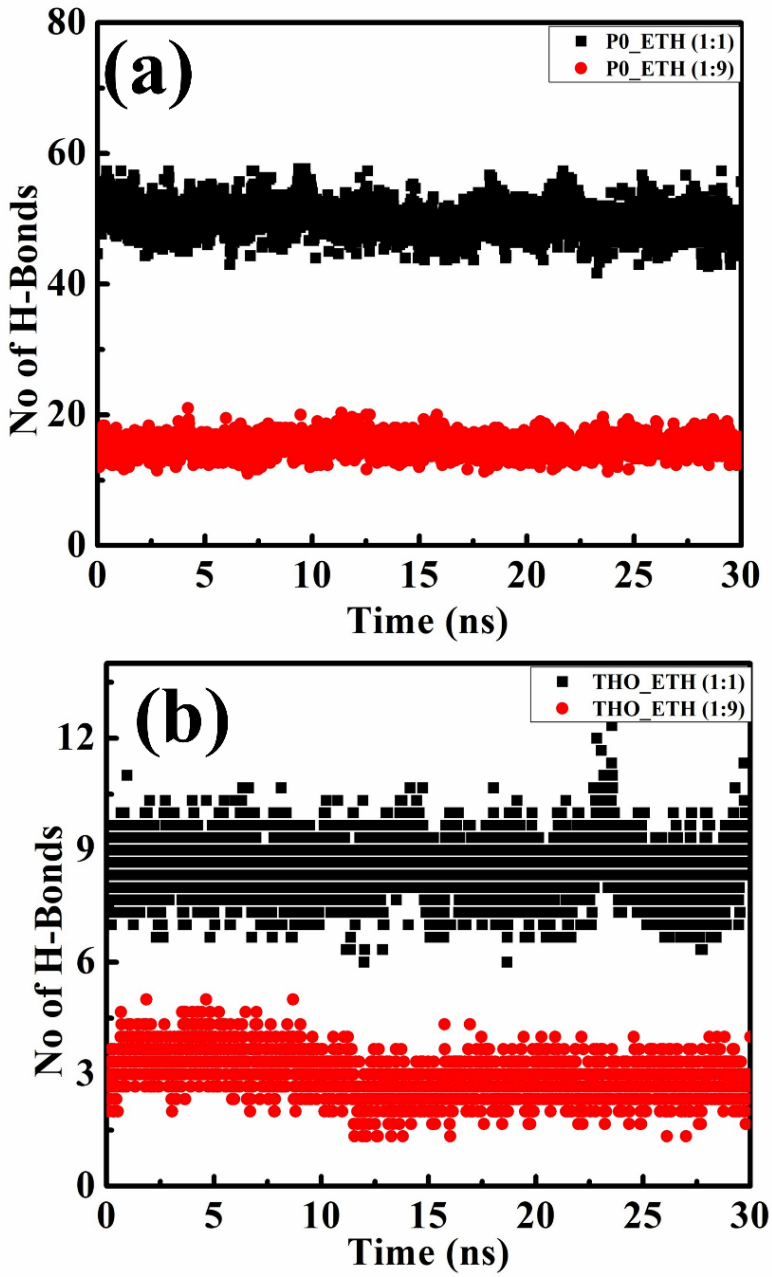


Figure. S25 (a). Number of Hydrogen bonds in procyanidins in ethanol-based system and (b) Number of Hydrogen bonds in theobromine in ethanol-based system

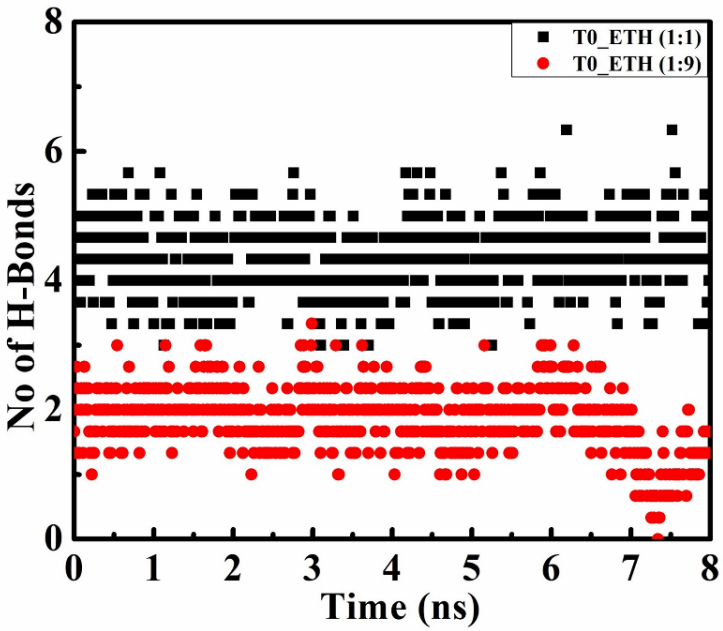


Figure. S26 Number of Hydrogen bonds in Theophylline in ethanol-based system

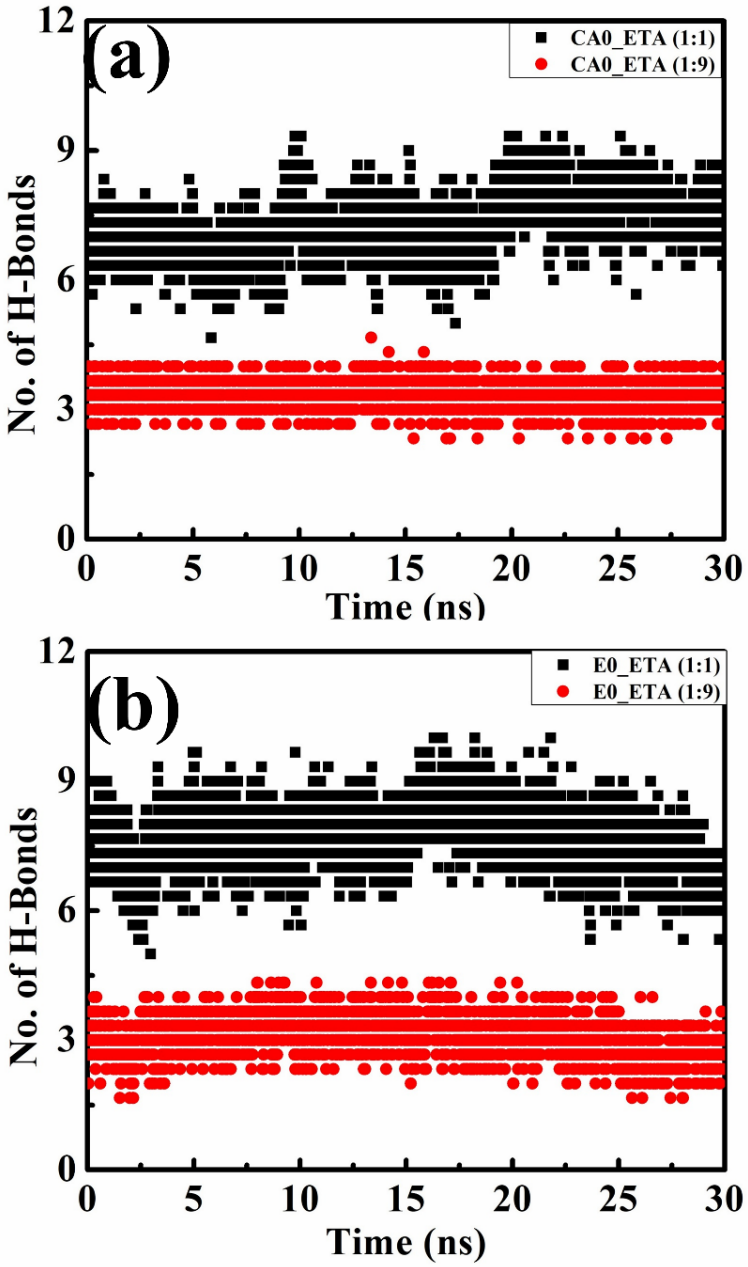


Figure. S27 (a). Number of Hydrogen bonds in catechin in ethyl acetate-based system and (b) Number of Hydrogen bonds in epicatechin in ethyl acetate -based system

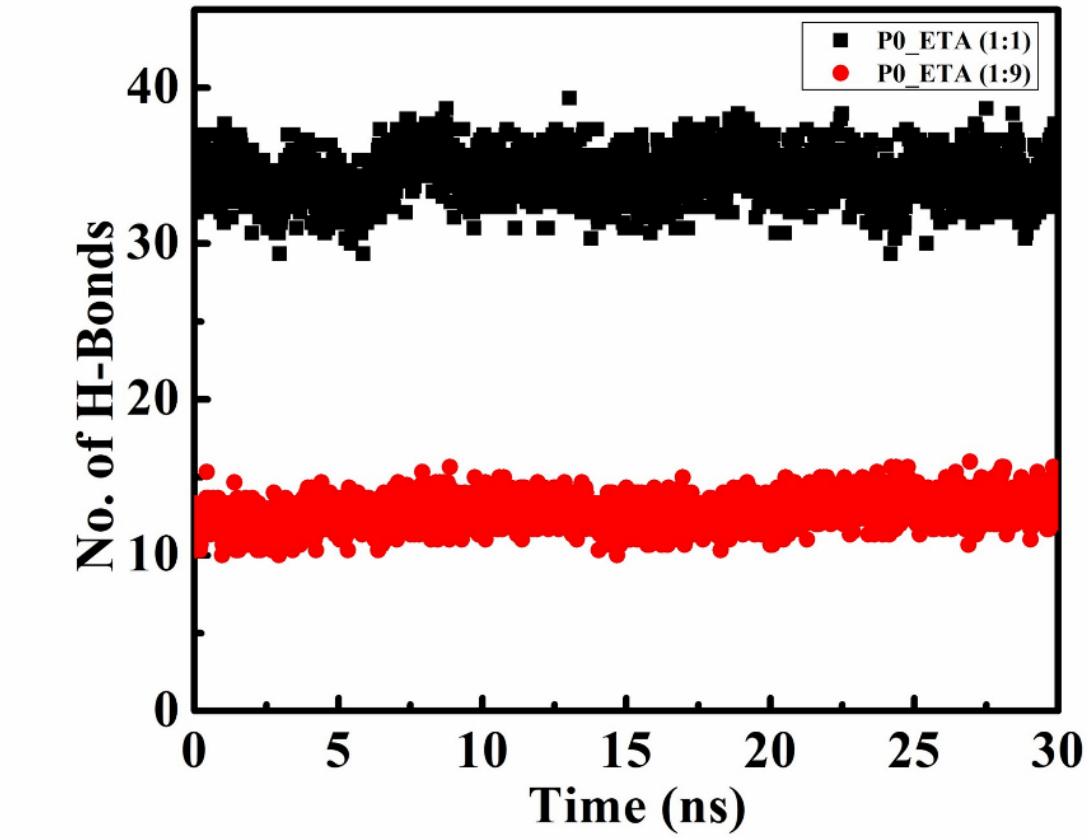


Figure. S28. Number of Hydrogen bonds in procyanidin in ethyl acetate-based system

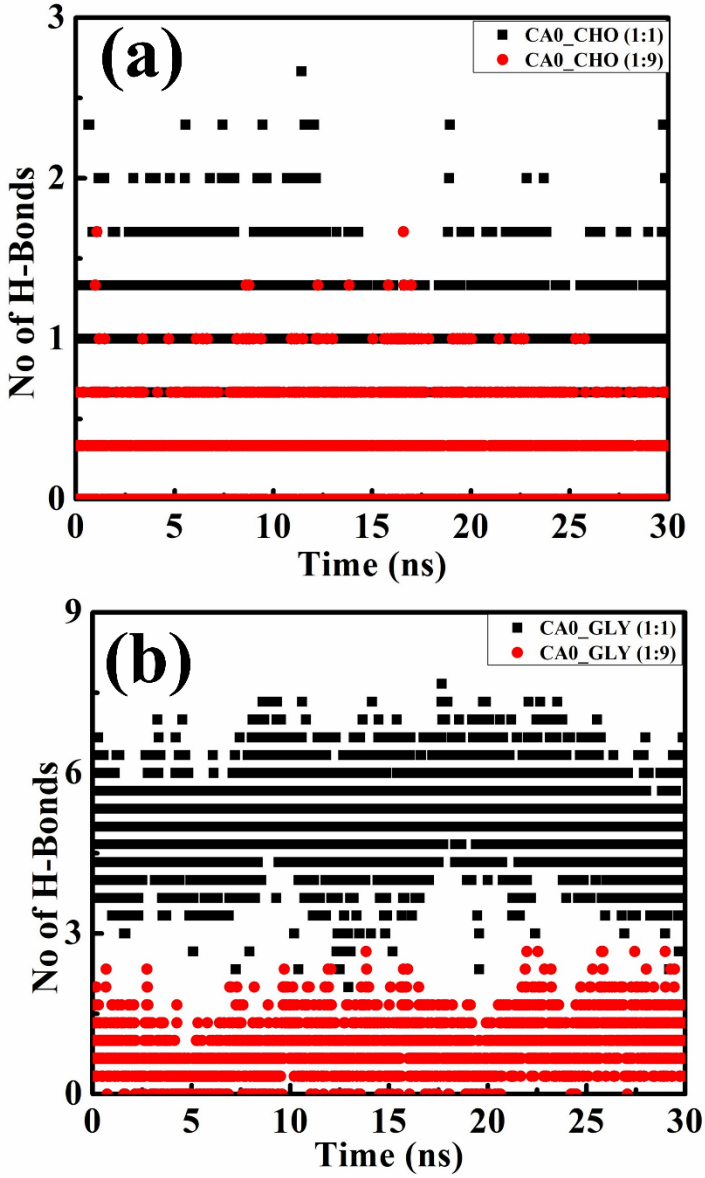


Figure. S29 (a). Number of Hydrogen bonds in Catechin in Choline chloride-based system and (b) Number of Hydrogen bonds in catechin in glycerol-based system

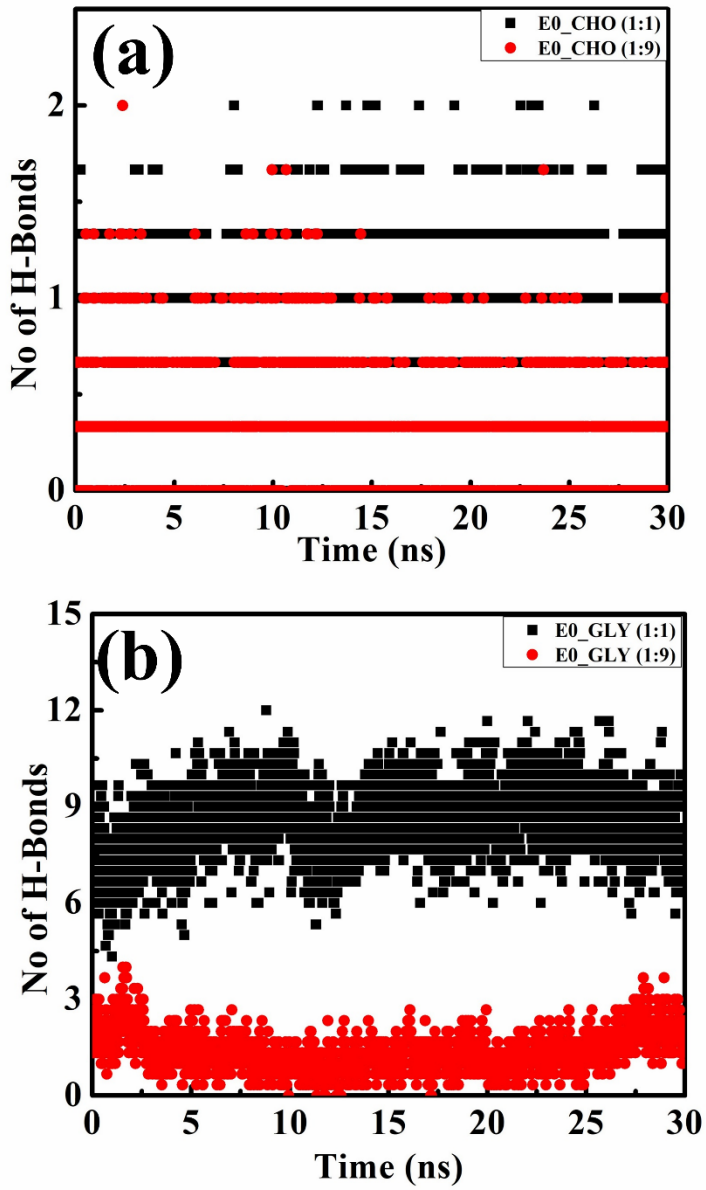


Figure. S30 (a). Number of Hydrogen bonds in epicatechin in Choline chloride-based system and (b) Number of Hydrogen bonds in epicatechin in glycerol-based system

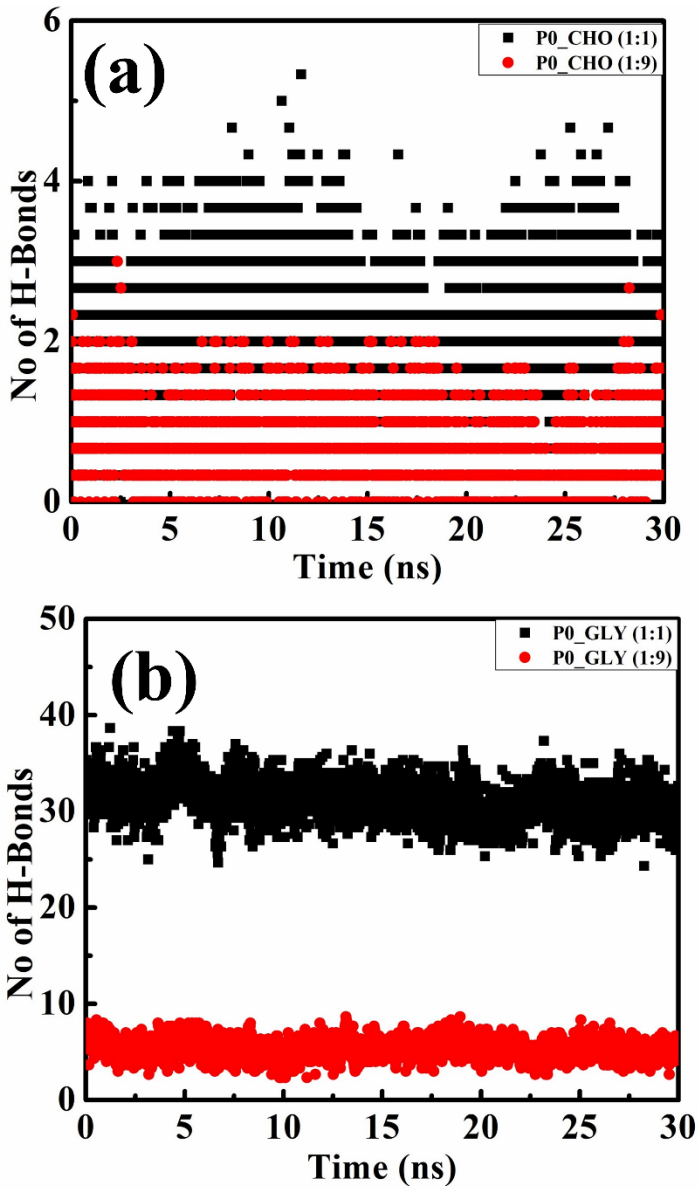


Figure. S31 (a). Number of Hydrogen bonds in Procyanidin in Choline chloride-based system and (b) Number of Hydrogen bonds in Procyanidin in glycerol-based system

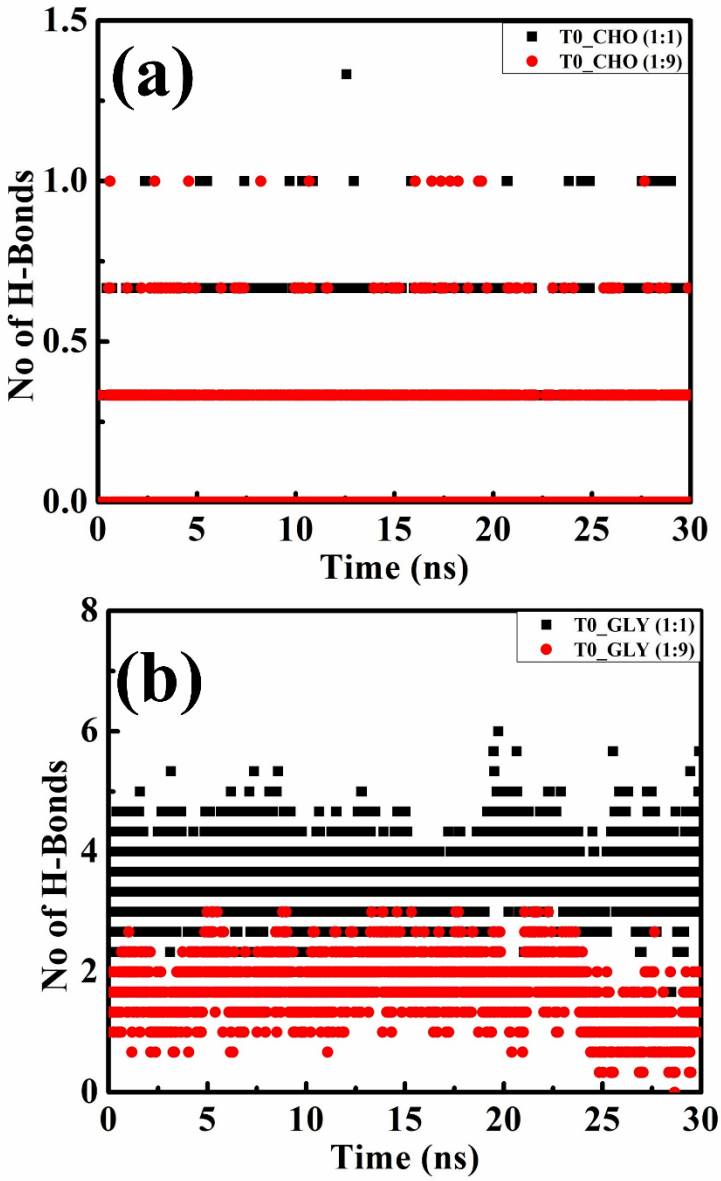


Figure. S32 (a). Number of Hydrogen bonds in Theophylline in Choline chloride-based system and (b) Number of Hydrogen bonds in Theophylline in glycerol-based system

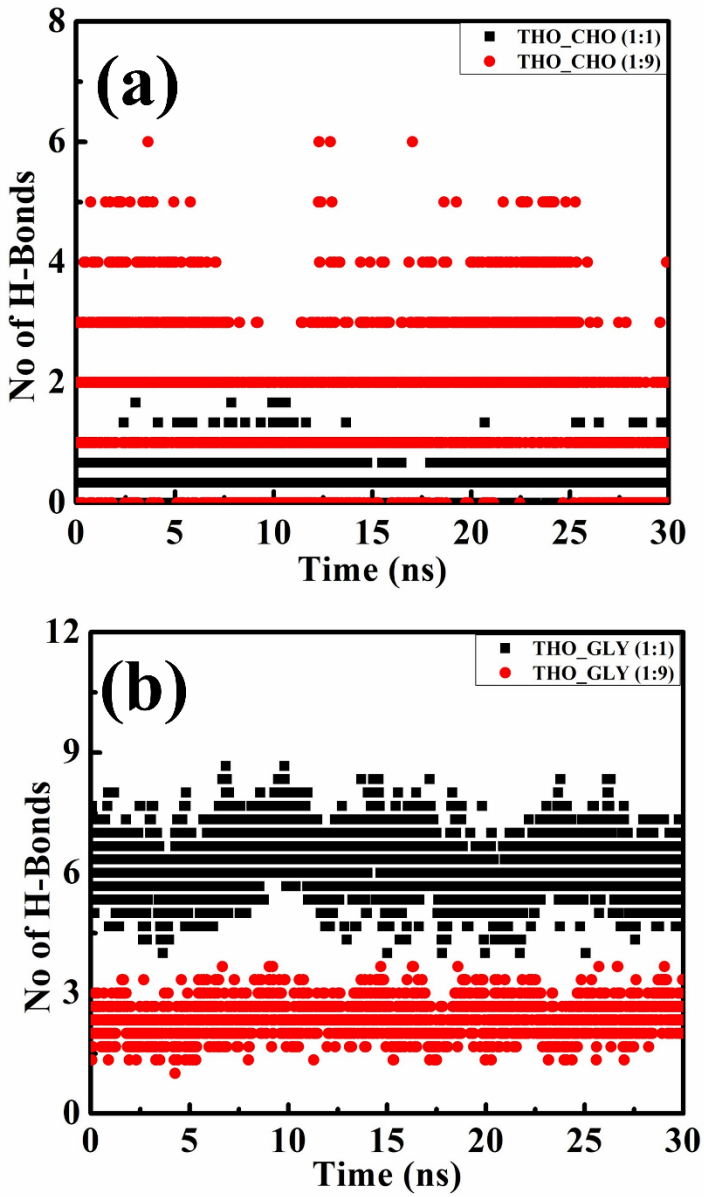


Figure. S33 (a). Number of Hydrogen bonds in Theobromine in Choline chloride-based system and (b) Number of Hydrogen bonds in Theobromine in glycerol-based system

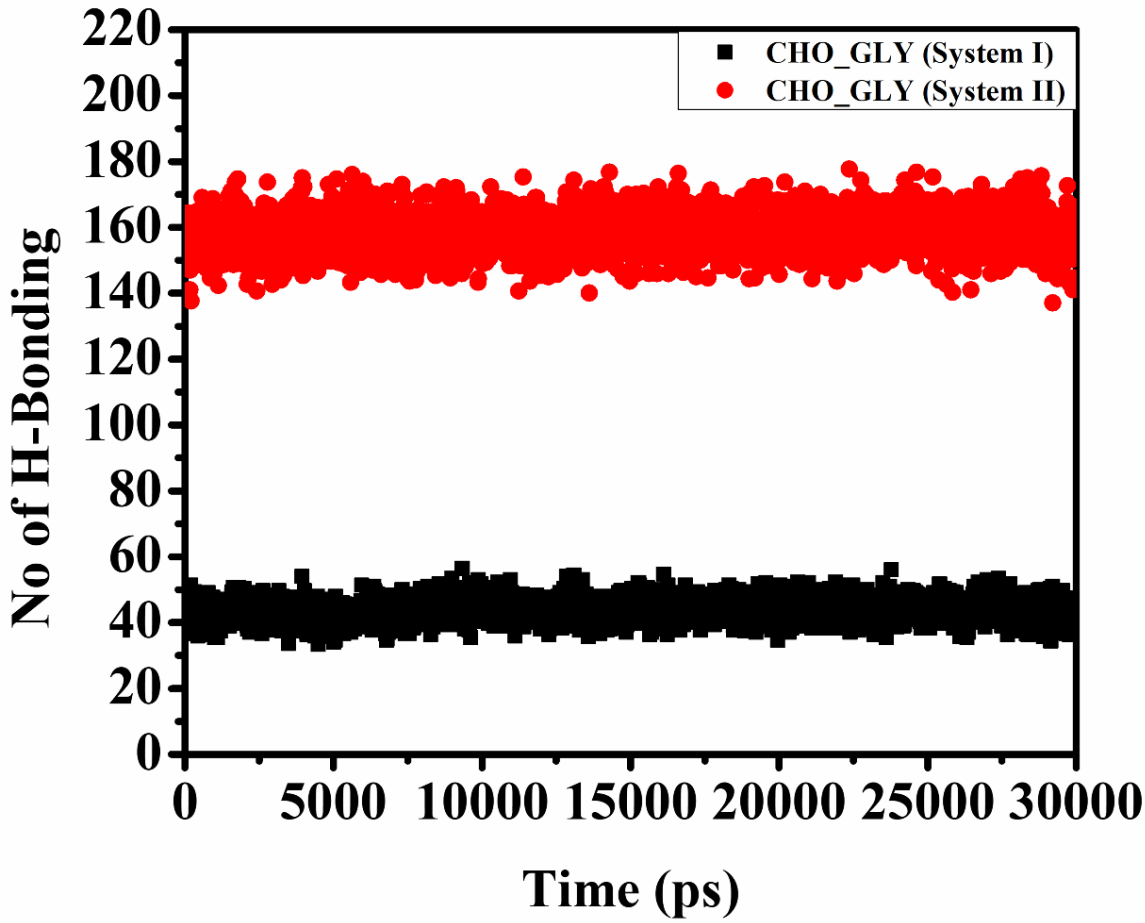


Figure. S34 Number of Hydrogen bonds between Choline chloride and glycerol

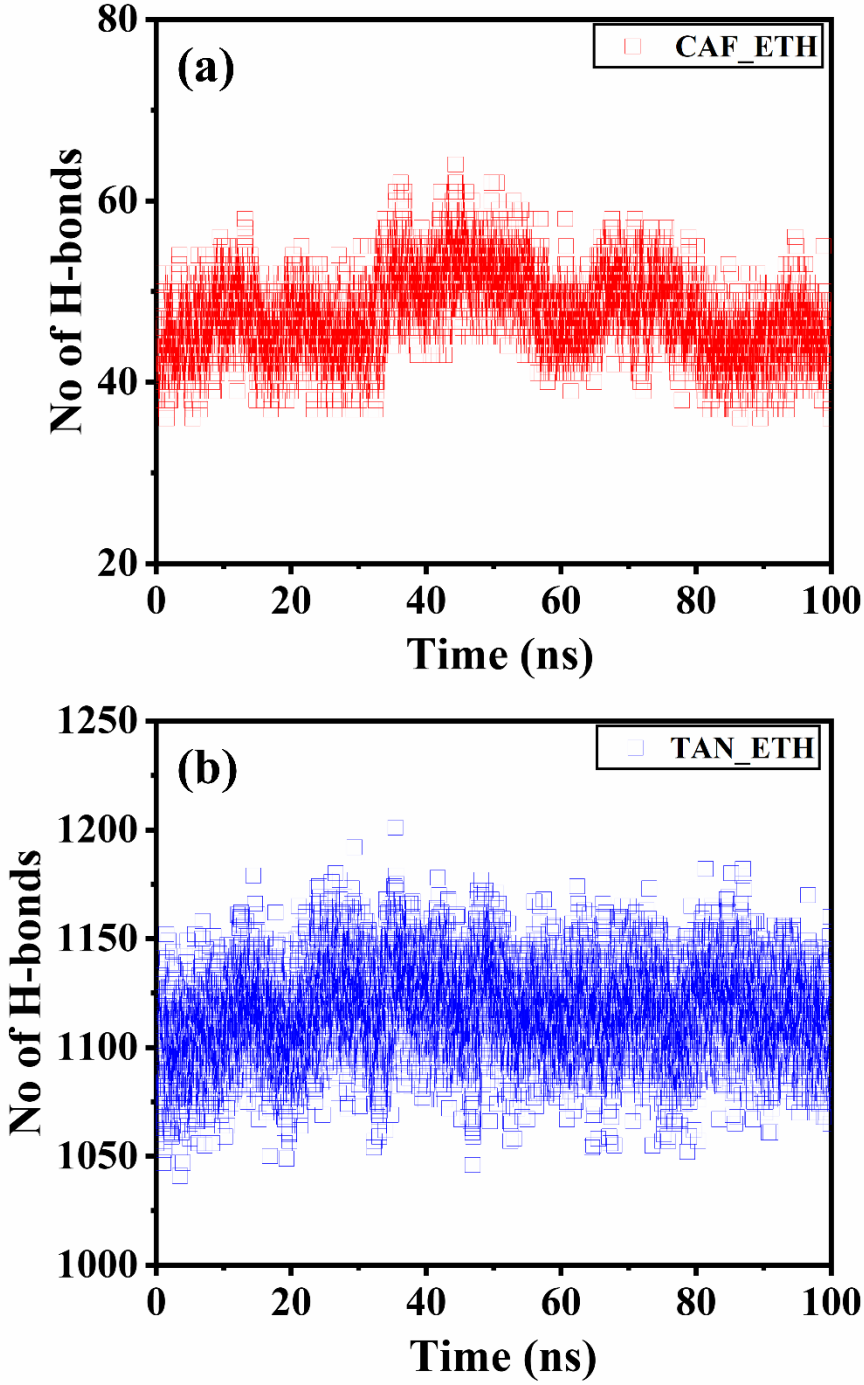


Figure S35. Number of H-bonds (1:1) ethanol system (a) Caffeine to ethanol (b) Tannic acids to ethanol

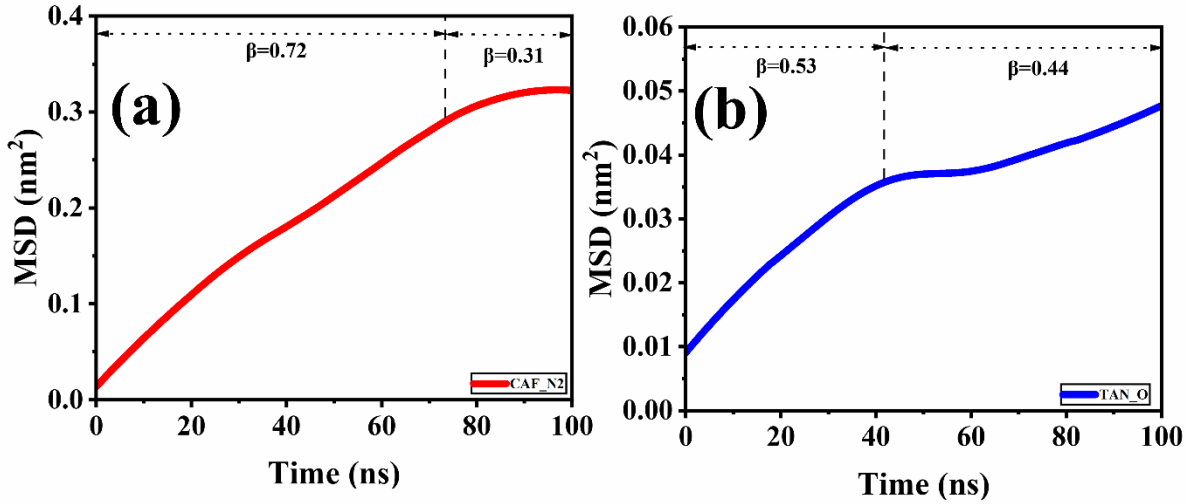


Figure S36. MSD in the ethanol (1:1) system (a) N2 of caffeine (b) H3 of tannic acids

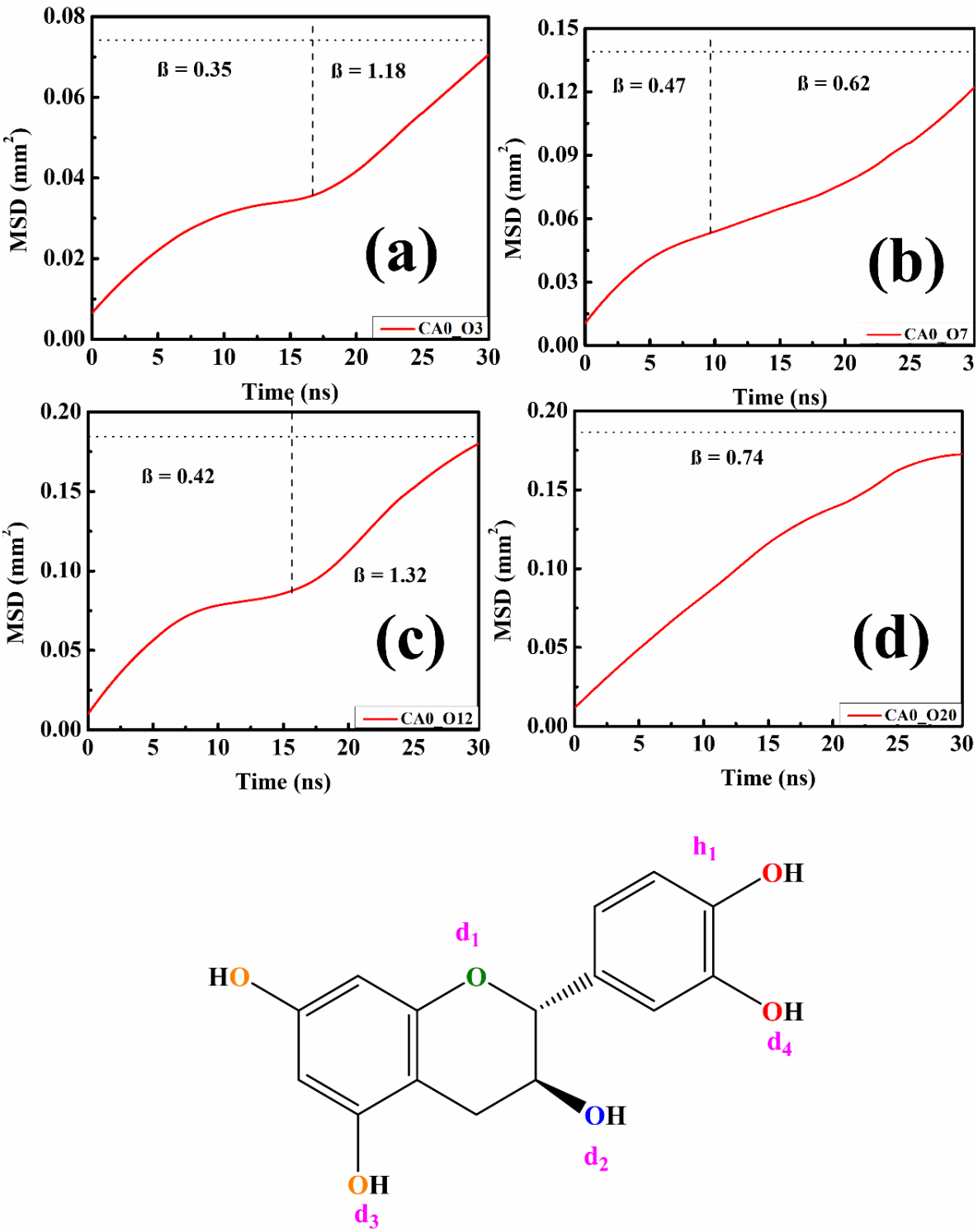


Figure. S37 MSD curve of catechin in ethanol-based system I (1:1) (a)O3 (denoted by d_1) (b) O7 (denoted by d_2) (c) O12 (denoted by d_3) and (d) O20 (denoted by d_4).

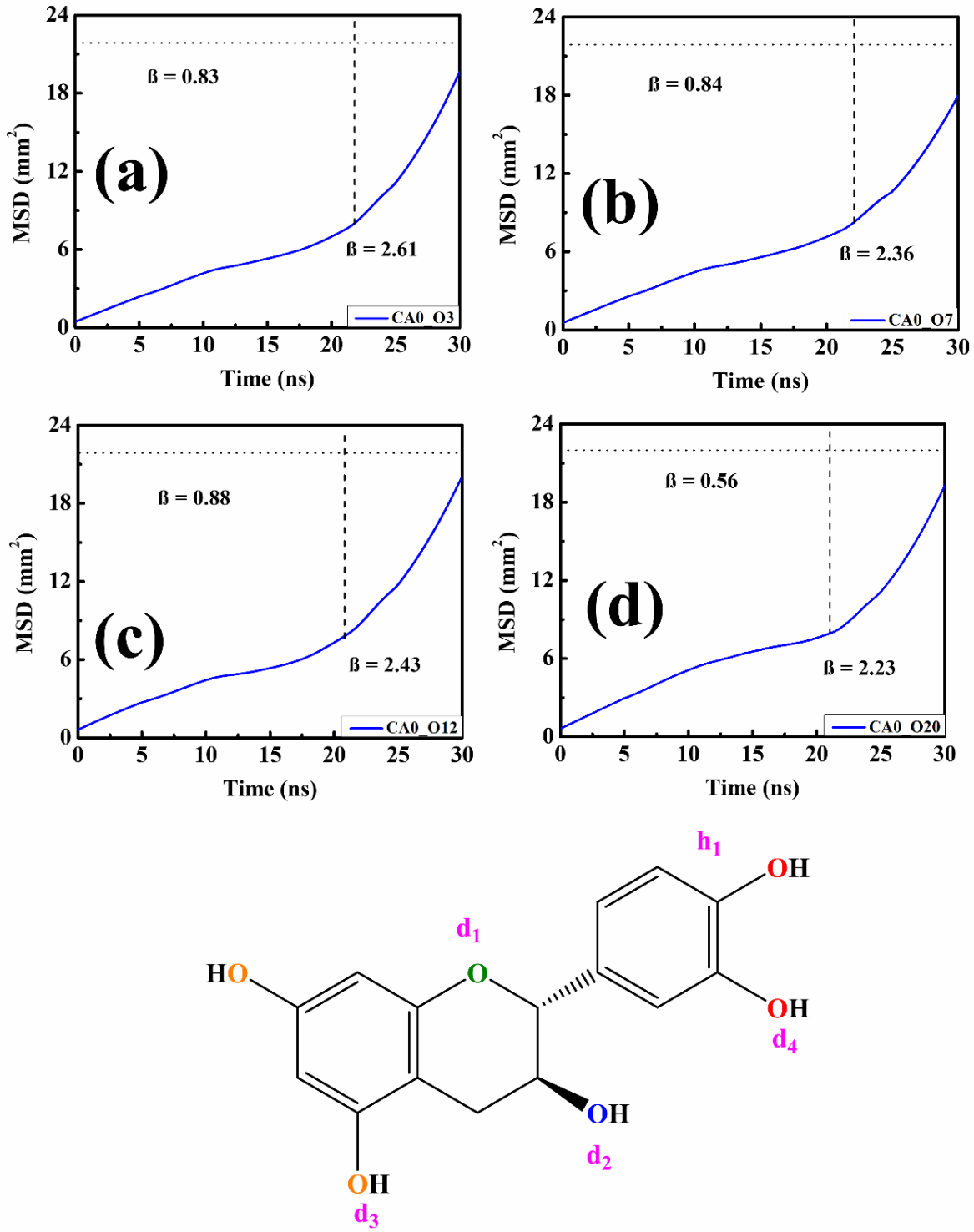
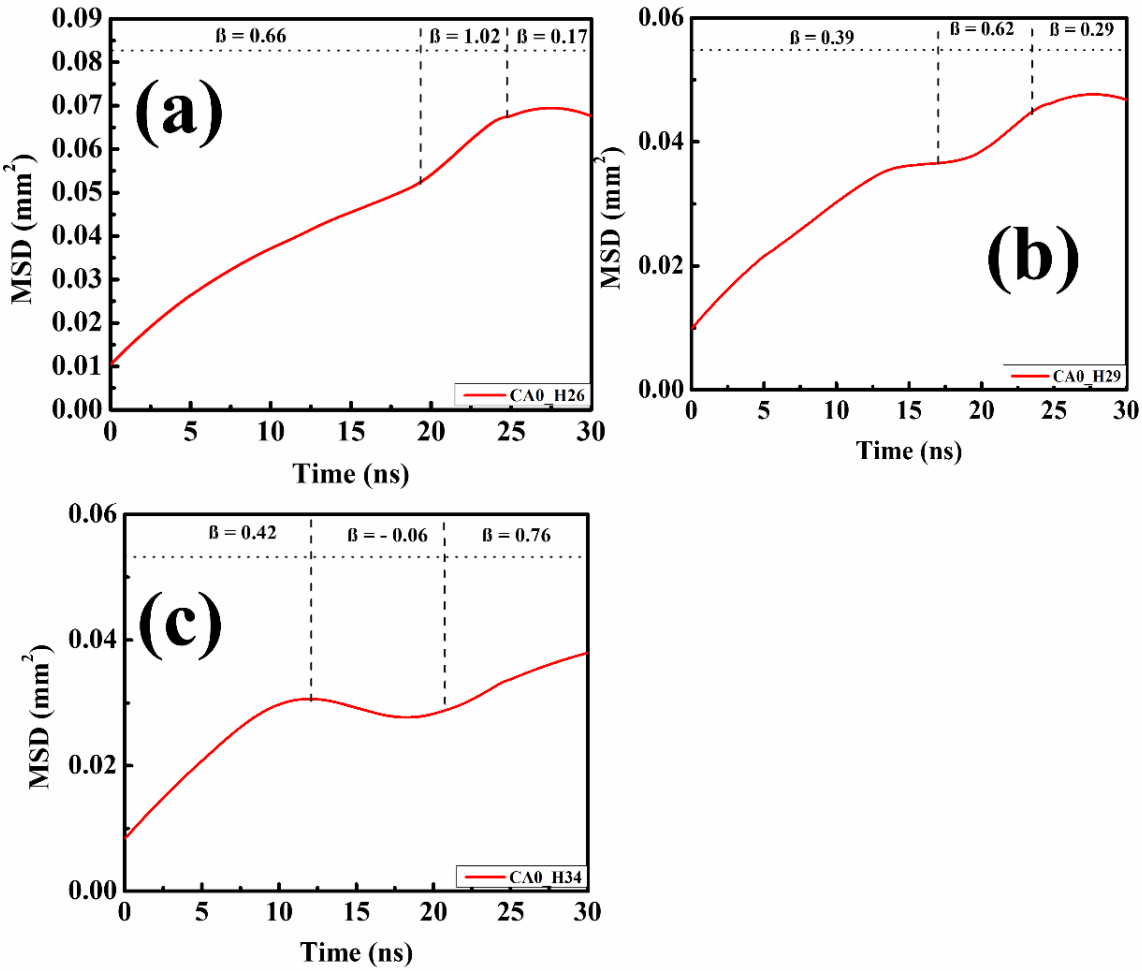


Figure. S38 MSD curve of catechin in ethanol-based system II (1:9) (a)O3 (denoted by d_1) (b) O7 (denoted by d_2) (c) O12 (denoted by d_3) and (d) O20 (denoted by d_4).



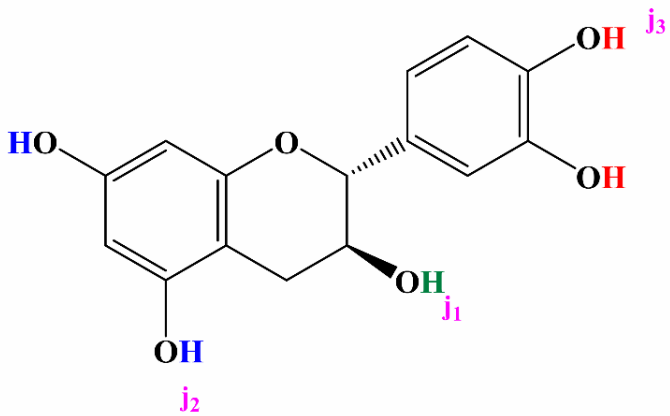
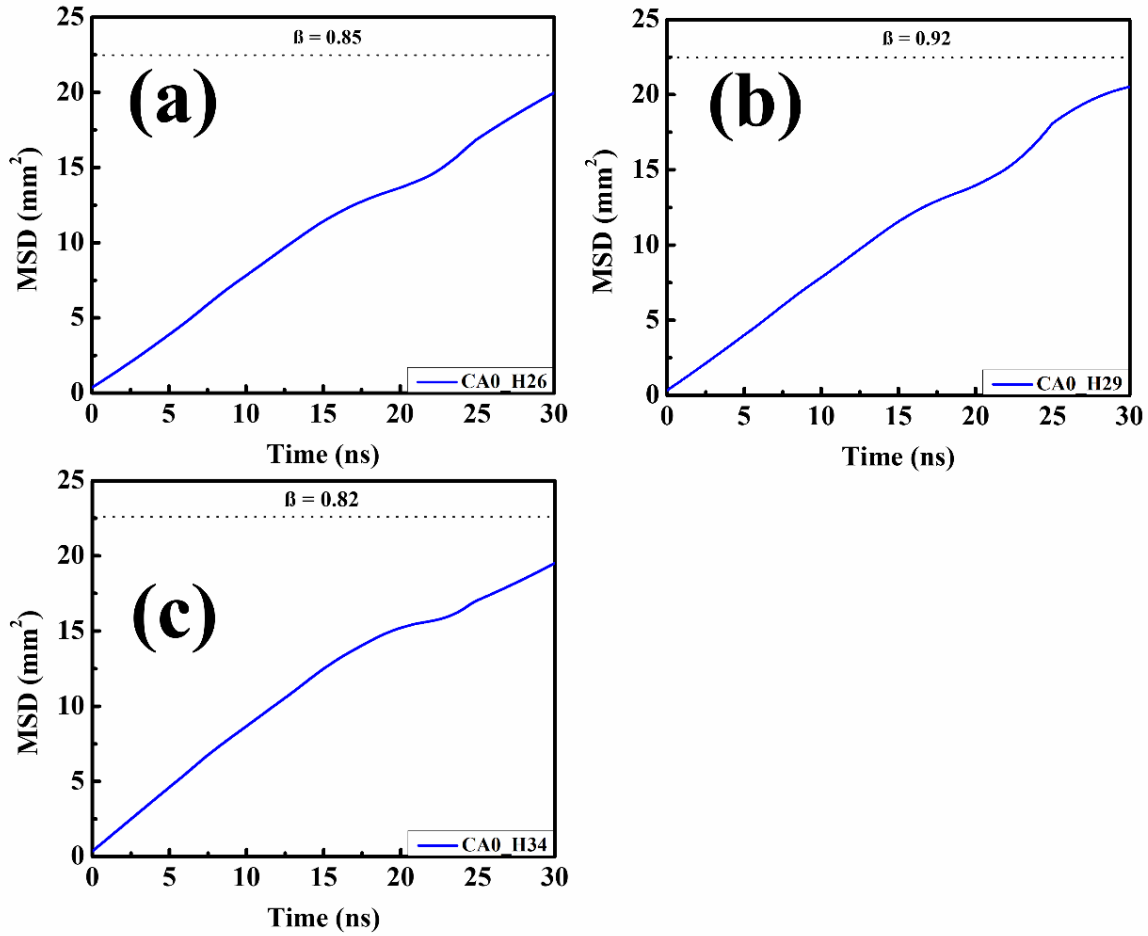


Figure. S39. MSD curve of catechin in ethyl acetate-based system I (a)H26 (denoted by j_1) (b) H29 (denoted by j_2) and (c)H34 (denoted by j_3) .



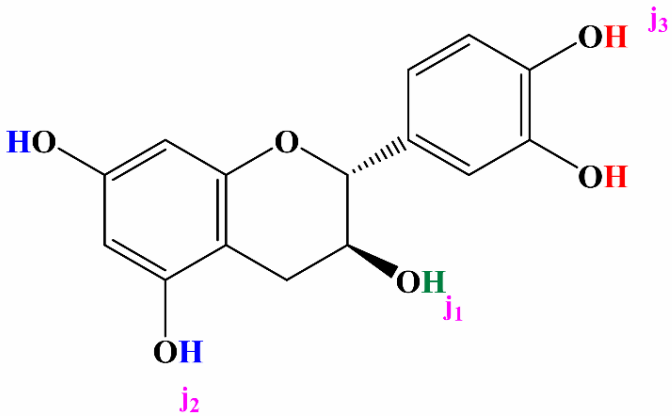
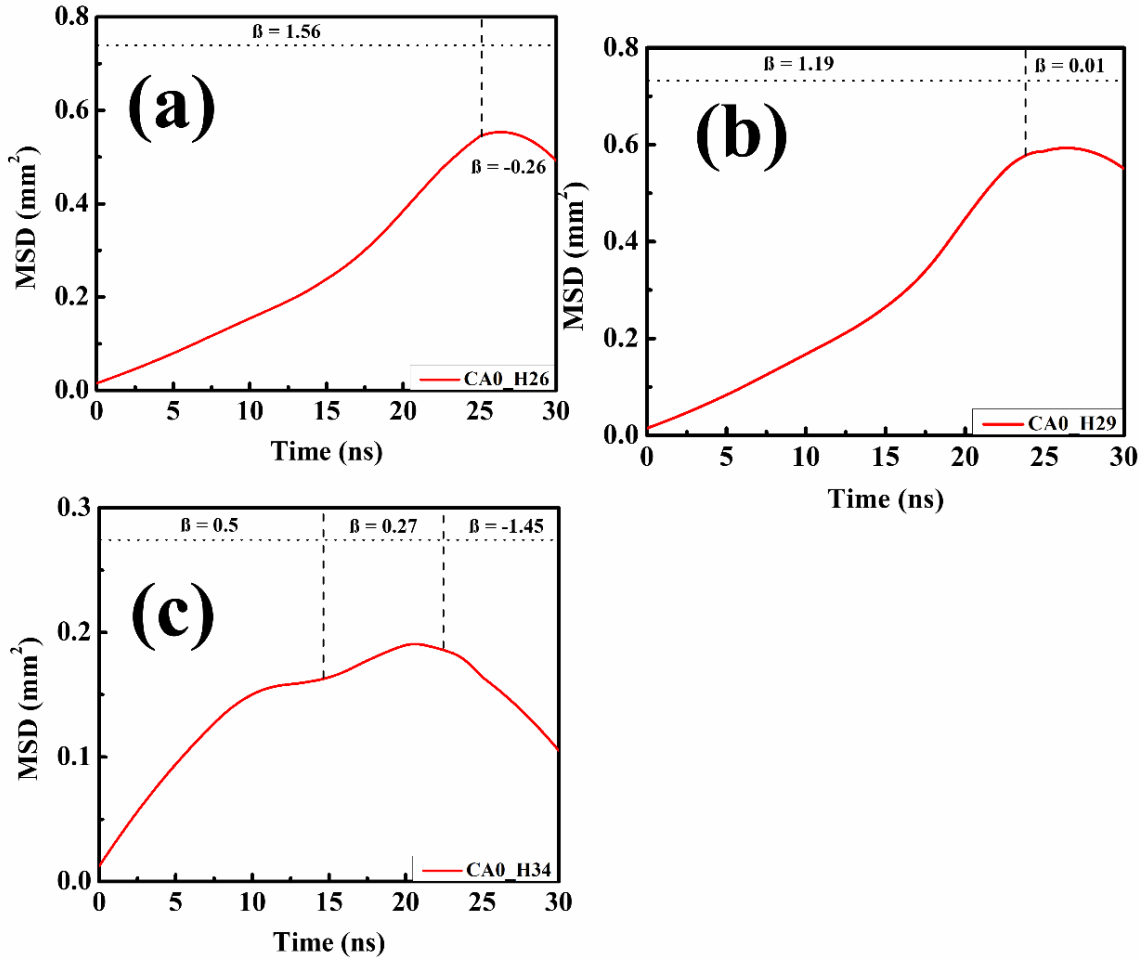


Figure. S40. MSD curve of catechin in ethyl acetate-based system II (a)H26 (denoted by j_1) (b) H29 (denoted by j_2) and (c)H34 (denoted by j_3) .



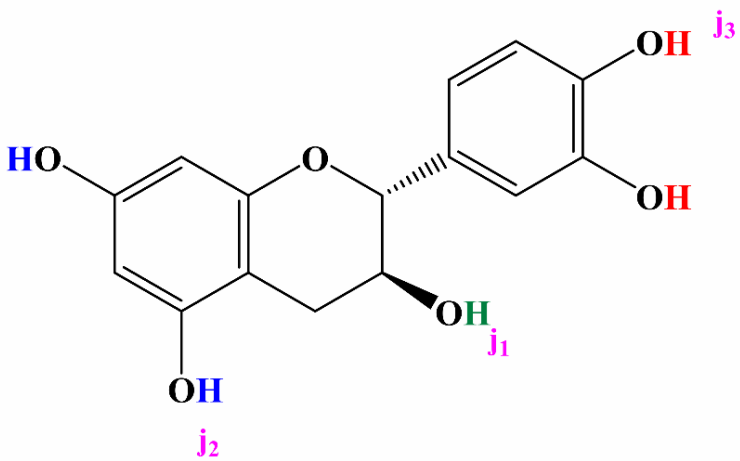


Figure.S41. MSD curve of catechin in DES based system I (a)H26 (denoted by j_1) (b) H29 (denoted by j_2) and (c)H34 (denoted by j_3) .

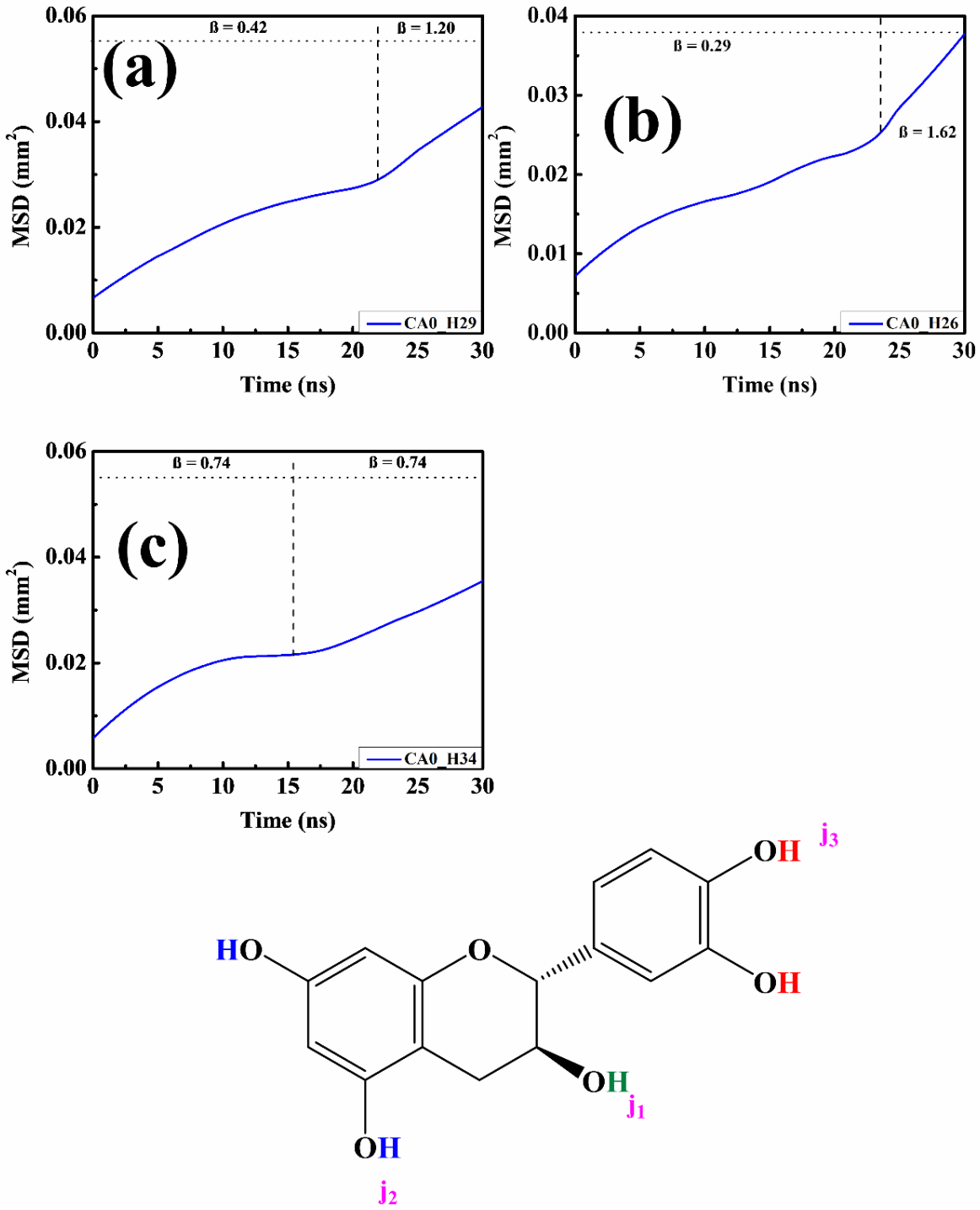
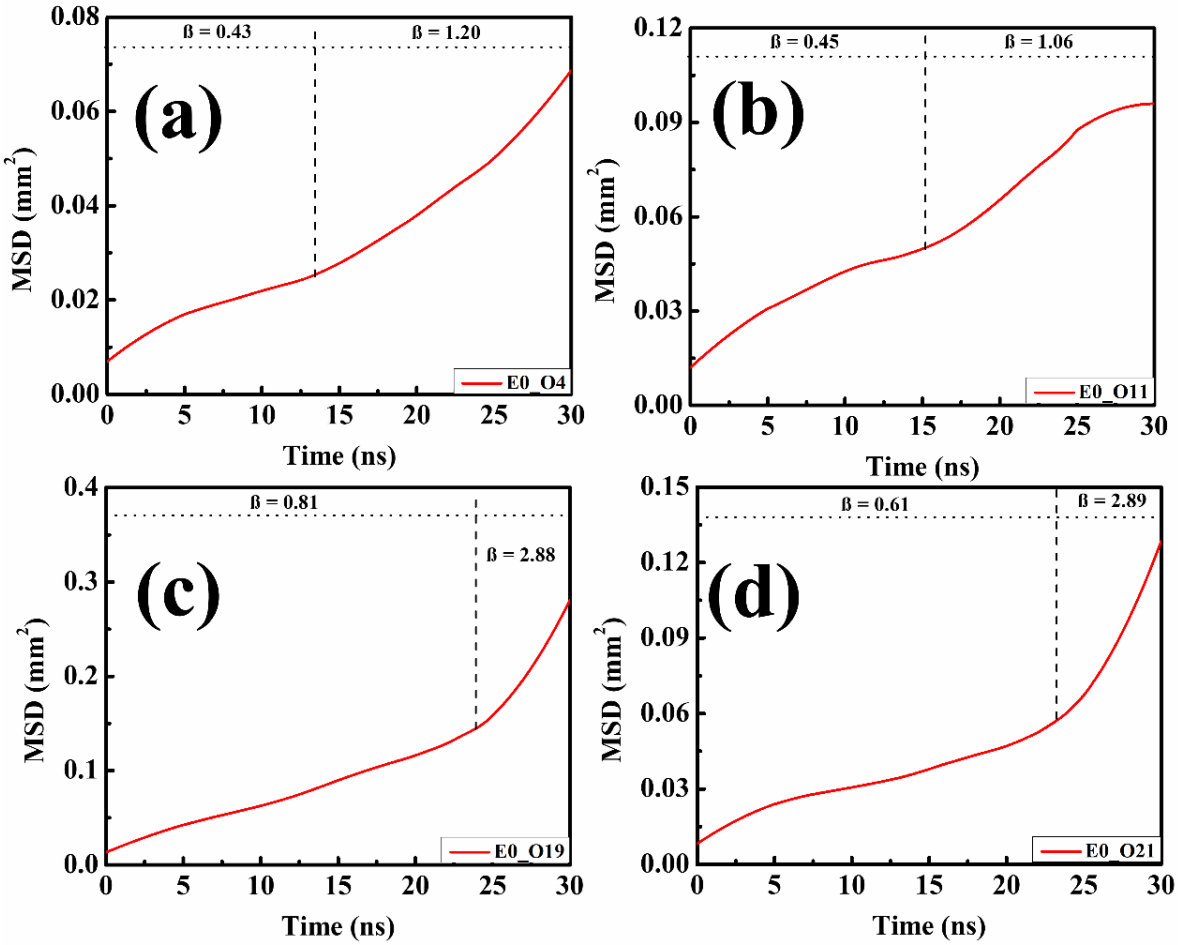


Figure.S42. MSD curve of catechin in DES based system II (a)H26 (denoted by j_1) (b) H29 (denoted by j_2) and (c)H34 (denoted by j_3) .



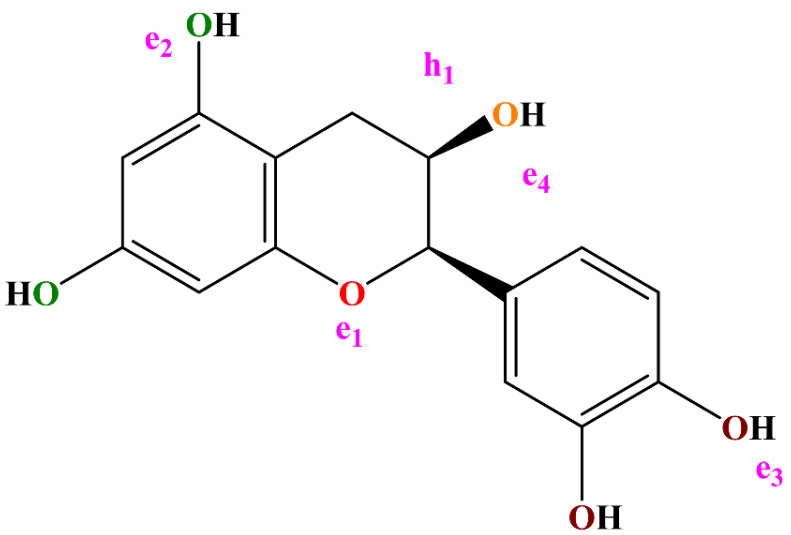


Figure. S43 MSD curve of epicatechin in ethanol-based system I (1:1) (a)O4 (denoted by e_1) (b) O11 (denoted by e_2) (c) O19 (denoted by e_3) and (d) O21(denoted by e_4).

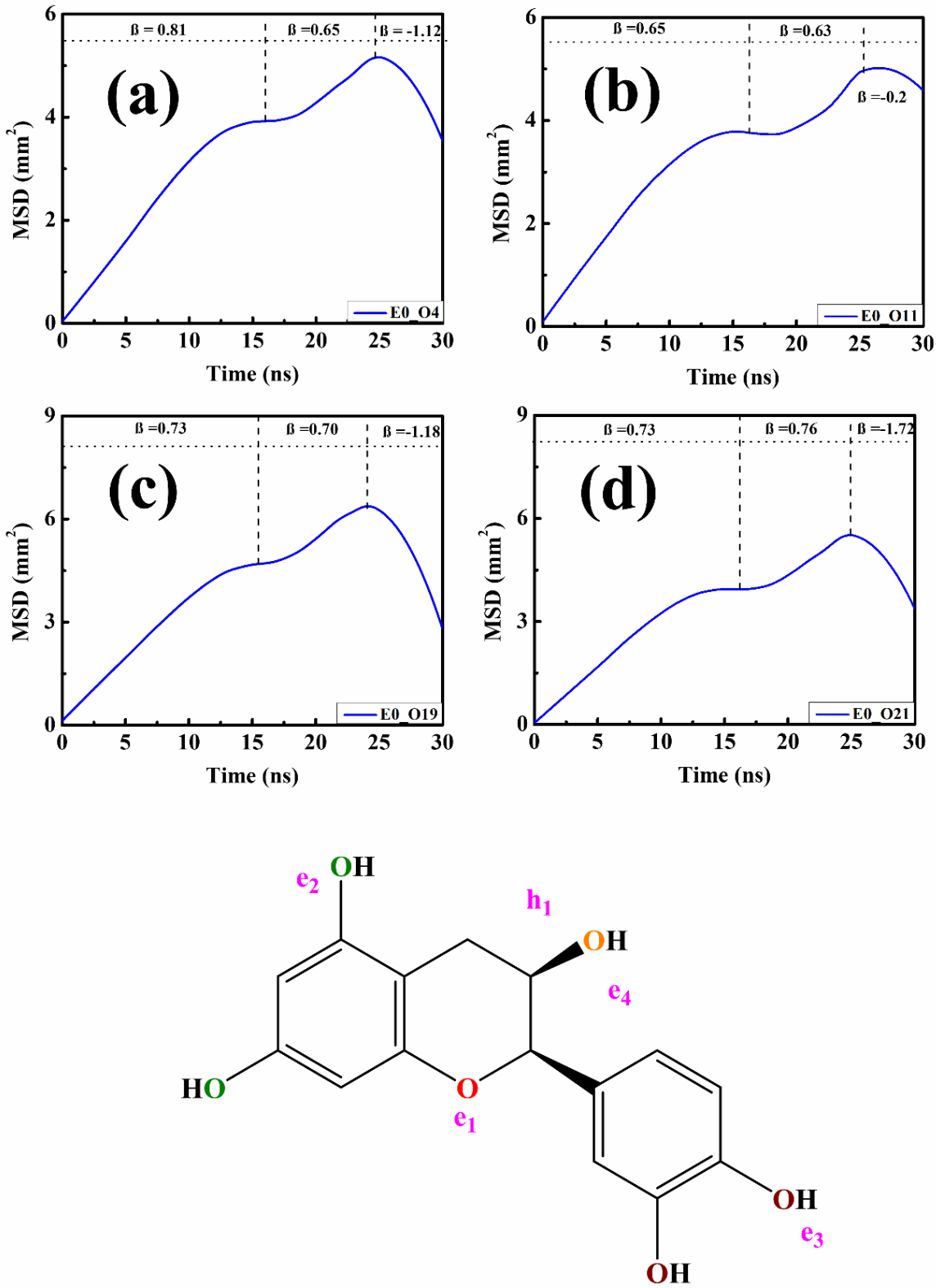


Figure. S44 MSD curve of epicatechin in ethanol-based system II (1:9) (a)O4 (denoted by e_1) (b) O11 (denoted by e_2) (c) O19 (denoted by e_3) and (d) O21(denoted by e_4).

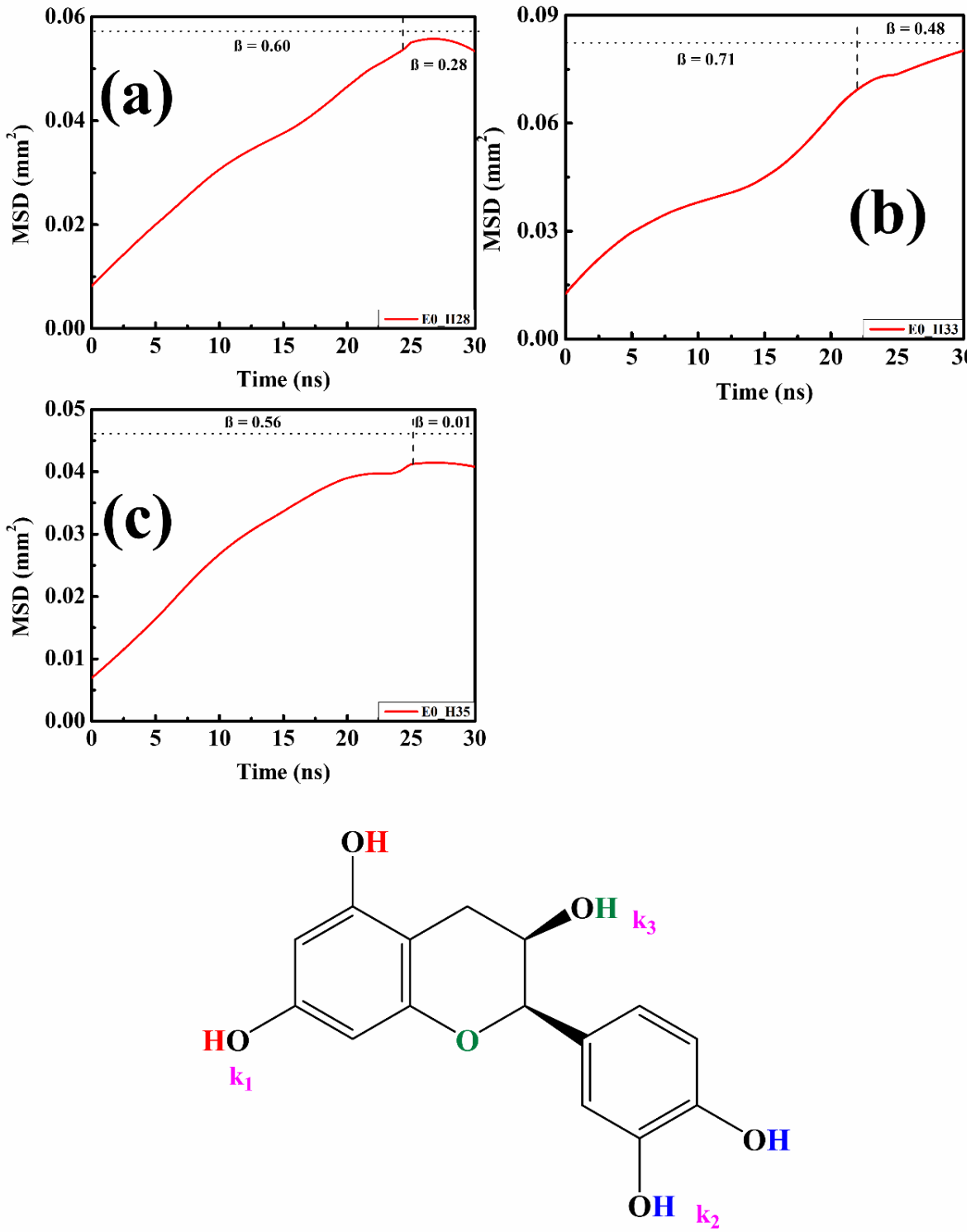
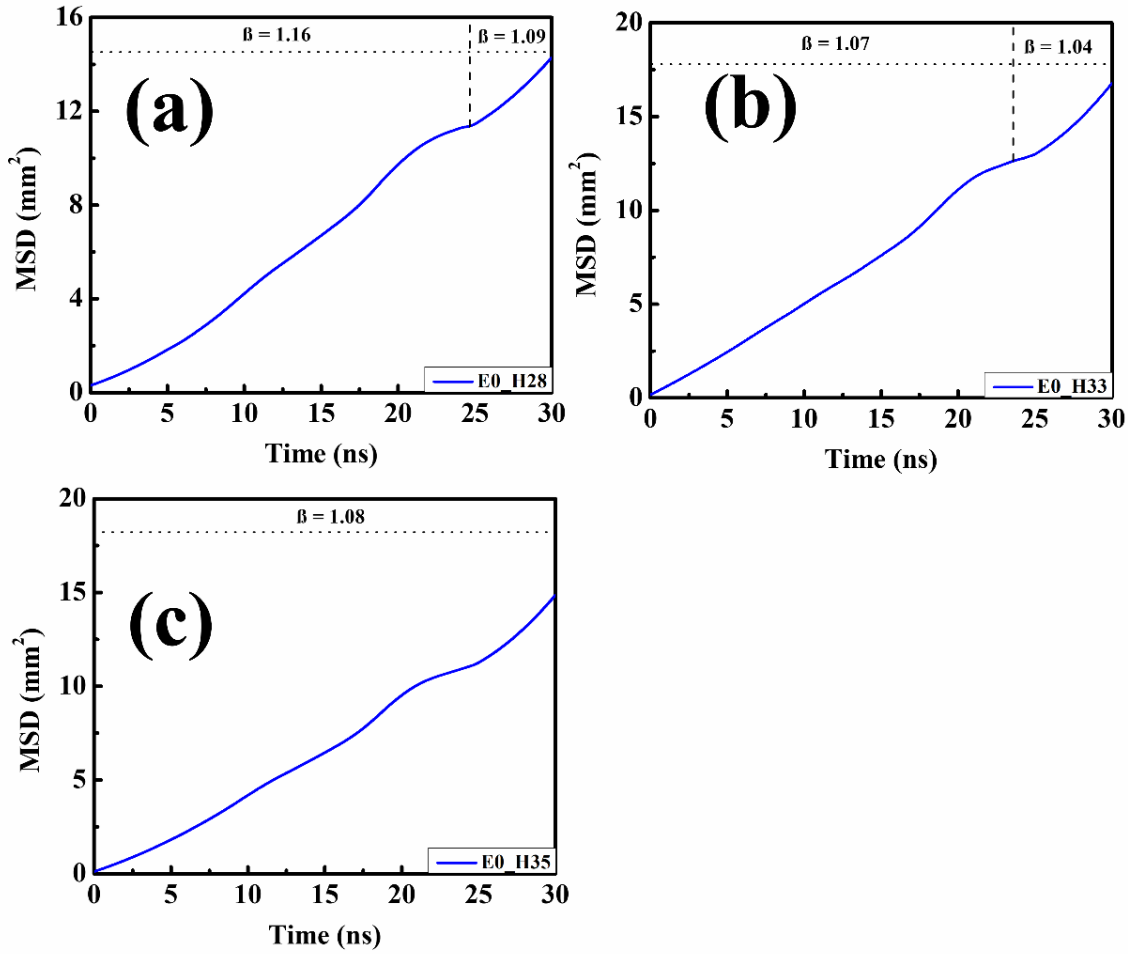


Figure. S45. MSD curve of epicatechin in ethyl acetate-based system I (a)H28 (denoted by k_1) (b) H33 (denoted by k_2) and (c)H35 (denoted by k_3).



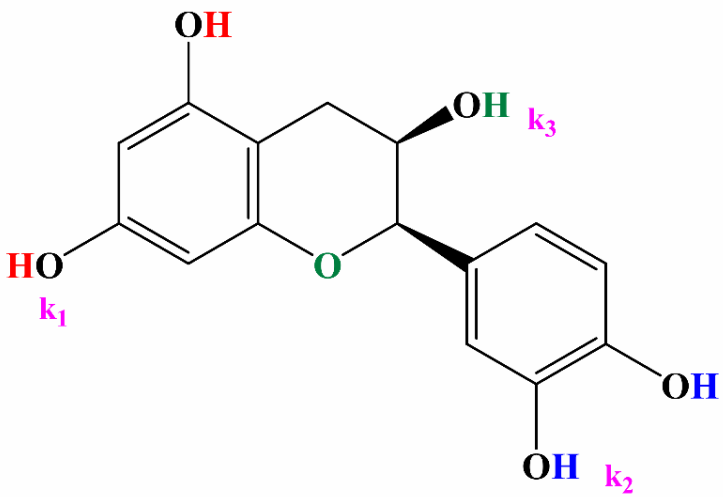


Figure. S46. MSD curve of epicatechin in ethyl acetate-based system II (a)H28 (denoted by k_1) (b) H33 (denoted by k_2) and (c)H35 (denoted by k_3).

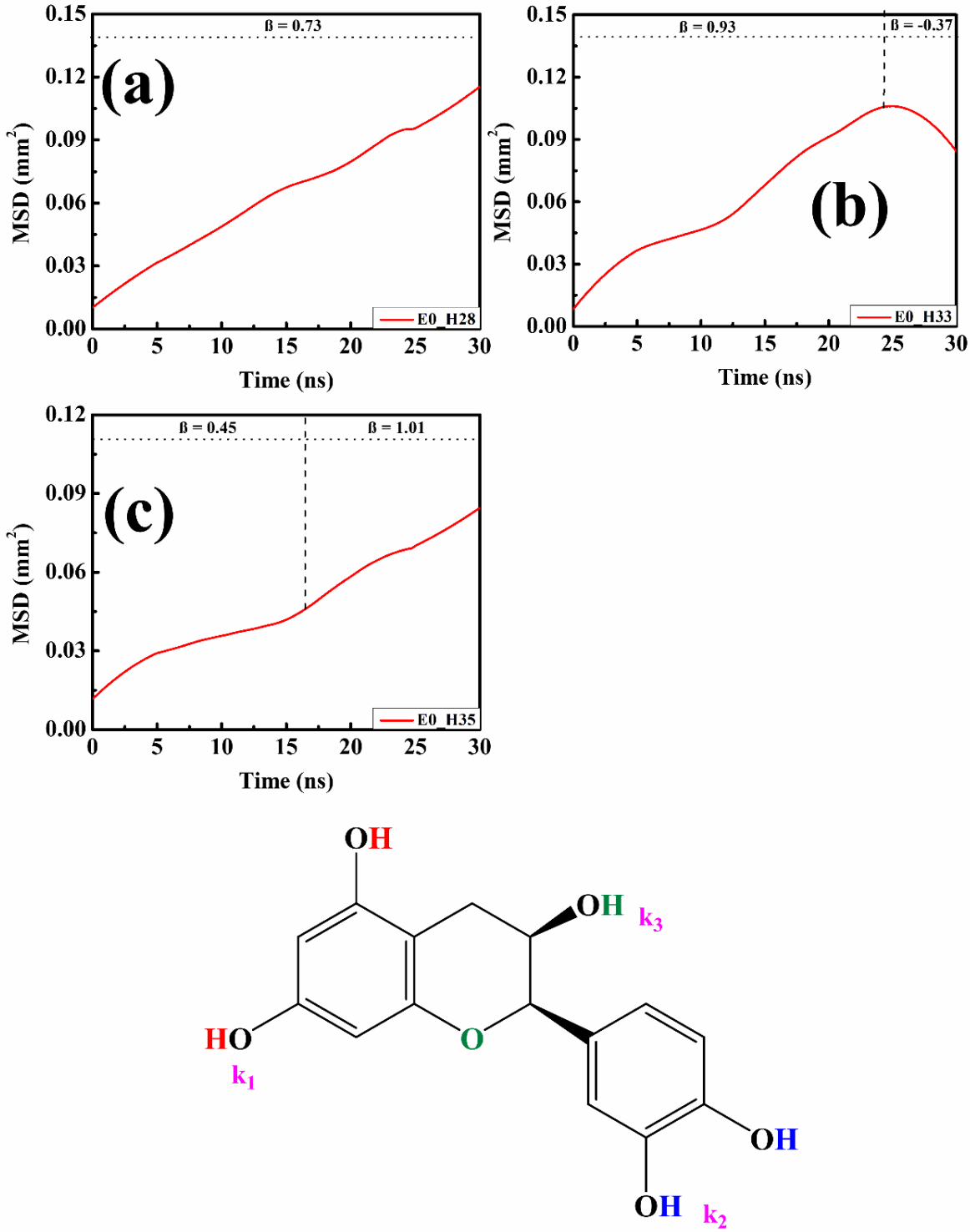


Figure.S47. MSD curve of epicatechin in DES based system I (a)H28 (denoted by k_1) (b) H33 (denoted by k_2) and (c)H35 (denoted by k_3).

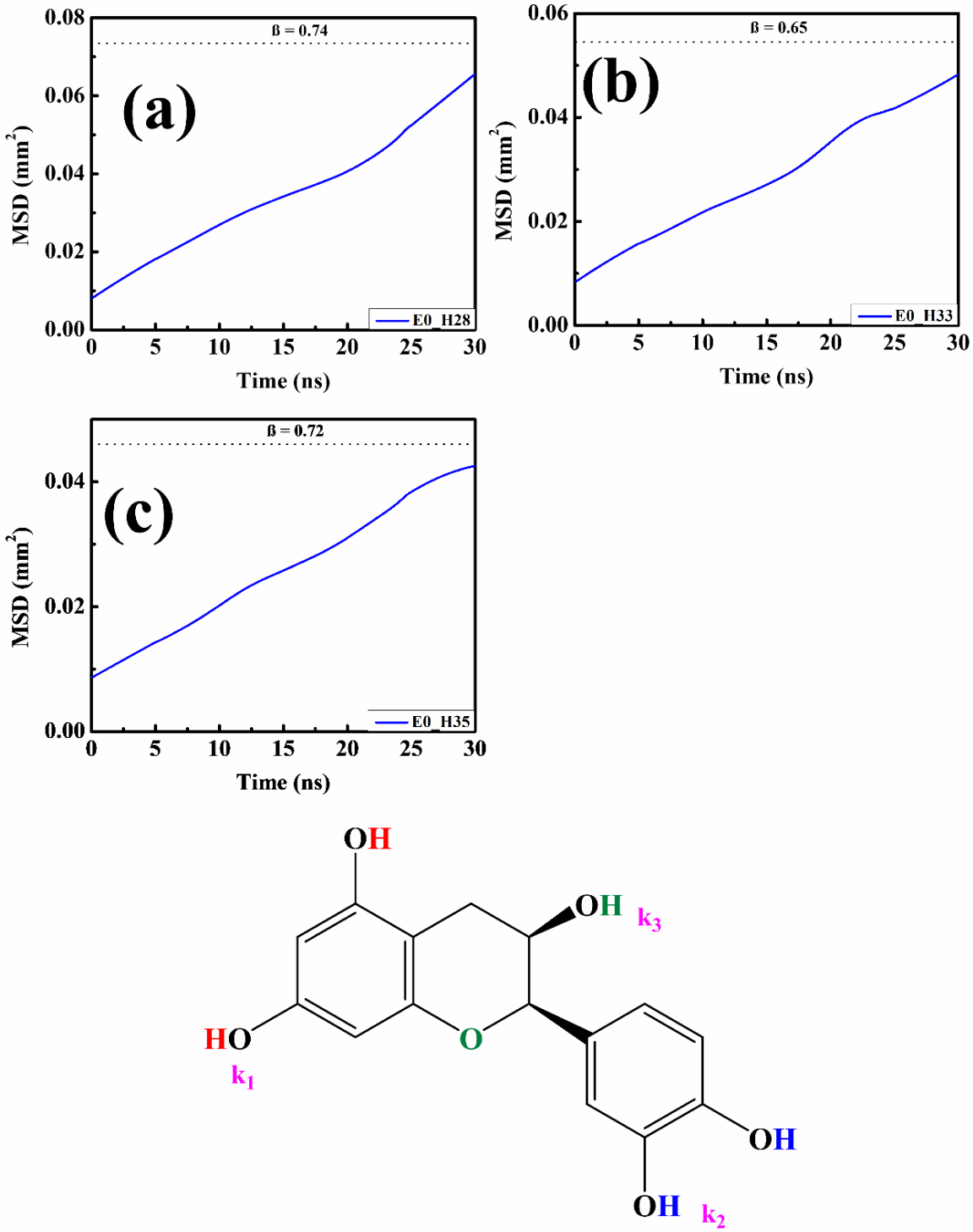
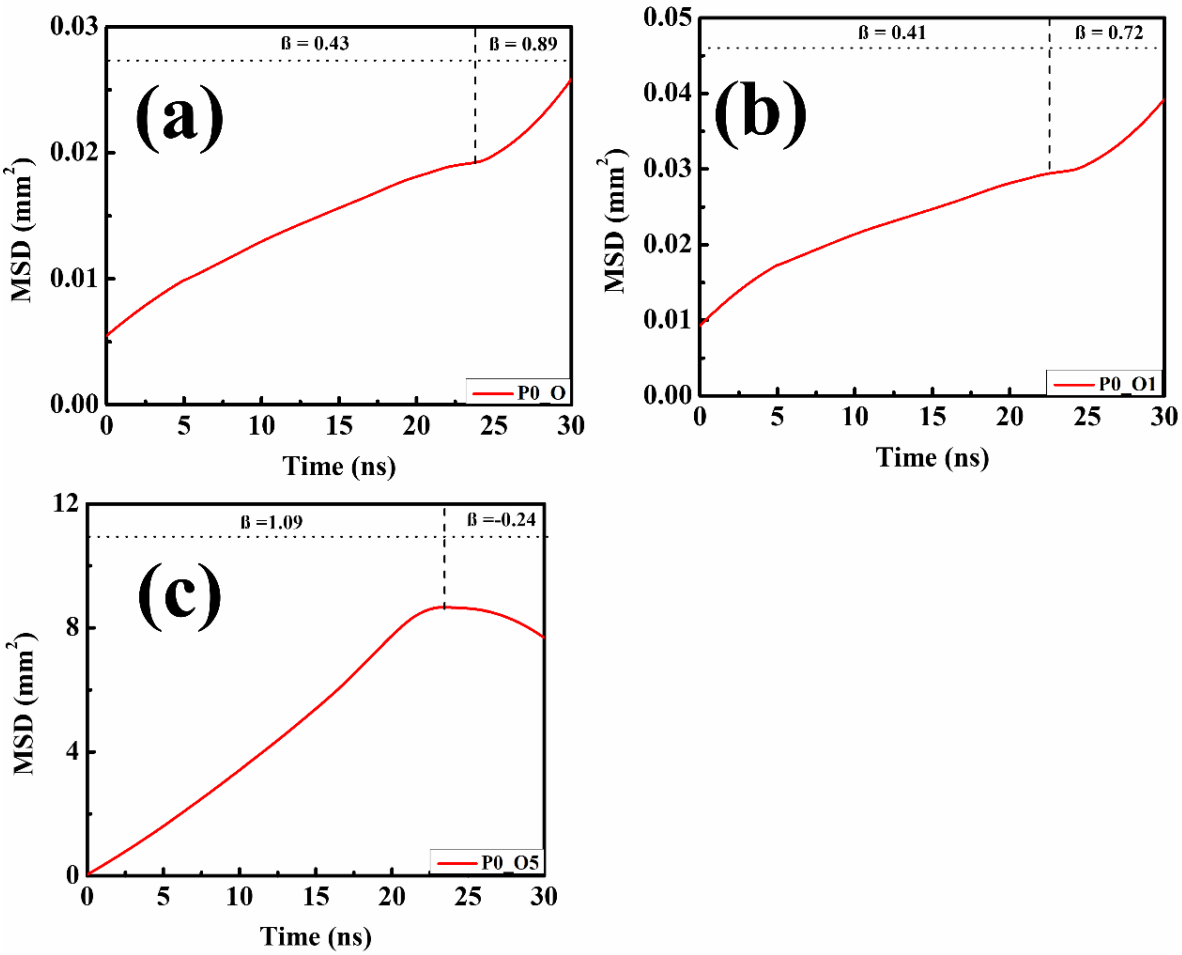


Figure.S48. MSD curve of epicatechin in DES based system II (a)H28 (denoted by k_1) (b) H33 (denoted by k_2) and (c)H35 (denoted by k_3).



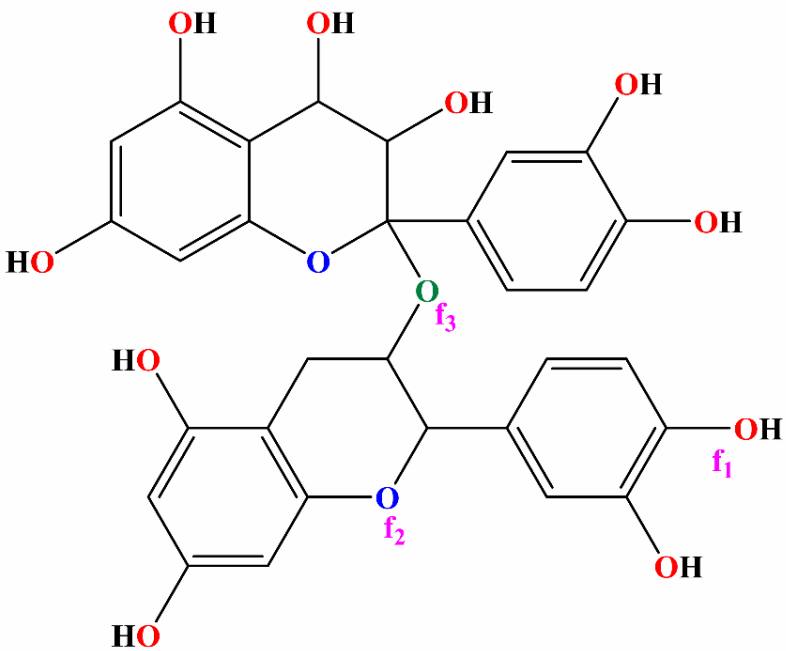
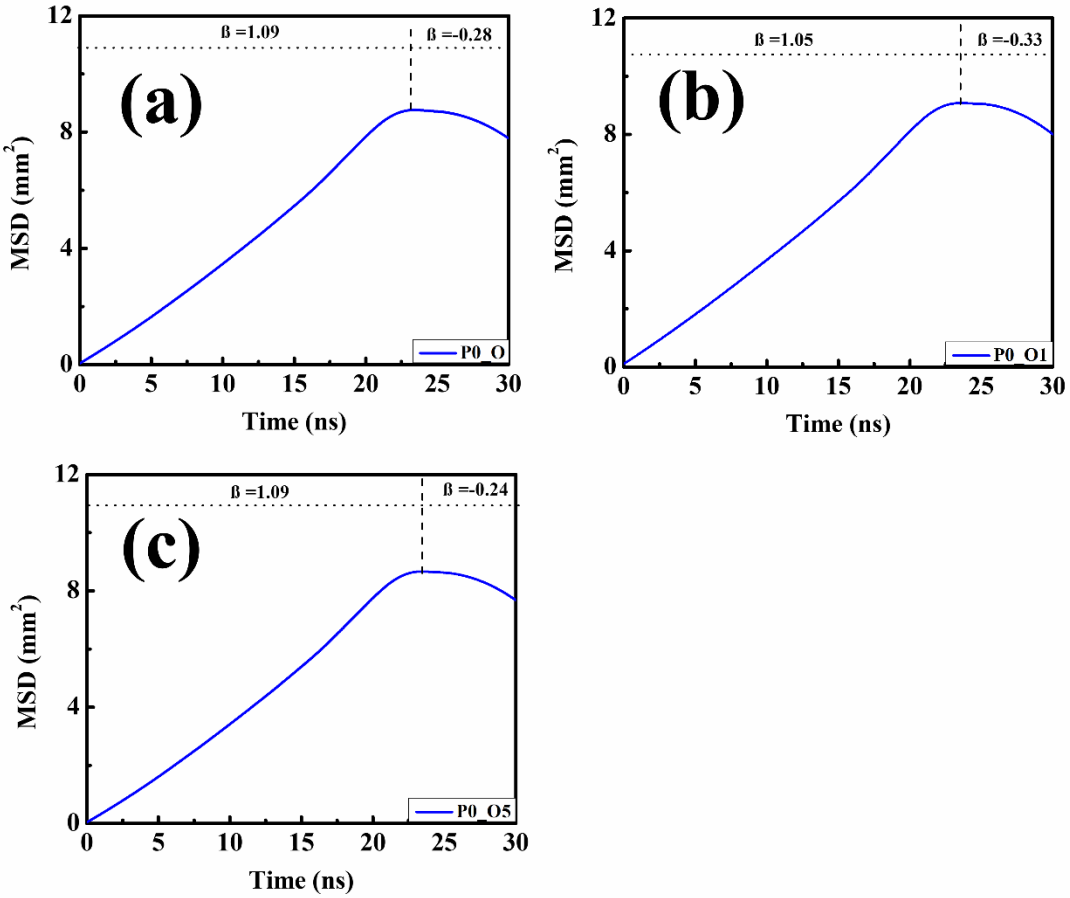


Figure. S49 MSD curve of procyanidins in ethanol-based system I (1:1) (a)O (denoted by f_1) (b) O1 (denoted by f_2) and (c) O5 (denoted by f_3).



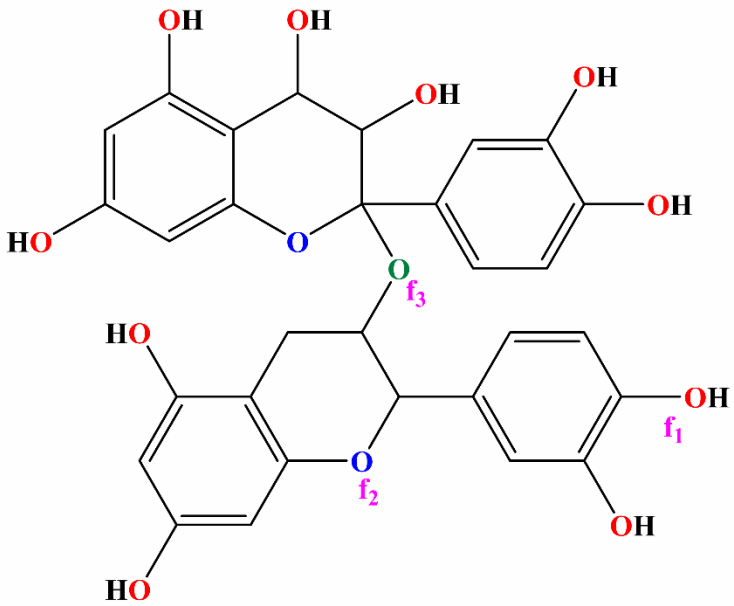


Figure. S50 MSD curve of procyanidins in ethanol-based system II (1:9) (a)O (denoted by f_1) (b) O1 (denoted by f_2) and (c) O5 (denoted by f_3).

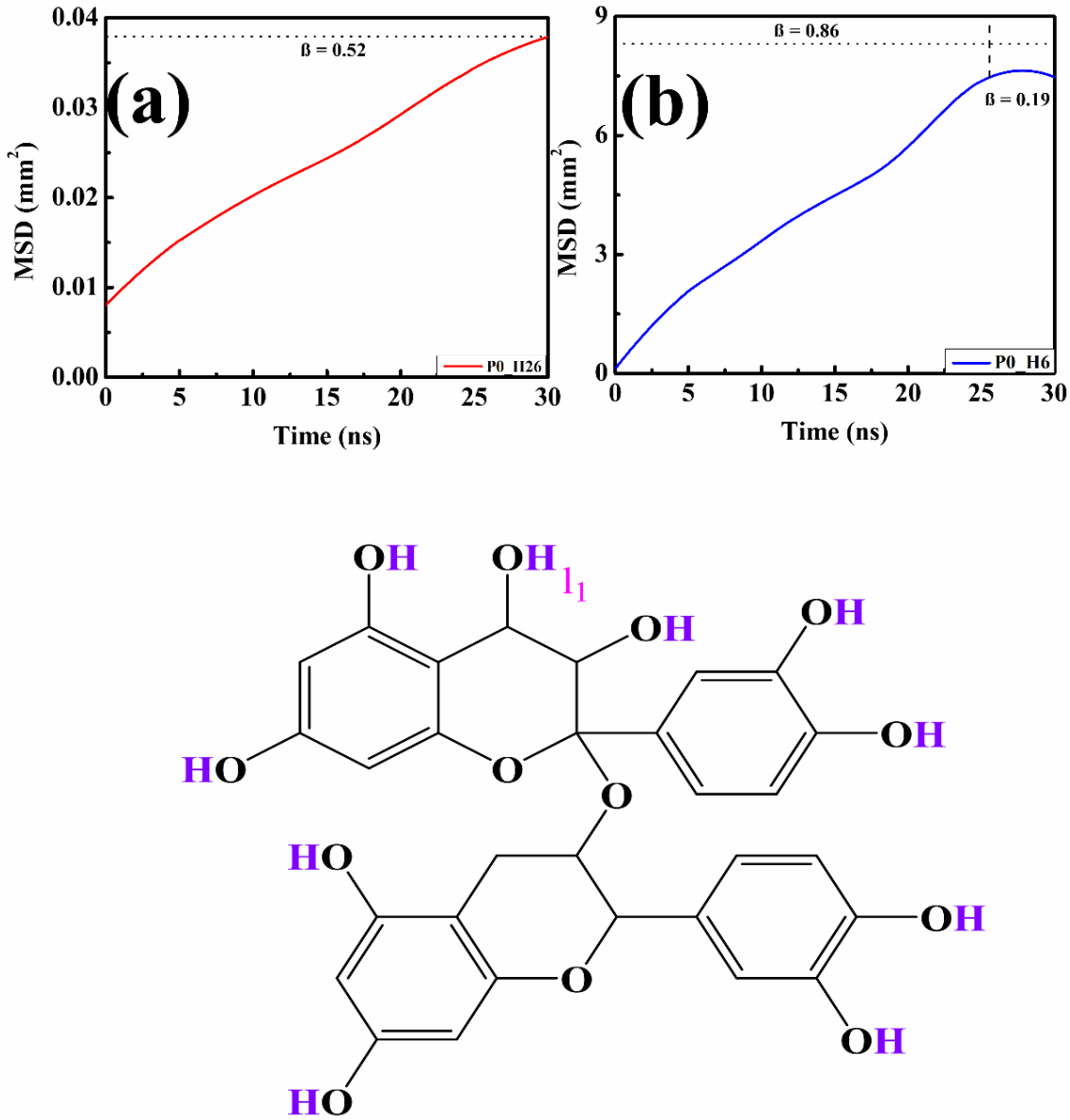


Figure. S51. MSD curve of Procyanidins in ethyl acetate-based system (a)H6 (system I) (denoted by l_1) (b) H6 (system II) (denoted by l_1).

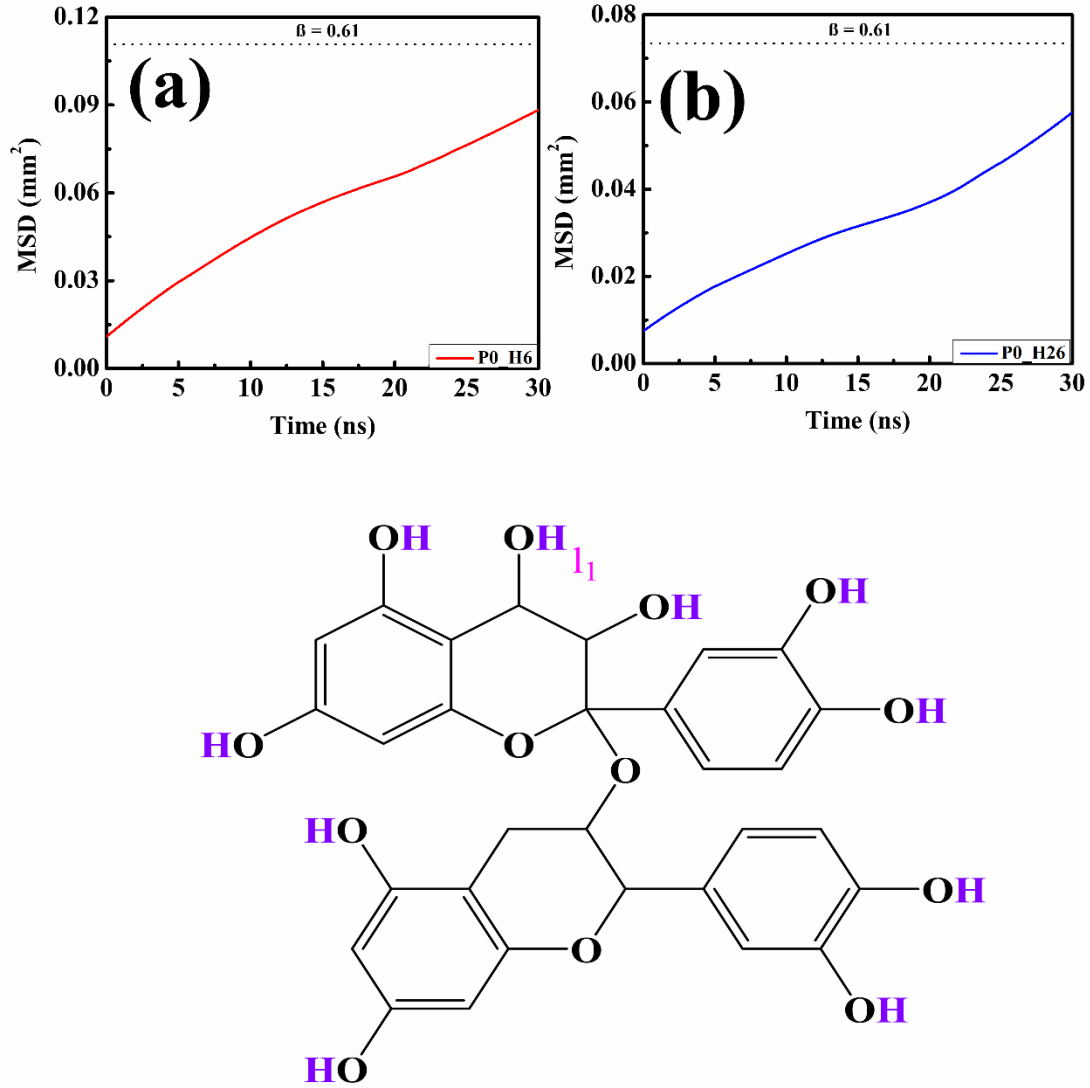


Figure. S52. MSD curve of Procyanidins in DES based system (a)H6 (system I) (denoted by l_1) (b) H6 (system II) (denoted by l_1).

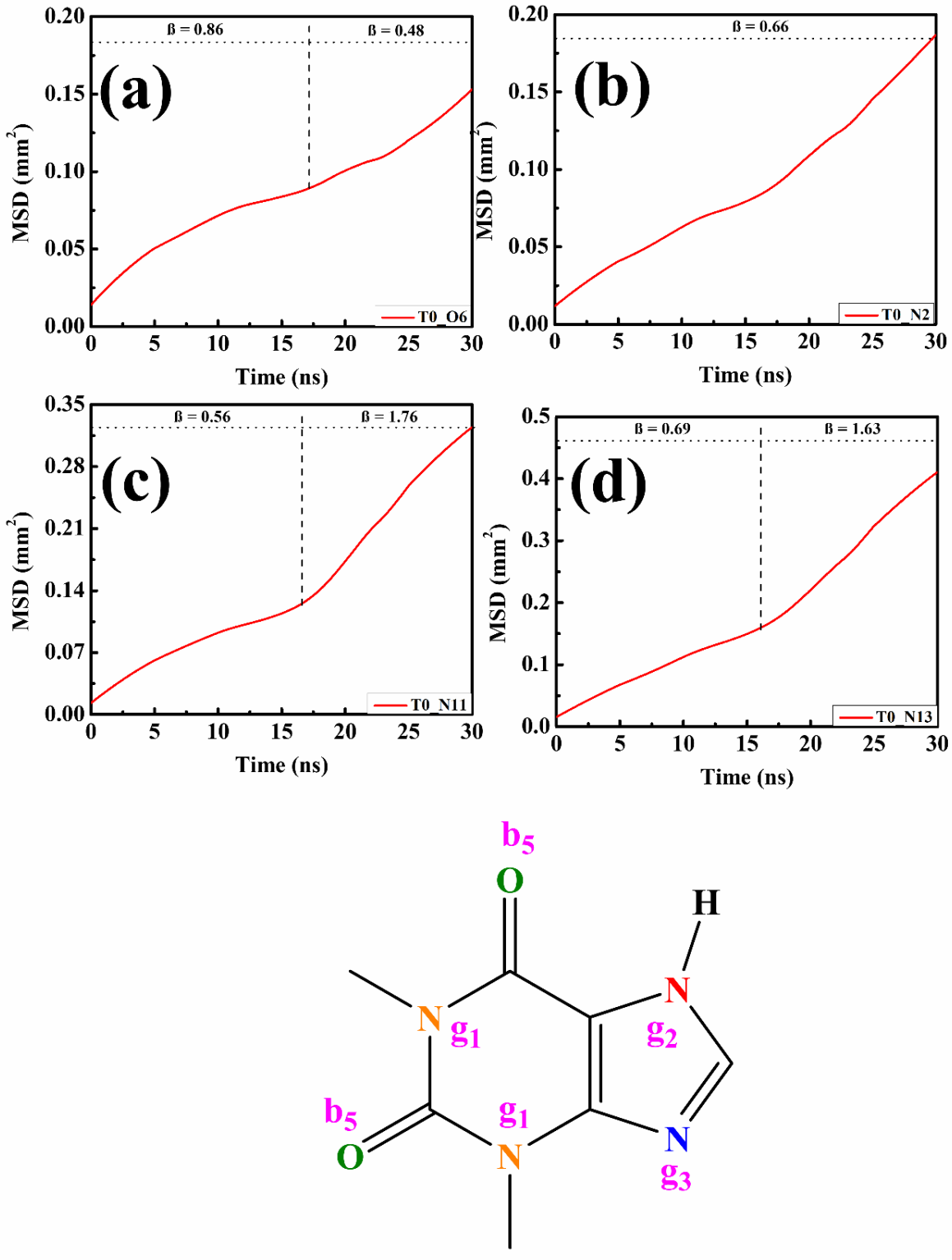


Figure. S53 MSD curve of Theophylline in ethanol-based system I (1:1) (a)O6 (denoted by g_1) (b) N2 (denoted by g_2) (c)N11 (denoted by g_3) and (c) N13 (denoted by g_4).

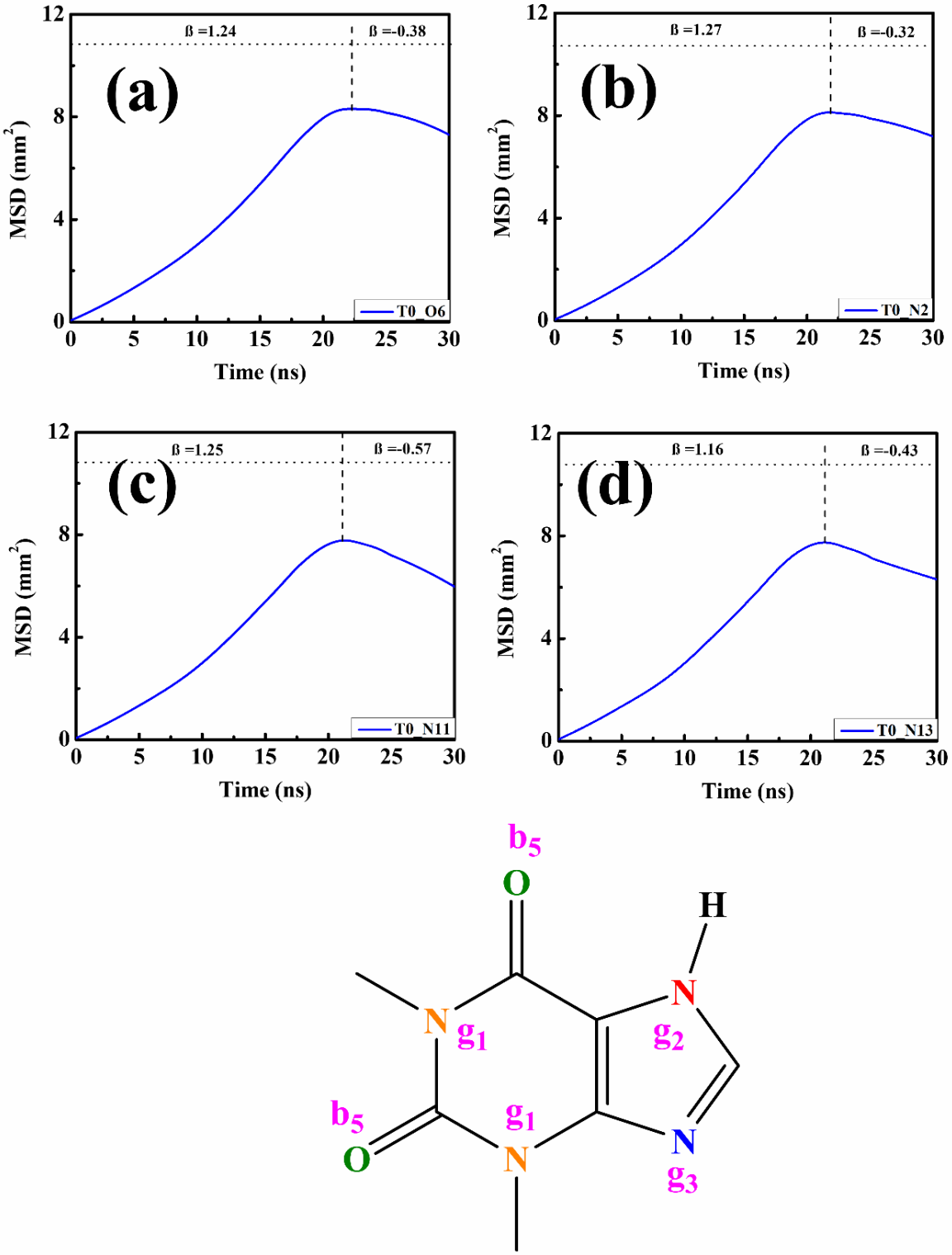


Figure. S54 MSD curve of Theophylline in ethanol-based system II (1:9) (a)O6 (denoted by g_1) (b) N2 (denoted by g_2) (c)N11 (denoted by g_3) and (c) N13 (denoted by g_4).

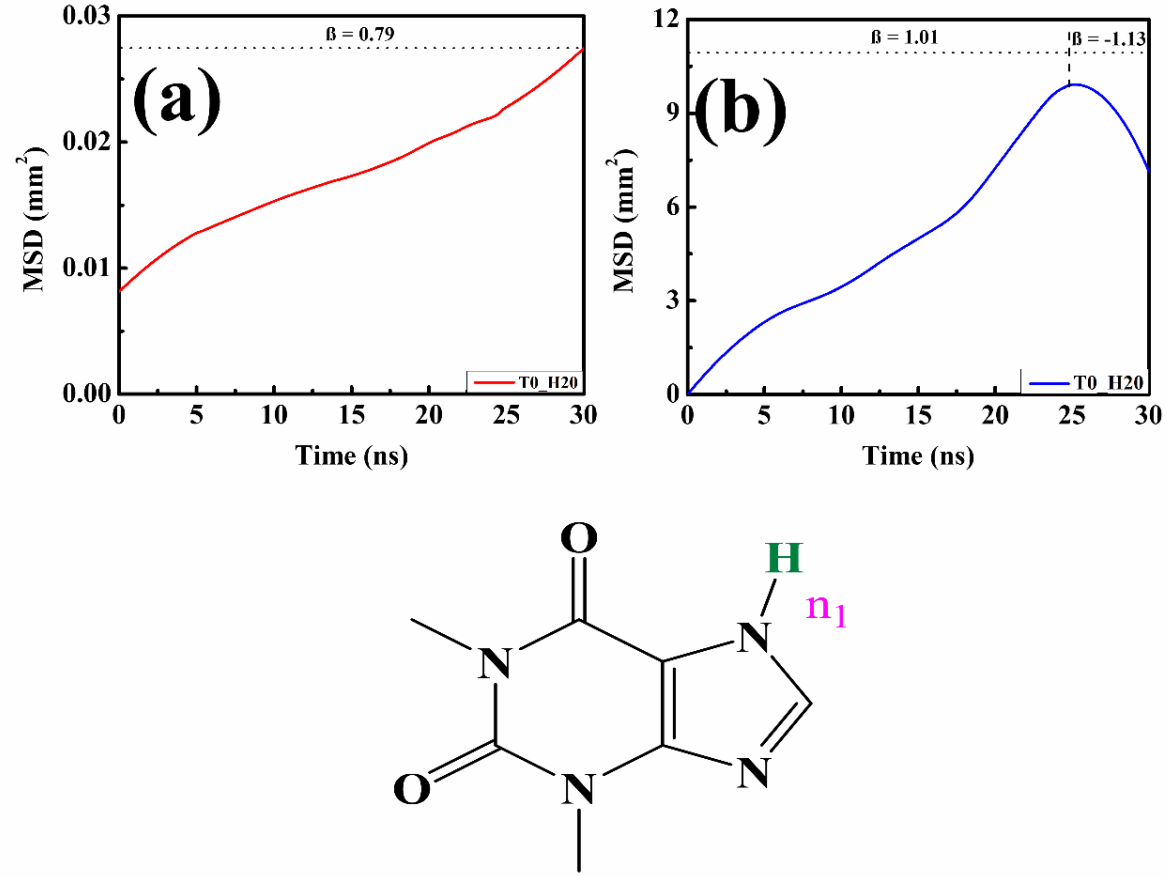


Figure. S55. MSD curve of Theophylline in ethyl acetate-based system (a)H6 (system I)(denoted by n_1) (b) H6 (system II) (denoted by n_1).

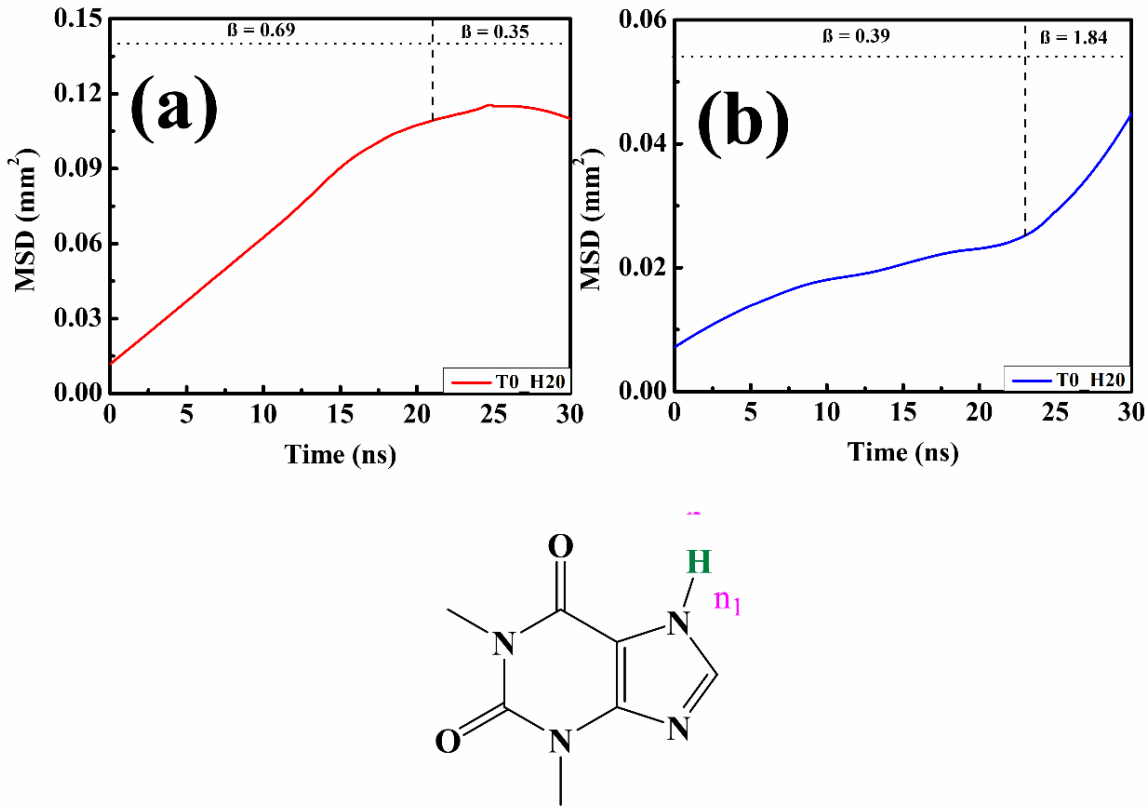
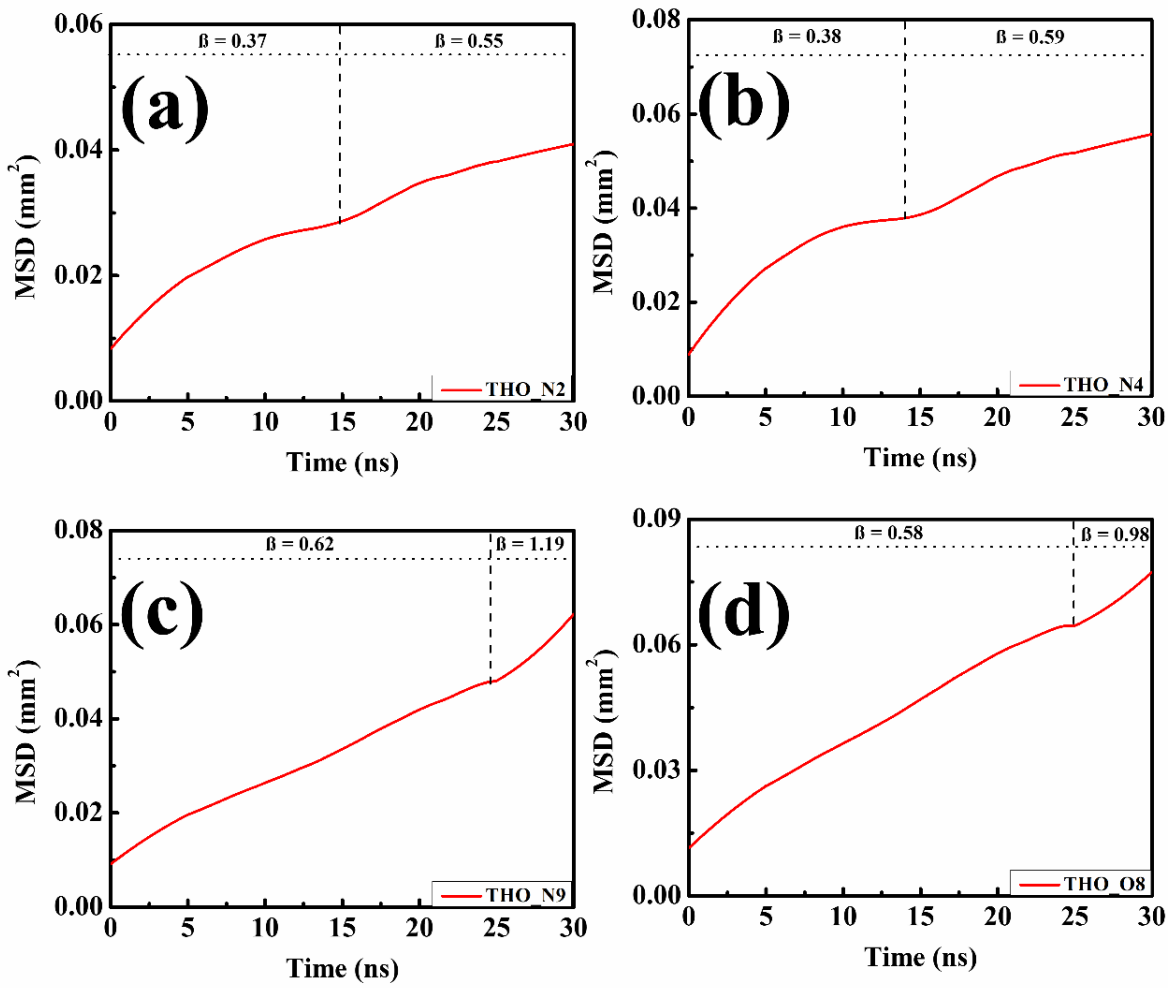


Figure. S56. MSD curve of theophylline in DES based system (a)H6 (system I) (denoted by n_1) (b) H6 (system II) (denoted by n_1).



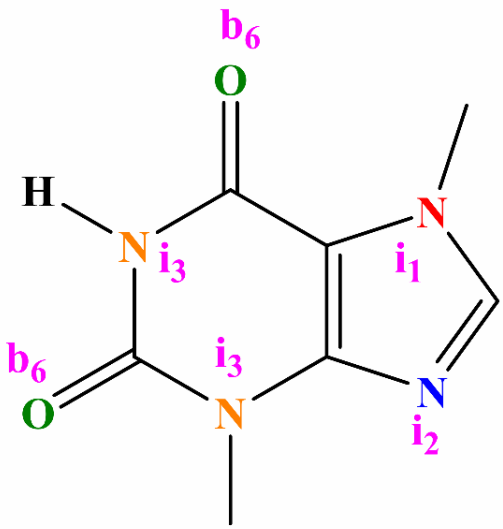
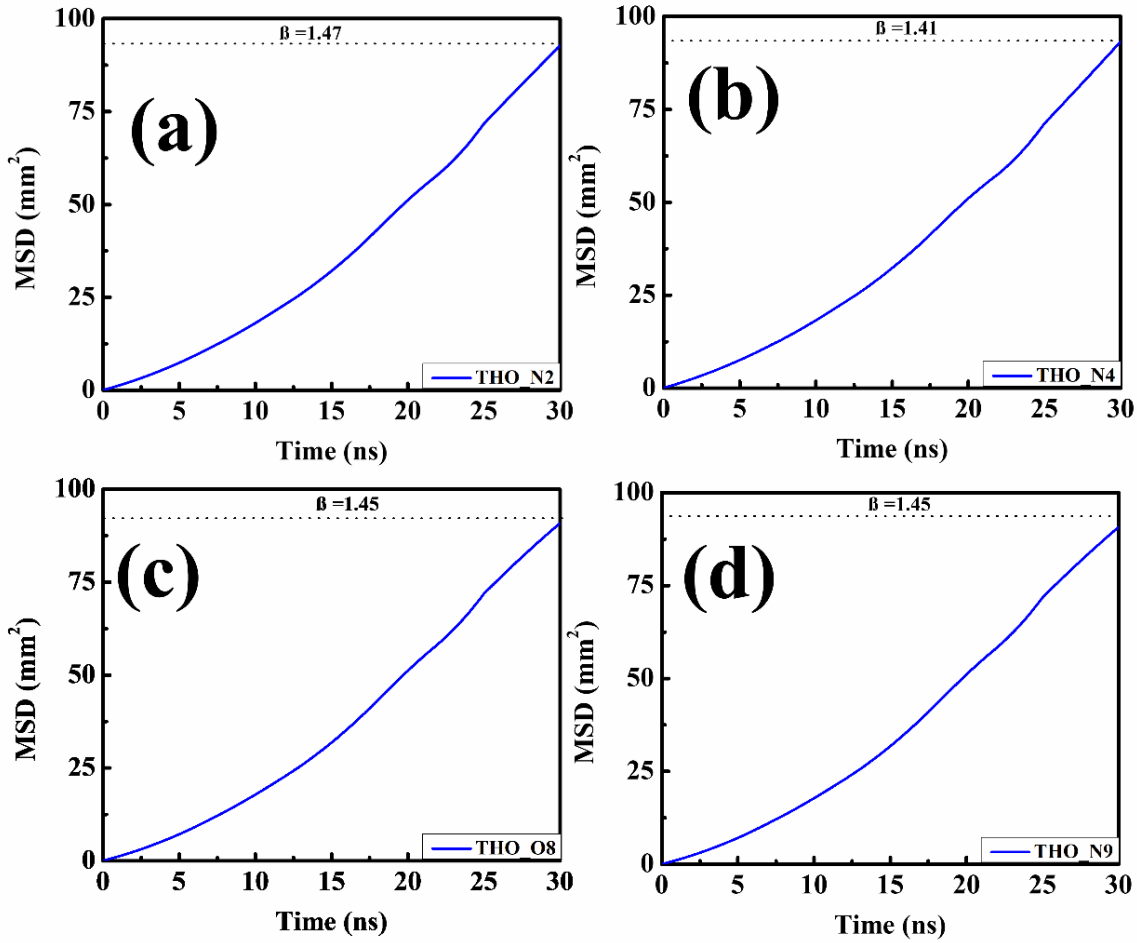


Figure. S57 MSD curve of Theobromine in ethanol-based system I (1:1) (a)N2 (denoted by i_1) (b) N4 (denoted by i_2) (c)N9 (denoted by i_3) and (d) O8 (denoted by b_6).



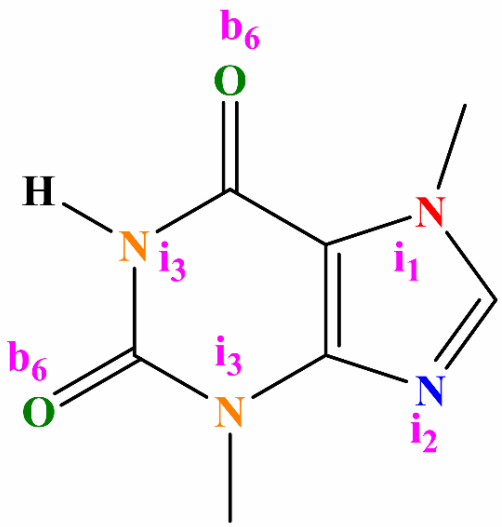


Figure. S58 MSD curve of Theobromine in ethanol-based system I (1:1) (a)N2 (denoted by i_1) (b) N4 (denoted by i_2) (c)N9 (denoted by i_3) and (d) O8 (denoted by b_6).

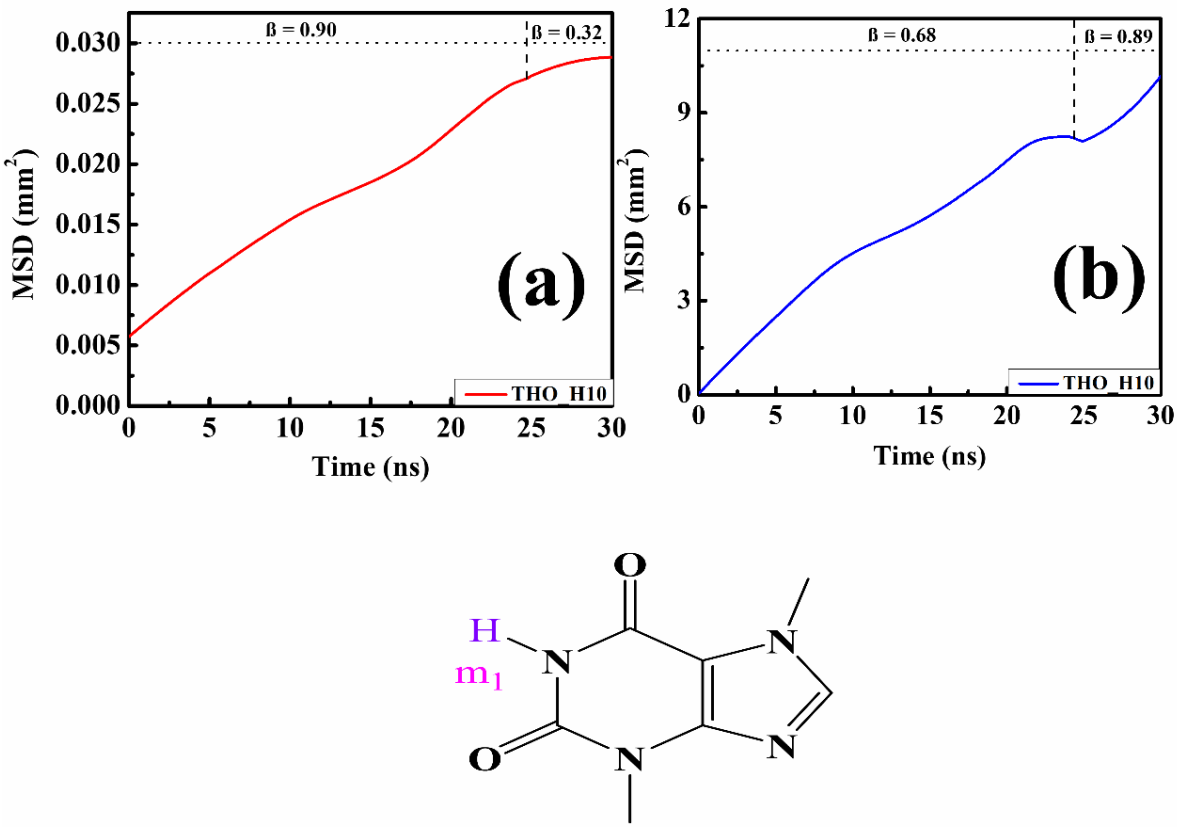


Figure. S59. MSD curve of Theobromine in ethyl acetate-based system (a)H20 (system I) (denoted by m_1) (b) H20 (system II) (denoted by m_1).

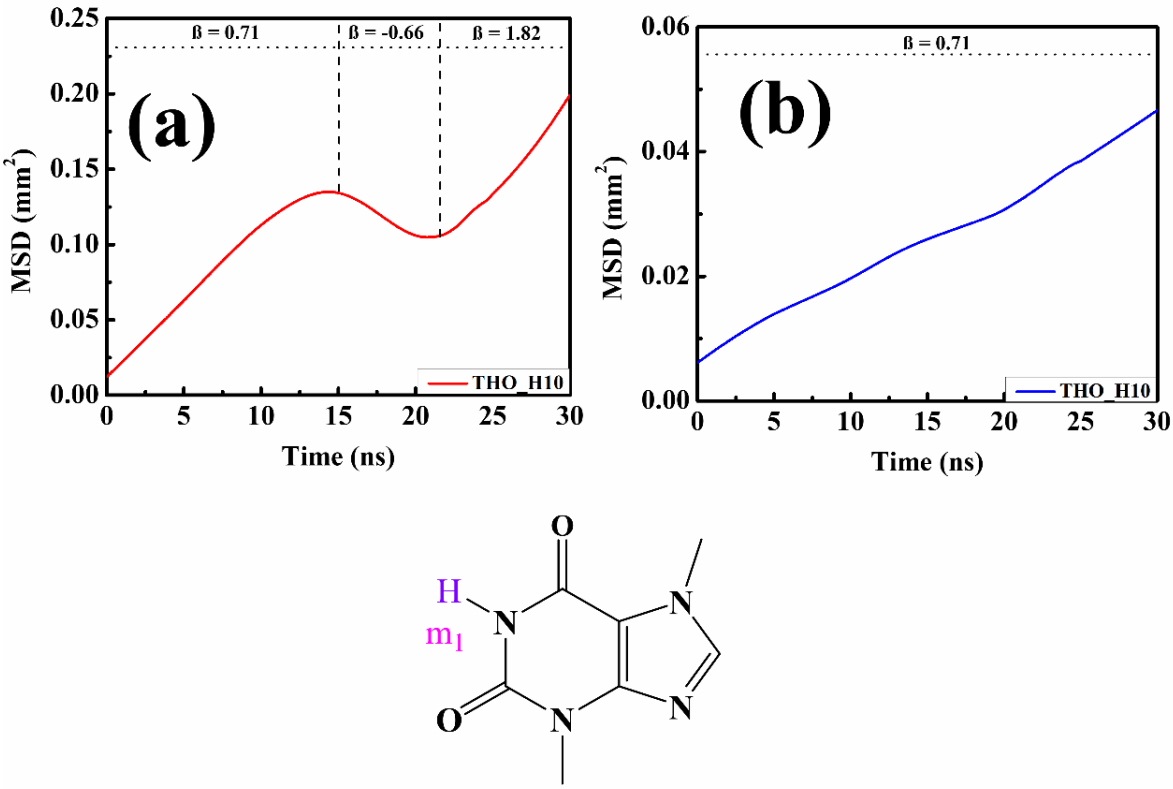


Figure.S60 MSD curve of Theobromine in ethyl acetate-based system (a)H20 (system I) (denoted by m_1) (b) H20 (system II) (denoted by m_1).